iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

nature biotechnology
ARTICLES
https://doi.org/10.1038/s41587-021-01139-4

Check for updates

Cell2location maps fine-grained cell types in spatial transcriptomics

Vitalii Kleshchevnikov[1], Artem Shmatko[1,2], Emma Dann[1], Alexander Aivazidis[1], Hamish W. King[1,3], Tong Li[1], Rasa Elmentaite[1], Artem Lomakin[4,5], Veronika Kedlian[1], Adam Gayoso[6], Mika Sarkin Jain[1,7], Jun Sung Park[1,4], Lauma Ramona[1], Elizabeth Tuck[1], Anna Arutyunyan[1], Roser Vento-Tormo[1], Moritz Gerstung[4,5], Louisa James[3], Oliver Stegle[1,5,8] ✉ and Omer Ali Bayraktar[1] ✉

Spatial transcriptomic technologies promise to resolve cellular wiring diagrams of tissues in health and disease, but comprehensive mapping of cell types in situ remains a challenge. Here we present cell2location, a Bayesian model that can resolve fine-grained cell types in spatial transcriptomic data and create comprehensive cellular maps of diverse tissues. Cell2location accounts for technical sources of variation and borrows statistical strength across locations, thereby enabling the integration of single-cell and spatial transcriptomics with higher sensitivity and resolution than existing tools. We assessed cell2location in three different tissues and show improved mapping of fine-grained cell types. In the mouse brain, we discovered fine regional astrocyte subtypes across the thalamus and hypothalamus. In the human lymph node, we spatially mapped a rare pre-germinal center B cell population. In the human gut, we resolved fine immune cell populations in lymphoid follicles. Collectively, our results present cell2location as a versatile analysis tool for mapping tissue architectures in a comprehensive manner.

The cellular architecture of tissues, where distinct cell types are organized in space, underlies cell–cell communication, organ function and pathology. Emerging spatial transcriptomics technologies provide key opportunities to map resident cell types and cell signaling in situ in a scalable manner. Despite existing proof-of-concept applications of these technologies[1–6], it remains a challenge to define spatial transcriptomics workflows that allow comprehensive mapping of resident cell types in tissues. One reason is the enormous cellular diversity across organs, including myriad fine-grained cell types such as immune[7,8], stromal[7,9] and neuronal[10,11] subpopulations. These fine cell types, marked by subtle transcription differences, are often specialized across tissues and yield large potential for discoveries. Another reason is the diversity of spatial tissue architecture, ranging from the brain featuring discrete anatomical regions with distinct cell types to immune organs featuring dynamically modified microenvironments with fine-grained cell types. To create comprehensive cellular wiring diagrams of tissues with spatial transcriptomics, it is necessary to establish experimental and computational methods that are sufficiently sensitive to resolve fine-grained differences in cell types across diverse tissues.

Strategies that generate coupled single-cell and spatially resolved transcriptomics offer a scalable approach to address these challenges. The key principle is to first identify resident cell types based on single-cell RNA sequencing (scRNA-seq) from dissociated tissues and then map the identified cell types to their tissue positions in situ based on spatial transcriptomic profiles[12]. Here, spatial RNA-seq technologies, such as Visium[13], HDST[14] and Slide sequencing (Slide-seq)[2], where mRNAs are positionally captured from thin tissue sections using microarray or bead array grids, provide transcriptome-wide data at high throughput and rely on simple histology and molecular biology protocols.

A key limitation of existing grid-based profiling methods is the lack of single-cell resolution, as spatial RNA-seq measurements combine multiple cells (Visium and Tomo-Seq)[13,15] or fractions of multiple cells (Slide-seq and HDST)[2,14] at each tissue location. This challenge can be overcome by the computational integration of spatial RNA-seq with the reference transcriptome signatures of cell types obtained from coupled scRNA-seq profiles. Although, recently, several computational methods have been proposed for this task[1–3,5,6], it is not clear whether these methods offer sufficient sensitivity to resolve fine-grained cell types in complex tissues, which is critical to make full use of the discovery potential of spatial transcriptomic data. Additionally, different confounding sources of variation need to be accounted for, which enables more complex integration tasks of data that consist of multiple experiments, including variable cell numbers across tissue positions and variable total mRNA content across cell types and individual cells[16,17].

To address the aforementioned challenges, we present cell2location, a Bayesian model that is designed to resolve fine-grained cell types in spatial transcriptomic data and create cellular maps of diverse tissues. Cell2location can handle complex experimental settings, performing joint analysis of multiple scRNA-seq and spatial transcriptomic datasets. Unique to the model is a hierarchical design to borrow statistical strength across locations, which yields

[1]Wellcome Sanger Institute, Hinxton, Cambridge, UK. [2]Moscow State University, Leninskie Gory, Moscow, Russia. [3]Centre for Immunobiology, Blizard Institute, Queen Mary University of London, London, UK. [4]European Molecular Biology Laboratory, European Bioinformatics Institute (EMBL-EBI), Hinxton, UK. [5]European Molecular Biology Laboratory, Genome Biology Unit, Heidelberg, Germany. [6]Center for Computational Biology, University of California, Berkeley, Berkeley CA, USA. [7]Theory of Condensed Matter, Department of Physics, Cavendish Laboratory, University of Cambridge, Cambridge, UK. [8]Division of Computational Genomics and Systems Genetics, German Cancer Research Center (DKFZ), Heidelberg, Germany.
✉e-mail: oliver.stegle@embl.de; ob5@sanger.ac.uk

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

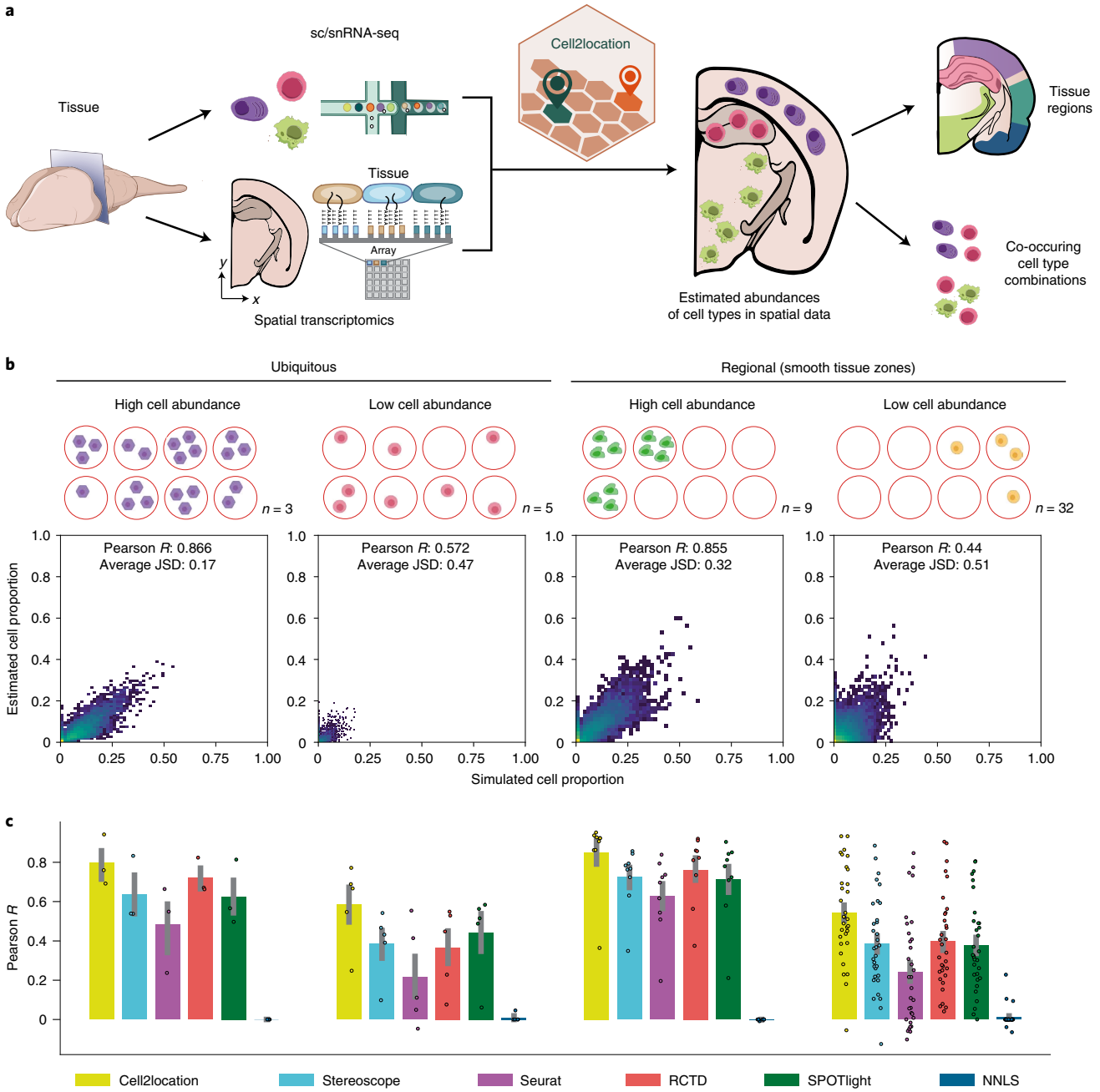


Fig. 1 | Cell2location overview and validation using simulated data. a, Overview of the spatial mapping approach and the analysis workflow of cell2location. From left to right: scRNA-seq and spatial transcriptomics profiles are generated from the same or related tissues. Cell2location takes reference cell type signatures derived from scRNA-seq and spatial transcriptomics data as input. The model then decomposes spatially resolved multi-cell RNA count matrices into the reference cell type signatures, thereby estimating cell abundance of individual cell types across locations. **b**, Model validation using simulated data. A benchmark dataset is constructed by combining cells drawn from 49 reference cell types (obtained from mouse brain scRNA-seq; Fig. 2b), using alternative cell type abundance patterns. Top: four alternative cell type abundance patterns considered for data simulation (see also Supplementary Fig. 2) and the number of cell types assigned to each pattern (Methods). Bottom: 2D histogram plots, displaying the concordance between simulated (*x* axis) and estimated (*y* axis) cell type proportions across 2,500 locations, aggregated for cell types assigned to the respective pattern. Color denotes 2D histogram count (50 bins along both the *x* axis and *y* axis). Pearson *R* denotes Pearson correlation, and JSD denotes Jensen–Shannon divergence. **c**, Assessment of cell2location and alternative methods for estimation of cell type proportions for 49 cell types from the four respective abundance patterns (columns, *x* axis, 2,500 locations and 49 cell types). Shown are bar plots of Pearson correlation coefficients (*y* axis, *R*) between simulated and estimated cell type proportions for individual cell types across locations, considering alternative methods (color, *x* axis). Bar height denotes the average correlation coefficient across cell types; error bars denote standard errors.

improved resolution and sensitivity, in particular for resolving fine-grained cell types in complex tissues. Here we assess cell2location in three different tissues, where we consistently show improved sensitivity for mapping fine-grained cell types. Moreover, we map rare B cell states in the human lymph nodes and identify regional astrocyte subtypes in the mouse brain.

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

Results

Cell2location: a Bayesian model for spatial mapping of cell types. Cell2location maps the spatial distribution of cell types by integrating single-cell or nuclei RNA-seq and spatial transcriptomic data from a given tissue (Fig. 1a). The first step of the cell2location workflow is to estimate reference cell type signatures from single-cell and single-nucleus RNA-seq profiles, which correspond to gene expression profiles for a set of user-provided cell types and subpopulations identified using conventional clustering or other methods. By default, cell2location uses negative binomial regression for this step, which allows robust combination of data across technologies and batches (Methods and Supplementary Fig. 1). In a second step, cell2location uses these reference signatures and one or multiple spatial transcriptomic datasets as input to decompose mRNA counts at individual spatial locations into reference cell types (Fig. 1a and Supplementary Fig. 1).

Although cell2location is related to existing transcriptomic decomposition methods[1,3,5,6,18], it offers features to (1) borrow statistical strength across locations with similar cell composition, (2) account for batch variation across slides as well as variation in mRNA detection sensitivity and (3) estimate absolute cell type abundances by incorporating prior information about the analyzed tissues (see Supplementary Methods for details). In addition, cell2location is computationally efficient, owing to variational approximate inference and GPU acceleration. The cell2location software is integrated with the scvi-tools framework[19] and comes with a suite of downstream analysis tools, including the identification of groups of co-located cell types.

To validate cell2location, we initially considered simulated data that were generated to reflect diverse cell abundance patterns across ubiquitous and spatially restricted cell types (Fig. 1b). In brief, we created a spatial transcriptomics dataset with 2,500 locations by sampling cells from 49 reference cell types, using a combination of ubiquitous abundance patterns and cell types distributed according to smooth 'tissue zones' as observed in real data (Fig. 1b, Supplementary Fig. 2 and Methods). Reference cell type signatures and other simulation parameters were adapted from a mouse brain scRNA-seq dataset (Methods). Of note, unlike most benchmark datasets on real data (see below), this synthetic dataset provides ground truth information on the presence and absence of cell types as well as on quantitative abundance.

We then used the cell2location to estimate cell type proportions for different cell types at individual locations. Comparison of these estimates to the simulated (true) cell type proportions revealed that cell2location mapped cell types with high accuracy, irrespective of the cell abundance pattern (Fig. 1b), including low abundant and regional cell types.

Next, we compared cell2location to recently proposed methods for the estimation of relative cell type abundance from spatial transcriptomics data (Stereoscope[3], Seurat[1], RCTD[5], NNLS (AutoGeneS)[18,20] and SpotLIGHT[6]). Across the considered cell abundance patterns, we found that cell2location yielded more accurate estimates of the cell type proportions across locations than other methods (Fig. 1c). In addition to this quantitative assessment of cell type proportions, we also assessed the ability of alternative methods to determine locations where individual cell types are present—that is, have non-zero cell abundance—which yielded a consistent ranking of methods (Supplementary Fig. 3b).

We note that, in addition to estimating relative cell type proportions, cell2location can incorporate prior information about the tissue to estimate absolute cell type abundance, which, again, were highly concordant with the simulated ground truth (Supplementary Fig. 4a). Finally, we explored how specific design choices of cell2location contribute to its performance (Supplementary Fig. 4b), and we assessed the robustness of the model to incomplete reference cell type sets (Supplementary Fig. 4c) as well as variation in hyper-parameter settings (Supplementary Fig. 5).

Cell2location accurately maps mouse brain cell types. To validate cell2location in a real tissue, we first applied the model to data from the mouse brain, which features diverse neural cell types organized in a well-characterized spatial architecture across brain areas, thus presenting a canonical use case to test spatial genomics[1,3,5,6]. We generated matched single nucleus (sn) and Visium spatial RNA-seq (10x Genomics) profiles of adjacent mouse brain sections that contain multiple regions from the telencephalon and diencephalon (Fig. 2a). To assess the biological and intra-organ technical variation in spatial mapping, we assayed two mouse brains and serial tissue sections from each brain (total of three and two matched sections from two animals, respectively, and an extra section for snRNA-seq), creating a rich multi-modal and replicated transcriptomic dataset.

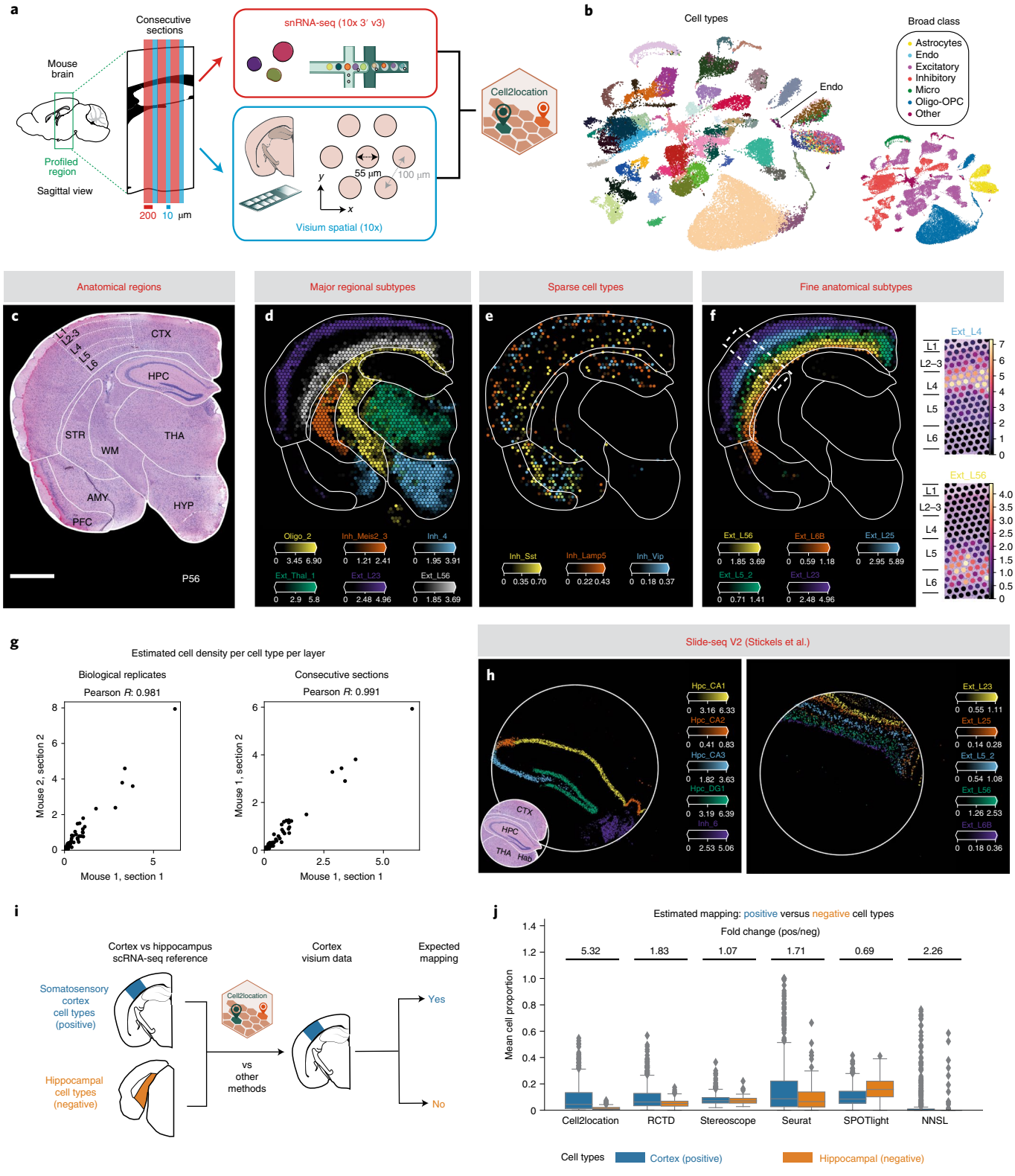
To define reference cell type signatures, we pooled snRNA-seq data across replicate animals and tissue sections (40,532 nuclei) and applied a conventional scRNA-seq workflow followed by Louvain clustering (Methods). This identified 51 putative cell type clusters, which were well represented across replicates (Supplementary Fig. 6). Of these, 44 clusters could be identified as expected cell types in the brain (Fig. 2b, Supplementary Fig. 7 and Methods), including astrocytes, excitatory or inhibitory neurons or glia subtypes, annotated by markers from prior literature[10,16,21]. The two astrocyte clusters

Fig. 2 | Comprehensive and accurate spatial mapping of cell types in the mouse brain. a, Experimental workflow. Adjacent tissue sections from two mice were used to generate single-nucleus and spatial data, across a total of five pairings of Visium and snRNA-seq, respectively (and an extra section for snRNA-seq). **b**, UMAP representation (*x* and *y* axis) of broad cell type classes (right) and 59 cell subtypes (left; see Supplementary Fig. 6a for labels), as identified by Louvain clustering of the integrated snRNA-seq dataset across sections from **a**. **c**, Mouse brain regions present in the assayed section. H&E image of the tissue section and the outlines of brain regions are shown. Scale bar, 1 mm. **d**, Estimated cell abundances (color intensity) of major cell types (color) across regions. **e**, Estimated cell abundances (color intensity) of sparse inhibitory neurons (color). **f**, Estimated cell abundances (color intensity) of cortical excitatory neurons (color). Small panels show the estimated cell abundance (color) for two select subtypes on the region indicated with the white bounding box. **g**, Consistency of cell2location cell abundance estimates across replicate slides (*x* and *y* axis), considering either replicate animals (left) or consecutive serial sections from the same animal (right). Considered are cell abundance estimates across 59 reference cell types across cortical layers; *R* denotes Pearson correlation. **h**, Spatial mapping in the Slide-seq V2 data. Left, estimated cell abundances (color intensity) of hippocampal neuron types (Hpc subtypes) and a thalamic habenular neuron type (Inh_6). Right, estimated cell abundances (color intensity) of cortical excitatory neurons. **i**, Benchmarking workflow for assessing the spatial mapping specificity of cell2location and alternative methods. scRNA-seq reference cell types[11] from manually dissected SSp cortex layers (top, positive set) and the par-/post-/pre-subiculum divisions of the hippocampus (bottom, negative set) were mapped to Visium data from the SSp cortex. **j**, Distribution of estimated cell type proportions at locations with high abundance of each cell type in the spatial data for positive cell types (SSp cortex, orange, *x* axis, $n = 98$) and negative cell types (hippocampus, blue, *x* axis, $n = 23$) in the SSp cortex region for alternative methods. Box plots show the distribution of cell proportions (99.8% quantile across locations within the SSp cortex) for each cell type and each replicate section (Supplementary Fig. 11e; $n = 5$ sections). Fold changes denote the difference in average 99.8% quantile of cell type proportions between positive and negative cell types. Box center, bounds and whiskers show median, quartiles and data points that lie within 1.5× interquartile range of the lower and upper quartiles, respectively. The observations that fall outside this range are displayed as dots.

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

could be further subdivided into ten transcriptional subtypes (see next section). Collectively, this resulted in reference cell type signatures for 59 distinct and annotated cell subtypes, which comprehensively cover the profiled brain areas (Supplementary Fig. 7).

We applied cell2location to map these cell types to spatial locations of the mouse brain Visium data, across replicate animals and sections. The resulting cell type mapping was markedly consistent with anatomical locations expected from prior literature (Fig. 2c–f and Supplementary Files 1–6). Our results captured diverse cell type patterns, resolving broad regional cell types such as excitatory neurons of thalamus and oligodendrocytes of white matter (Fig. 2d)[10,22], rare subtypes such as cortical interneurons (Fig. 2e)[23,24] and sub-regional subtypes such as excitatory neurons across cortical layers (Fig. 2f and Supplementary Fig. 8e)[23,24]. The cell type



iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

abundance estimates were highly concordant across adjacent tissue sections and replicate animals, as quantified across cortical layers (Fig. 2g and Supplementary Fig. 9). Finally, we clustered tissue locations based on their estimated cell type abundance profiles (Methods), which yielded clusters that recapitulate the anatomical brain areas, with the reference cell types mapping to the expected tissue regions (Supplementary Fig. 10).

Next, to assess the specificity of cell2location and alternative methods, we mapped a large existing scRNA-seq dataset that includes cell types from multiple brain regions[11], only some of which are present in our spatial data (Fig. 2i). Consequently, this dataset spans 'positive' cell types from the somatosensory (SSp) cortex ($n = 98$) as well as 'negative' cell types that are exclusively present in the hippocampus ($n = 23$)[11]. When mapping these cell types to our SSp cortex Visium data, we would expect that negative cell types are excluded and only positive cell types are assigned substantial cell type proportions.

Indeed, cell2location mapping yielded substantially lower cell proportion estimates for the negative compared to the positive cell types (Fig. 2j; positive/negative fold change = 5.1). By contrast, other methods yielded a substantially weaker contrast between negative and positive cell types (Fig. 2j; positive/negative fold change = 0.7–1.9 for other methods; Supplementary Fig. 11). This indicates that cell2location accurately excludes reference cell types that should not be present in a spatial dataset, whereaas other methods tend to 'overfit' reference cell types. We also used cell type proportion estimates derived from the scRNA-seq data, which were generated from manually dissected cortical layers, to confirm the quantitative cell type proportions estimated by cell2location (Supplementary Fig. 11 and Supplementary File 7).

Finally, to show the versatility of cell2location across spatial transcriptomic technologies, we mapped our mouse brain snRNA-seq reference to a Slide-seq dataset of the mouse brain that offers 10-µm spatial resolution[25]. Cell2location resolved cell types in Slide-seq spatial data, including fine anatomical subtypes across hippocampal divisions and cortical layers (Fig. 2h, Supplementary Fig. 12a and Supplementary Files 8 and 9). Moreover, cell2location revealed that more than 50% of Slide-seq 'beads' (that is, 10-µm-diameter spatial locations) contained more than one cell type (Supplementary Fig. 12b), consistent with previous observations[5].

Collectively, our results show that cell2location maps diverse brain cell types with high accuracy and reproducibility and that the method is broadly applicable to spatial transcriptomic technologies.

Identification of regional astrocyte subtypes. Next, we assessed cell2location for mapping fine-grained cell types in the mouse brain. We focused on the heterogeneity of astrocytes, which is the largest class of glial cells that intimately regulate neuronal circuit development and function[26]. Although astrocytes were historically considered to be a homogeneous cell type, recent evidence suggests that astrocytes are regionally diversified across the brain[26], including fine anatomical divisions such as cortical layers[27–29]. However, the extent of the regional heterogeneity of astrocytes is largely unknown, as existing studies have primarily compared their prevalence across selected brain areas[30,31] or described broad astrocyte subtypes across major brain divisions (for example, telencephalon versus diencephalon)[10,32].

To identify regionally enriched astrocytes, we annotated and spatially mapped ten molecularly and spatially distinct astrocyte subtypes that were identified based on our snRNA-seq reference. In brief, we considered 3,013 cells annotated as astrocytes in our initial analysis (Fig. 2b). We then applied BBKNN[33] to further refine the integration of these cells across all six snRNA-seq datasets (Methods), followed by Leiden clustering, which identified 17 transcriptomically defined subclusters (Supplementary Fig. 13). Next, we performed an initial mapping of these putative cell subtypes using cell2location, merging clusters that were not sufficiently distinct in both their mapped spatial locations and marker gene expression profiles (Supplementary Figs. 13 and 14 and Methods). This resulted in ten molecularly and spatially distinct astrocyte subtypes (Fig. 3a,b and Supplementary Fig. 15), which we considered for further analysis. The extent of molecular heterogeneity across these subtypes (measured by the number of marker genes) was substantially lower than across neuronal subpopulations, confirming the transcriptionally fine-grained nature of these cell populations (Supplementary Fig. 13d).

We identified gray matter astrocyte subtypes that were spatially enriched in the thalamus (THAL), hypothalamus (HYPO), cortex (CTX), amygdala (AMY), hippocampus (HPC) and striatum (STR) as well as white matter astrocytes (Fig. 3b and Supplementary Fig. 15). These subtypes expressed expected regional astrocyte markers[10], such as *Foxg1* in the telencephalon (CTX, AMY, HPC and STR) and *Slc6a11* and *Agt* in the diencephalon (THAL and HYPO) (Supplementary Fig. 15). Moreover, they stratified to finer regional subtypes based on the expression of subtype-specific genes such as *Angpt1* and combinatorially expressed markers such as *Gria1* (Supplementary Fig. 15; see below).

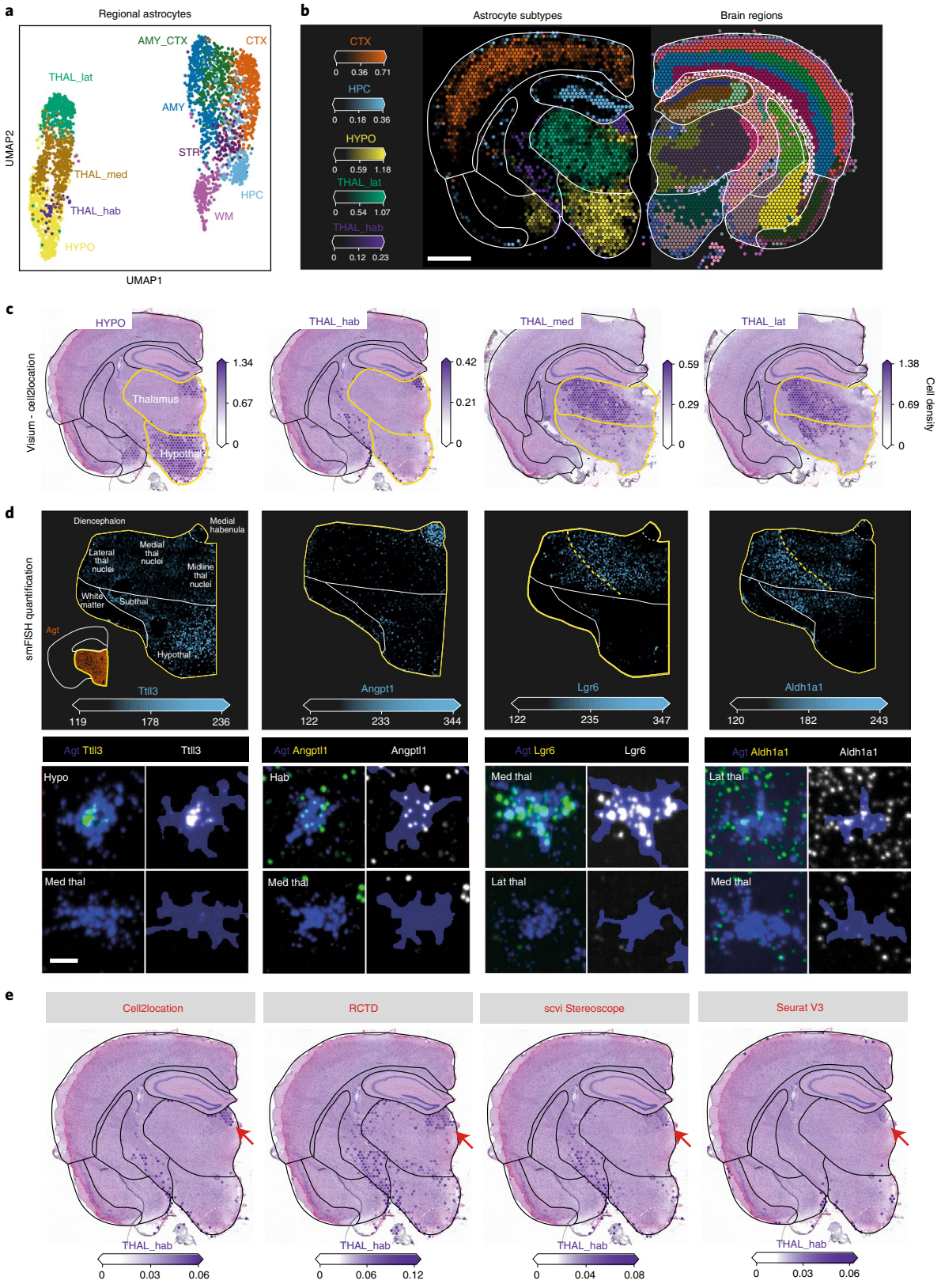
We focused on diencephalic astrocyte subtypes, as only one type of gray matter astrocyte subtype has been described in this brain area to date[10]. Notably, our data revealed four regionally distinct gray matter astrocyte subpopulations across the diencephalon (Fig. 3c and Supplementary Fig. 15). HYPO astrocytes were mainly located in the hypothalamus and, to a lesser extent, in the midline thalamic nuclei and the lateral habenula, and they were marked by *Ttll3* expression. Three THAL subtypes occupied different regions of the thalamus. THAL_hab astrocytes, a rare cell population in our reference (Supplementary Fig. 6d), were restricted to the medial habenula and ventral white matter tracts and specifically expressed *Angpt1*. THAL_med astrocytes were enriched in the medial thalamic nuclei and were marked by *Lgr6* expression. Finally, THAL_lat astrocytes, although present throughout the thalamus and the subthalamus, were most prevalent in ventrolateral thalamic nuclei and were marked by elevated *Aldh1a1* expression compared to other subtypes (Fig. 3c and Supplementary Fig. 15).

We validated the identity and distribution of diencephalic astrocyte subtypes using single-molecule fluorescence in situ hybridization (smFISH)[27]. We used *Agt* smFISH to specifically label and

Fig. 3 | Spatial discovery of regional astrocyte subtypes. a, UMAP representation (*x* and *y* axis) of ten molecularly and spatially distinct astrocyte subtypes (color) identified in our snRNA-seq reference data. **b**, Estimated cell abundances (color intensity) of selected regional astrocyte subtypes (left hemisphere) and brain regions as identified using clustering of locations (right hemisphere). Scale bar, 1 mm. **c**, Estimated cell abundances (color intensity) of four diencephalic astrocyte subtypes. Thal_med and Thal_lat are shown from a more representative section from the same mouse brain. **d**, smFISH validation. Top panels: Small map shows locations of diencephalic astrocytes segmented based on Agt smFISH. Large maps show quantified subtype marker expression (color intensity) at a single-astrocyte level (dots). Yellow dashed line indicates the medial boundary of high Aldh1a1 expression. Bottom panels, close-up images of diencephalic astrocyte subtypes. First columns show smFISH for astrocyte marker Agt and subtype markers. Second columns show astrocyte segmentation masks (blue) transparently overlaid on subtype genes. $n = 1$ mouse and $n = 3$ technical replicate tissue sections for each marker independently imaged. Scale bar, 10 µm. **e**, Estimated cell proportions (color intensity) for mapping THAL_hab astrocyte subtypes, considering the four best-performing methods (columns). Red arrows indicate the location of the medial habenula.

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

segment astrocytes throughout the diencephalon[10] (Supplementary Figs. 16 and 17) and then quantified the expression of co-stained subtype markers at single-astrocyte resolution (Fig. 3d). The in situ expression patterns of regional astrocyte markers (*Ttll3*, *Angpt1*, *Lgr6* and *Aldh1a1*) validated the spatial mapping of their respective astrocyte subtypes by cell2location (Fig. 3d). Finally, we also considered alternative tools for spatially mapping regional astrocytes, finding that cell2location was the only transcriptomic decomposition method that accurately mapped THAL_hab astrocytes to their validated tissue location (Fig. 3e and Supplementary Fig. 18).



iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

These results validate diencephalic astrocyte subtypes and show that cell2location can spatially map fine-grained cell types.

Resolving the cellular compartments of the human lymph node. To test cell2location on diverse tissue architectures, we applied the model to spatially map tissue microenvironments of the human lymph node. Unlike the mouse brain, the human lymph node is characterized by dynamic microenvironments with many spatially interlaced cell populations. We analyzed a publicly available Visium dataset of the human lymph node from 10x Genomics[34] and spatially mapped a comprehensive atlas of 34 reference cell types derived by integration of scRNA-seq datasets from human secondary lymphoid organs composed of 73,260 cells[35–37] (Fig. 4a,b, Supplementary Fig. 19 and Methods).

Histological examination of the lymph node Visium sample revealed multiple germinal centers (GCs). Correspondingly, after mapping with cell2location, the expected major regions within the lymph node tissue could be readily identified, including T cell and B cell zones and GCs with follicular dendritic cells (FDCs) (Fig. 4c, right panel, and Supplementary Files 10 and 11).

Next, we used the cell2location mapping results to identify the spatial co-occurrence of cell types to better understand the tissue organization and predict cellular interactions. We performed non-negative matrix factorization (NMF) of the cell type abundance estimates from cell2location (Fig. 4d). Similarly to the established benefits of applying NMF to conventional scRNA-seq[38,39], the additive NMF decomposition yielded a grouping of spatial cell type abundance profiles into components that capture co-localized cell types. This NMF-based decomposition naturally accounts for the fact that multiple cell types and microenvironments can co-exist at the same Visium locations (Supplementary Fig. 20) while sharing information across tissue areas (for example, individual GCs).

We observed that the groups of cell types identified by NMF corresponded to known functionally relevant cellular compartments of the lymph node (Fig. 4d). Furthermore, the NMF output enabled the dissection of the spatial compartments of B cell maturation. Specifically, naive B cells were mapped to a B follicle zone distinct from GCs (Fig. 4e, panel 1). We also identified the GC dark zone where B cells clonally expand and proliferate (Fig. 4e, panel 2) as well as the GC light zone where they undergo selection by T follicular helper cells and FDCs to differentiate into antibody-producing plasmablasts (Fig. 4e, panel 3).

In addition to confirming known biology, the cell2location combined with the NMF decomposition enabled the fine-grained spatial mapping of a rare secondary activation B cell state, which was recently reported as primed to undergo class switch recombination preceding the GC entry[37,40]. Our data indicate that this population (termed here 'preGC') occupies a different tissue compartment from the upstream activated cells and downstream GC cells (Fig. 4e, panel 4), suggesting that it is more likely to initiate new GC reactions rather than to join existing ones. Finally, our analysis was capable of distinguishing blood vessel zones containing endothelial and vascular smooth muscle cells (Supplementary Fig. 21a, panel 5) from the adjacent perivascular zone enriched in T helper and B memory cells (Supplementary Fig. 21a, panel 6).

Taken together, our results show that cell2location can map complex tissues characterized by spatially interlaced cell types and aid biological interpretation by identifying putatively interacting cell types.

Benchmarking cell2location for mapping fine immune cell types. Cell atlasing studies have greatly expanded the catalog of fine-grained cell types across human tissues, which are often of key interest for understanding tissue function and pathology[7,9,11]. To quantitatively benchmark cell2location and alternative methods for mapping transcriptionally fine-grained cell types, we considered immune cell populations across two human tissues where many fine subtypes with specialized immune functions can be identified[7–9]. For example, T follicular helper cells differ from the classical CD4$^+$ T cells in their ability to enter the GCs and guide B cell differentiation into plasma cells and memory B cells to ensure long-lasting immunity[41].

Initially, we examined GC-specific immune cell types in the human lymph node (Fig. 4). We classified six known GC cell types as either transcriptionally fine or transcriptionally distinct, based on the Euclidean distance of their expression signatures to the closest other cell type (Supplementary Fig. 22a and Methods). We annotated the anatomical locations of GCs using Visium histology images (Fig. 4f) and assessed the accuracy of alternative methods in mapping the six GC cell types to the expected GC locations. Although all considered methods achieved high accuracy for mapping distinct cell types such as FDCs (Fig. 4h), cell2location was markedly more accurate in mapping fine-grained cell types such as GC T follicular helper cells (Fig. 4f–h and Supplementary Figs. 22–24).

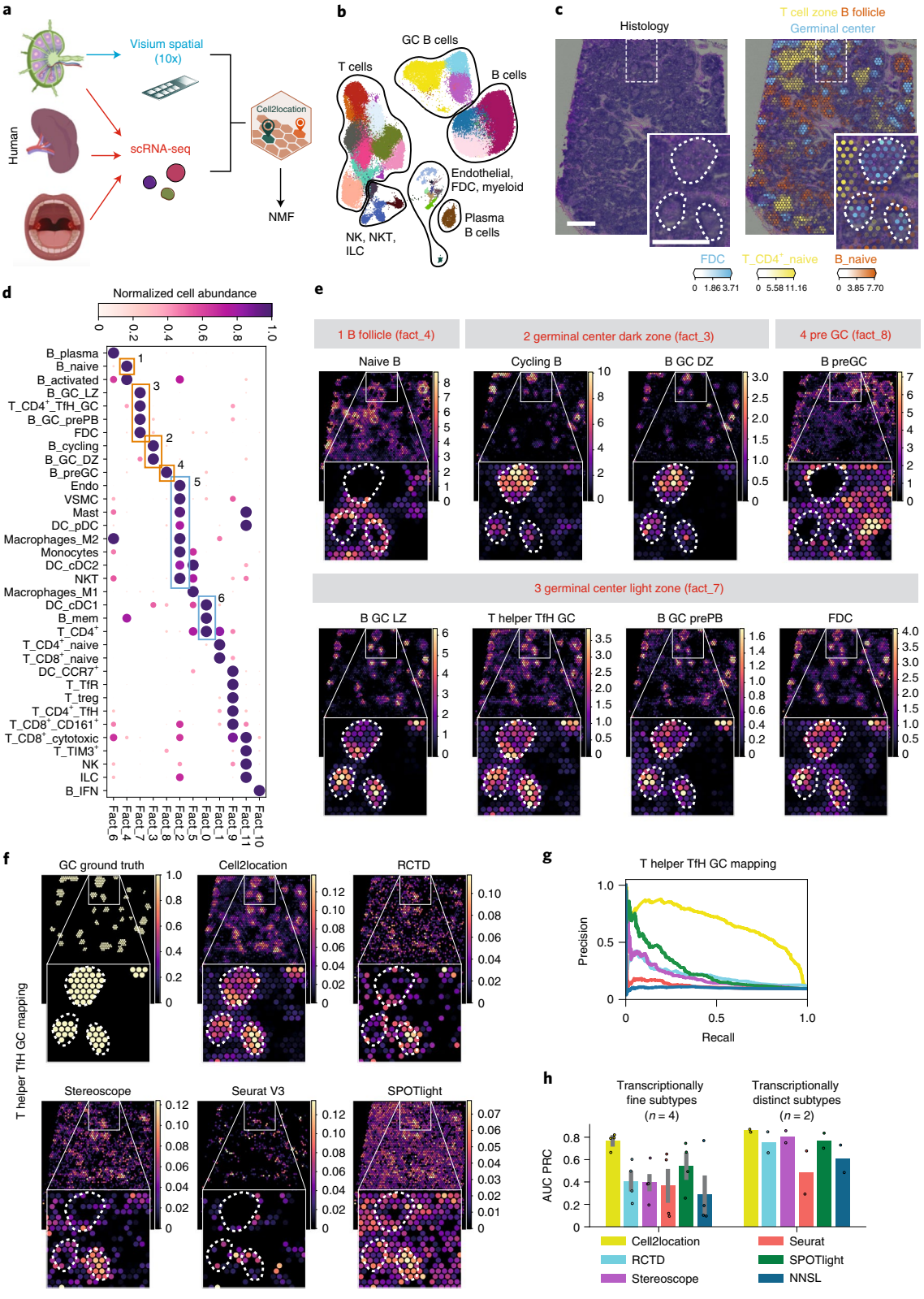
To extend our benchmarking to a second tissue with a larger number of fine-grained cell types, we examined immune cell populations in the human gut (Fig. 5a and Supplementary Figs. 25 and 26). The human gut presents a challenging application for spatial mapping as it contains a larger number of transcriptionally fine-grained immune and non-immune cell types, with the immune cells enriched in relatively small tissue areas across lymphoid follicles (for example, ~1% of tissue locations in our dataset).

We defined comprehensive cell type reference signatures by combining scRNA-seq datasets from pediatric and adult human intestinal tract, which was enriched for immune (CD45$^+$) cells to

Fig. 4 | Mapping the cellular compartments of the human lymph node. a, Analysis workflow and downstream analysis. **b**, UMAP representation (*x* and *y* axis) of 34 immune and non-immune cell subtypes (color), combining scRNA-seq data from three studies used to define reference gene expression signatures of cell types (Methods). **c**, Major tissue zones of the lymph node, including T cell and B cell zones, and GC reactions with FDCs. Left, H&E image of the tissue section. Right, estimated cell abundance (color intensity) of cell types (color) shown over the H&E image of the lymph node sample ($n=1$ donor). Scale bar, 1 mm. **d**, Identification of lymph node cell compartments using NMF. Shown is a dot plot of the estimated NMF weights of cell types (rows) across NMF components (columns), which correspond to cellular compartments. Shown are relative weights, normalized across components for every cell type. **e**, Estimated cell abundance of selected cell types that are associated with highlighted cellular compartments (Fig. 4d) is shown in panels 1–4. Spatial plots (*x* and *y* axis) show cell abundance (color intensity) for each cell type (subpanels). Boxes highlight the tissue regions shown in inset panels. Dashed circles indicate GCs. **f**, First panel: annotations of germinal center GC locations based on paired histology image (positive locations in yellow; negative locations in black). Other panels: spatial plots (*x* and *y* axis) showing estimated cell proportion of the T_CD4$^+$_TfH_GC subpopulation (color intensity) for alternative methods (subpanels). **g**, Quantitative assessment of the mapping accuracy for the T_CD4$^+$_TfH_GC cell subpopulation using the histology annotation of GC positive and negative locations as in **f**. Shown are precision recall curves for alternative methods. **h**, Assessment of spatial mapping of six subtypes that are expected to be present in annotated GC (positive locations) and absent from other locations (as in **g**). Shown are macro-average precision recall scores (bar height, *y* axis) for alternative methods (color, *x* axis), considering either transcriptionally fine (left, $n=4$) or transcriptionally distinct (right, $n=2$) subtypes. Error bars denote standard errors. Transcriptionally fine subtypes (B_GC_DZ, B_GC_LZ, B_GC_prePB and T_CD4$^+$_TfH_GC); transcriptionally distinct subtypes (B_Cycling and FDC).

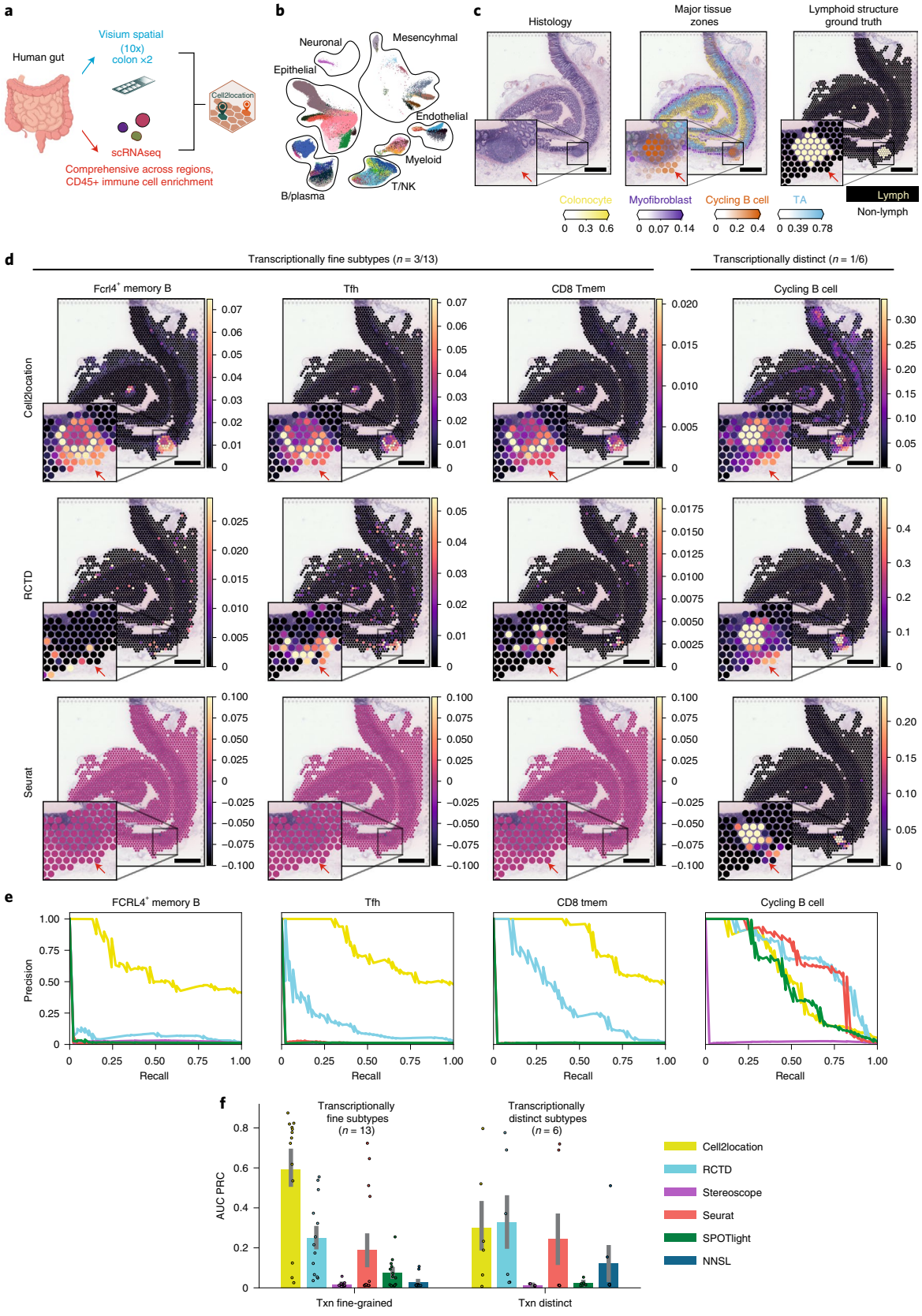
iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

represent them at equal proportion with non-immune (CD45⁻) subtypes, spanning 76 cell types in total across 153,901 cells (Fig. 5b)[42]. We mapped these reference cell types to a Visium dataset of the adult colon consisting of two consecutive tissue sections[43], where we annotated the locations of lymphoid follicles using histology and marker gene expression in Visium data (Fig. 5c and Supplementary Fig. 25b). Correspondingly, after mapping with cell2location, the expected major regions within the colon could be readily identified, including lymphoid follicles with cycling B cells, epithelial compartments with colonocytes and transit-amplifying progenitor cells as well as the muscularis mucosa layer with myofibroblasts (Fig. 5c and Supplementary Files 12 and 13)[43].



iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

We identified 19 immune cell types in our gut scRNA-seq reference that are known to be enriched in lymphoid follicles[44,45], which we again classified into transcriptionally fine versus transcriptionally distinct cell types (Supplementary Fig. 25c) and assessed their mapping to lymphoid follicles in the Visium data by cell2location and other methods. Although all considered methods achieved high accuracy for distinct cell types, such as cycling B cells, cell2location again achieved markedly higher accuracy for mapping fine-grained



iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

Fig. 5 | Cell2location outperforms alternative methods in mapping fine immune subtypes in the human gut. a, Analysis workflow. **b**, UMAP representation (*x* and *y* axis) of 76 immune and non-immune reference cell subtypes (color), which were obtained by combining scRNA-seq data from gut tissue and tertiary lymphoid follicles and used to define reference cell type signatures (Methods). **c**, Major tissue zones of the gut, including epithelial compartments with mature colonocytes and transit amplifying cells, muscularis mucosa with myofibroblasts and tertiary lymphoid follicle with immune cells (zoomed-in panel). Left, H&E image of the tissue section. Middle, estimated cell abundance (color intensity) of cell types (color) shown over the H&E image of the same gut sample. Right, ground truth histology-based annotation of lymphoid follicle (color) shown for the corresponding H&E image. $n=1$ donor and $n=2$ tissue sections. Scale bar, 1 mm. **d**, Spatial plots (*x* and *y* axis), depicting estimated cell type proportions (color intensity) for three selected transcriptionally fine subtypes (out of 13) and one transcriptionally distinct subtype (out of six). Cell types are shown in columns, and alternative methods to estimate the corresponding cell type proportions are shown in rows. Boxes highlight the tissue regions shown in inset panels. **e**, Precision recall curves for cell2location and alternative methods for mapping the indicated cell types for cell2location and alternative methods (methods in color; see legend in **f**). These cell types are expected to be present in annotated lymphoid follicles (positive set of locations) and absent from other locations (negative set). **f**, Summary of precision recall (AUC score, *y* axis) for cell2location and alternative methods (color, *x* axis). Cell types are stratified by the extent of transcriptional distinctiveness (fine versus distinct). These cell types are expected to be present in annotated GC (positive set of locations) and absent from other locations (negative set) for alternative methods. Shown are average AUC scores across cell types (bar height) with error bars corresponding to standard errors. AUC, area under the curve.

cell types (Fig. 5d–f; $n=13$ cell types). Alternative methods either failed to locate several of the fine immune cell types, such as FCRL4$^+$ memory B cells and Tfh cells in Visium data, or mapped them with low specificity (Fig. 5d–f and Supplementary Figs. 25d and 26).

Finally, we sought to explore the specific modeling choices in cell2location that are most relevant for the model performance (Supplementary Figs. 24 and 25e), which in particular confirmed the benefits of hierarchical modeling to borrow statistical strength across locations for mapping fine cell types.

Taken together, these results show that cell2location outperforms other methods in mapping transcriptionally fine-grained cell types in complex tissues.

Discussion

The integration of disassociated single-cell and spatial transcriptomics promises a high-throughput approach to map tissue cell types and cell signaling in situ. To best use the potential of these technologies and to create comprehensive cellular wiring diagrams of tissues, computational models for high-resolution mapping of cell types in spatial transcriptomic data are needed. Here we presented cell2location, a Bayesian model that provides sufficient sensitivity and resolution to spatially map a wide range of cell types, including transcriptionally fine subtypes, thereby allowing the creation of comprehensive atlases of diverse tissues in a versatile manner.

We performed extensive quantitative benchmarks across three tissues, consistently showing that cell2location enables a substantial advance for spatially mapping fine-grained cell types over existing tools. In the human lymph node and gut, cell2location provided higher accuracy to resolve fine immune cell populations. In the mouse brain, cell2location enabled accurate mapping of fine astrocyte subtypes.

Despite showing a small number of transcriptional differences, fine cell types often exhibit specialized and critical functions across tissues. In the immune system, we showed that cell2location can accurately resolve fine subtypes such as Tfh cells guiding B cell differentiation and FCRL4$^+$ B cells modulating immune responses at mucosal sites[44]. In the mouse brain, we resolved four regional astrocyte populations. In addition to their molecular and spatial identities, we can only speculate about the regionalized functions of these astrocytes. Astrocyte–neuron signaling in the lateral habenula regulates neuronal circuits implicated in depression[46]. Similarly, diencephalic astrocytes could shape the activity of local neuronal circuits, such as the lateral thalamic circuitry for sensory relay or medial thalamic circuits involved in attention control[47].

Although cell2locaiton is conceptually related to existing methods, specific design choices enable the benefits highlighted by our benchmarking results. Among the most important features, cell2location allows the borrowing of statistical strength across locations with similar cell composition, thereby improving sensitivity for mapping fine cell types. Additionally, the model can jointly process data across batches while accounting for technical differences as well as variation in mRNA detection sensitivity. Finally, we note that cell2location is implemented as robust, fast and versatile software, integrated with scvi-tools framework[19], enabling a wide range of different applications.

Although we showed that cell2location can deal with data from a range of existing technologies, upcoming technologies will motivate further refinements of the model. This includes support for higher-resolution spatial transcriptomics assays, for example by adapting the model to account for differences in the noise characteristics such as background probe binding or variable capture areas (for example, Nanostring[48,49] and LCM-RNA-seq[50]). The modular implementation of cell2location in pyro and pymc3 (refs. [51,52]) will facilitate such efforts.

A major future frontier for cell atlas projects is the three-dimensional (3D) mapping of tissues at cellular resolution. The spatial mapping approach described here can lay the foundations for 3D tissue atlases. Single-cell and spatial RNA-seq and imaging can be applied to consecutive tissue sections across an organ, integrating data from all technologies to build a multi-scale atlas. Regional cell type maps generated with cell2location can also guide gene selection for intact tissue volume imaging[53], leading to 3D models of cells in their native tissue context with whole transcriptome information.

In summary, we introduced here a versatile and flexible approach for spatial mapping of tissues. We expect that cell2location will have diverse applications in mapping tissue architecture across development, health and disease. In separate applications to the human endometrium[54], developing human gut[42] and human gonads[55], we further showed the versatility of cell2location to map fine-grained cell type differences across diverse tissues. Going forward, spatial cell type maps coupled with cell interaction analysis can help decipher developmental processes such as cellular differentiation and how it is regulated by extrinsic cues. These developmental cues can be recapitulated in vitro to refine tissue engineering approaches. Similarly, understanding how tissue architecture is perturbed in response to disease can enrich understanding of pathological processes and aid the design of therapeutic interventions.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41587-021-01139-4.

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

Received: 15 November 2020; Accepted: 28 October 2021;
Published online: 13 January 2022

References

1. Stuart, T. et al. Comprehensive integration of single-cell data. *Cell* **177**, 1888–1902 (2019).
2. Rodriques, S. G. et al. Slide-seq: a scalable technology for measuring genome-wide expression at high spatial resolution. *Science* **363**, 1463–1467 (2019).
3. Andersson, A. et al. Single-cell and spatial transcriptomics enables probabilistic inference of cell type topography. *Commun. Biol.* **3**, 565 (2020).
4. Moncada, R. et al. Integrating microarray-based spatial transcriptomics and single-cell RNA-seq reveals tissue architecture in pancreatic ductal adenocarcinomas. *Nat. Biotechnol.* **38**, 333–342 (2020).
5. Cable, D. M. et al. Robust decomposition of cell type mixtures in spatial transcriptomics. *Nat. Biotechnol.* https://doi.org/10.1038/s41587-021-00830-w (2021).
6. Elosua-Bayes, M., Nieto, P., Mereu, E., Gut, I. & Heyn, H. SPOTlight: seeded NMF regression to deconvolute spatial transcriptomics spots with single-cell transcriptomes. *Nucleic Acids Res.* **49**, e50 (2021).
7. Travaglini, K. J. et al. A molecular cell atlas of the human lung from single-cell RNA sequencing. *Nature* **587**, 619–625 (2020).
8. Conde, C. D. et al. Cross-tissue immune cell analysis reveals tissue-specific adaptations and clonal architecture in humans. Preprint at https://www.researchgate.net/publication/351185031_Cross-tissue_immune_cell_analysis_reveals_tissue-specific_adaptations_and_clonal_architecture_across_the_human_body (2021).
9. Vento-Tormo, R. et al. Single-cell reconstruction of the early maternal–fetal interface in humans. *Nature* **563**, 347–353 (2018).
10. Zeisel, A. et al. Molecular architecture of the mouse nervous system. *Cell* **174**, 999–1014 (2018).
11. Yao, Z. et al. A taxonomy of transcriptomic cell types across the isocortex and hippocampal formation. *Cell* **184**, 3222–3241 (2021).
12. Lein, E., Borm, L. E. & Linnarsson, S. The promise of spatial transcriptomics for neuroscience in the era of molecular cell typing. *Science* **358**, 64–69 (2017).
13. Ståhl, P. L. et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science* **353**, 78–82 (2016).
14. Vickovic, S. et al. High-definition spatial transcriptomics for in situ tissue profiling. *Nat. Methods* **16**, 987–990 (2019).
15. Junker, J. P. et al. Genome-wide RNA tomography in the zebrafish embryo. *Cell* **159**, 662–675 (2014).
16. Zeisel, A. et al. Brain structure. Cell types in the mouse cortex and hippocampus revealed by single-cell RNA-seq. *Science* **347**, 1138–1142 (2015).
17. Halpern, K. B. et al. Single-cell spatial reconstruction reveals global division of labour in the mammalian liver. *Nature* **542**, 352–356 (2017).
18. Aliee, H. & Theis, F. AutoGeneS: automatic gene selection using multi-objective optimization for RNA-seq deconvolution. *Cell Syst.* **12**, 706–715 (2021).
19. Gayoso, A. et al. scvi-tools: a library for deep probabilistic analysis of single-cell omics data. Preprint at https://www.researchgate.net/publication/351290216_scvi-tools_a_library_for_deep_probabilistic_analysis_of_single-cell_omics_data (2021).
20. Lawson, C. L. & Hanson, R. J. *Solving Least Squares Problems* (SIAM, 1995).
21. Tasic, B. et al. Adult mouse cortical cell taxonomy revealed by single cell transcriptomics. *Nat. Neurosci.* **19**, 335–346 (2016).
22. Phillips, J. W. et al. A repeated molecular architecture across thalamic pathways. *Nat. Neurosci.* **22**, 1925–1935 (2019).
23. Lim, L., Mi, D., Llorca, A. & Marín, O. Development and functional diversification of cortical interneurons. *Neuron* **100**, 294–313 (2018).
24. Tasic, B. et al. Shared and distinct transcriptomic cell types across neocortical areas. *Nature* **563**, 72–78 (2018).
25. Stickels, R. R. et al. Highly sensitive spatial transcriptomics at near-cellular resolution with Slide-seqV2. *Nat. Biotechnol.* **39**, 313–319 (2021).
26. Ben Haim, L. & Rowitch, D. H. Functional diversity of astrocytes in neural circuit regulation. *Nat. Rev. Neurosci.* **18**, 31–41 (2017).
27. Bayraktar, O. A. et al. Astrocyte layers in the mammalian cerebral cortex revealed by a single-cell in situ transcriptomic map. *Nat. Neurosci.* **23**, 500–509 (2020).
28. Batiuk, M. Y. et al. Identification of region-specific astrocyte subtypes at single cell resolution. *Nat. Commun.* **11**, 1220 (2020).
29. Lanjakornsiripan, D. et al. Layer-specific morphological and molecular differences in neocortical astrocytes and their dependence on neuronal layers. *Nat. Commun.* **9**, 1623 (2018).
30. Chai, H. et al. Neural circuit-specialized astrocytes: transcriptomic, proteomic, morphological, and functional evidence. *Neuron* **95**, 531–549 (2017).
31. Huang, A. Y.-S. et al. Region-specific transcriptional control of astrocyte function oversees local circuit activities. *Neuron* **106**, 992–1008 (2020).
32. John Lin, C.-C. et al. Identification of diverse astrocyte populations and their malignant analogs. *Nat. Neurosci.* **20**, 396–405 (2017).
33. Polański, K. et al. BBKNN: fast batch alignment of single cell transcriptomes. *Bioinformatics* **36**, 964–965 (2020).
34. V1_Human_Lymph_Node - Datasets - Spatial Gene Expression - Official 10x Genomics Support. https://support.10xgenomics.com/spatial-gene-expression/datasets/1.1.0/V1_Human_Lymph_Node
35. James, K. R. et al. Distinct microbial and immune niches of the human colon. *Nat. Immunol.* **21**, 343–353 (2020).
36. Park, J.-E. et al. A cell atlas of human thymic development defines T cell repertoire formation. *Science* **367**, eaay3224 (2020).
37. King, H. W. et al. Single-cell analysis of human B cell maturation predicts how antibody class switching shapes selection dynamics. *Sci. Immunol.* **6**, eabe6291 (2021).
38. Levitin, H. M. et al. De novo gene signature identification from single-cell RNA-seq with hierarchical Poisson factorization. *Mol. Syst. Biol.* **15**, e8557 (2018).
39. Kotliar, D. et al. Identifying gene expression programs of cell-type identity and cellular activity with single-cell RNA-seq. *eLife* **8**, e43803 (2019).
40. Roco, J. A. et al. Class-switch recombination occurs infrequently in germinal centers. *Immunity* **51**, 337–350 (2019).
41. Sun, B. T. *Helper Cell Differentiation and Their Function* (Springer, 2014).
42. Elmentaite, R. et al. Cells of the human intestinal tract mapped across space and time. *Nature* **597**, 250–255 (2021).
43. Fawkner-Corbett, D. et al. Spatiotemporal analysis of human intestinal development at single-cell resolution. *Cell* **184**, 810–826 (2021).
44. Spencer, J. & Sollid, L. M. The human intestinal B-cell response. *Mucosal Immunol.* **9**, 1113–1124 (2016).
45. Koboziev, I., Karlsson, F. & Grisham, M. B. Gut-associated lymphoid tissue, T cell trafficking, and chronic intestinal inflammation. *Ann. N. Y. Acad. Sci.* **1207**, E86–E93 (2010).
46. Cui, Y. et al. Astroglial Kir4.1 in the lateral habenula drives neuronal bursts in depression. *Nature* **554**, 323–327 (2018).
47. Ward, L. M. The thalamus: gateway to the mind. *Wiley Interdiscip. Rev. Cogn. Sci.* **4**, 609–622 (2013).
48. Merritt, C. R. et al. Multiplex digital spatial profiling of proteins and RNA in fixed tissue. *Nat. Biotechnol.* **38**, 586–599 (2020).
49. Roberts, K. et al. Transcriptome-wide spatial RNA profiling maps the cellular architecture of the developing human neocortex. Preprint at https://www.researchgate.net/publication/350237379_Transcriptome-wide_spatial_RNA_profiling_maps_the_cellular_architecture_of_the_developing_human_neocortex (2021).
50. Nichterwitz, S. et al. Laser capture microscopy coupled with Smart-seq2 for precise spatial transcriptomic profiling. *Nat. Commun.* **7**, 12139 (2016).
51. Salvatier, J., Wiecki, T. V. & Fonnesbeck, C. Probabilistic programming in Python using PyMC3. *PeerJ Comput. Sci.* **2**, e55 (2016).
52. Bingham, E. et al. Pyro: deep universal probabilistic programming. *J. Mach. Learn. Res.* **20**, 973–978 (2019).
53. Zhao, S. et al. Cellular and molecular probing of intact human organs. *Cell* **180**, 796–812 (2020).
54. Garcia-Alonso, L., Handfield, L. F. & Roberts, K. et al. Mapping the temporal and spatial dynamics of the human endometrium in vivo and in vitro. *Nat. Genet.* **53**, 1698–1711 (2021).
55. Vento-Tormo, R. et al. Single-cell roadmap of human gonadal development. Preprint at https://assets.researchsquare.com/files/rs-496470/v1_covered.pdf?c=1631867880 (2021).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2022

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

Methods

Cell2location model. For a complete derivation of the cell2location model, see the Supplementary Methods. In brief, cell2location is a Bayesian model that estimates the absolute abundance of cell types at each location by decomposing the spatial expression count matrix into a predefined set of reference cell type signatures. The model takes an untransformed spatial expression count matrix $d_{s,g}$ of genes $g = \{1, .., G\}$ at locations $s = \{1, .., S\}$ as input, as, for example, obtained from the 10x Space Ranger software (10x Visium data). Cell2location allows the joint modeling of multiple spatial datasets $e = \{1, .., E\}$ (also known as batch integration) where e denotes individual experiments obtained using one or multiple technologies. The second input is a matrix of reference cell type signatures $g_{f,g}$, which correspond to the expected mRNA count of genes g for each cell type $f = \{1, .., F\}$. By default, this matrix is estimated using a negative binomial regression model, which allows for combining data across batches and technologies (see below and Supplementary Methods).

Cell2location model. Cell2location models the elements of the spatial expression count matrix $d_{s,g}$ as negative binomial distributed, given an unobserved gene expression level (rate) $\mu_{s,g}$ and gene- and batch-specific over-dispersion $\alpha_{e,g}$:

$$d_{s,g} \sim NB\left(\mu_{s,g}, \alpha_{e,g}\right).$$

The expression rate of genes g at location s, $\mu_{s,g}$ in the mRNA count space is modeled as a linear function of reference cell types signatures $g_{f,g}$:

$$\mu_{s,g} = \left(\underbrace{m_g}_{\text{technology sensitivity}} \cdot \underbrace{\sum_f w_{s,f}\, g_{f,g}}_{\text{cell type contributions}} + \underbrace{s_{e,g}}_{\text{additive shift}} \right) \cdot \underbrace{y_s}_{\text{per-location sensitivity}} .$$

Here, $w_{s,f}$ denotes regression weight of each reference cell type f at location s, which can be interpreted as the abundance of cells expressing reference cell type signature f at location s; m_g is a gene-specific scaling parameter that adjusts for differences in sensitivity between technologies; $s_{e,g}$ captures gene-specific additive shift that allows the model to account for contaminating RNA; and y_s is a location-specific scaling parameter that models variation in RNA detection sensitivity across locations s across one or multiple batches e (by use of hierarchical priors; Supplementary Methods).

Because of the organization of tissues into zones, the cell type abundance estimates $w_{s,f}$ are not independent across locations. To borrow statistical strength and use information sharing across locations, $w_{s,f}$ is modeled using a hierarchical decomposition prior (factorization), assuming $r = \{1, .., R\}$ groups of cell types that partially share a common cell abundance profile (for example, due to tissue zones; Supplementary Methods). Unless stated otherwise, R is set to 50.

Approximate variational inference is used to estimate all model parameters, implemented in the scvi-tools framework[19], building on pyro and pymc3 probabilistic programming frameworks as backend[51,52], which supports GPU acceleration. For full details, see the Supplementary Methods.

Note on the construction of reference cell type signatures. It is important to aim for a comprehensive and detailed cell-type reference, which includes as many of the cell types and subpopulations that are present in situ as possible. This can be achieved by generating a paired snRNA-seq reference from the same tissue sample. However, imperfect matching of cell populations is often acceptable (Figs. 4 and 5 and Supplementary Fig. 4c).

Based on one or multiple snRNA-seq input datasets, reference cell type signatures (gene expression) are estimated based on a defined set of cell populations, which, in turn, can be identified using conventional clustering workflows. By default, the cell2location employs a negative binomial regression to estimate reference cell type signatures. This model-based approach allows robust combinination of data across technologies and batches (Supplementary Fig. 19), which results in improved spatial mapping accuracy (Supplementary Figs. 24a–c and 25e). Alternatively, if batch effects are not a concern, a hard-coded method that estimates per-cluster average mRNA counts without any model-based strategy can be sufficient. This strategy is simple and fast and yields similar levels of accuracy if batch effects are small (Supplementary Fig. 24a–c). The hard-coded method is also recommended for non-unique molecular identifier (UMI) technologies for which the negative binomial count model is often not appropriate, such as Smart-seq2.

Note on determining model hyper-parameters. The cell2locaiton model has two parameters that should be adjusted by the user, taking known experimental and biological characteristics of a given dataset into consideration:

1. Expected cell abundance per location $\hat{N}$
2. Regularization of within-experiment variation in RNA detection sensitivity α^y

Supplementary Fig. 27 provides a flowchart to guide the user to determine appropriate settings for these hyper-priors, which were used to determine the model parameters for all presented results.

The expected cell abundance $\hat{N}$ per location is provided as a single tissue-level estimate, which can be derived from histology images (H&E or DAPI), ideally paired to the spatial expression data or at least representing the same tissue type. This parameter can be estimated by manually counting nuclei in 10–20 locations in the histology image (for example, using the 10x Loupe browser; Supplementary Fig. 8) and computing the average cell abundance. An appropriate setting of this prior is essential to inform the estimation of absolute cell type abundance values; however, the model is robust to a range of similar values (Supplementary Fig. 5). In settings where suitable histology images are not available, the size of capture regions relative to the expected size of cells can be used to estimate $\hat{N}$ (Slide-seq V2; Supplementary Fig. 27). For all analyses in this manuscript, a single tissue-level estimate was used; however, as an advanced feature, cell2location can use the per-location number of cells.

Hyper-parameter α^y for regularizing within-experiment variation in RNA detection sensitivity. α^y is the shape parameter of the prior gamma distribution over y_s that represents the extent to which RNA detection sensitivity for individual locations deviates from the mean sensitivity of each batch. Large values of α^y correspond to small within-experiment variability (Supplementary Methods). By default, we assume small within-experiment variability in RNA detection sensitivity across locations ($\alpha^y = 200$), which results in values of y_s close to the mean sensitivity for each experiment. However, when strong gradients in mRNA detection sensitivity are observed, which is a common issue in adult human 10x Visium data, we recommend less strict regularization ($\alpha^y = 20$).

Although good choices of these hyper-parameters can have a positive effect on accuracy and sensitivity, the estimates are robust to a range of suboptimal choices (Supplementary Fig. 5). The estimation of absolute cell abundance requires appropriate settings of $\hat{N}$ in particular.

Validation of cell2location using synthetic data and comparison to alternative methods. *Constructing a synthetic spatial transcriptomics dataset.* Simulated spatial transcriptomics data were generated by combining expression profiles of cells drawn from each of 49 cell types in the mouse brain snRNA-seq reference data (see below), generating abundance profiles at 2,500 locations. snRNA data from the two most mouse brain snRNA-seq samples that had minimal batch effect were split into one dataset used to generate the synthetic data (50% of cells) and a second dataset used to evaluate cell2location and alternative methods (50% of cells), similarly to the strategy proposed by Andersson et al.[3]. Hyper-parameters for data simulation were chosen to mimic (the typically low) cell counts observed for cell types in real tissues, additionally matching empirically observed sparsity profiles. Cell type abundances were simulated according to either a spatially ubiquitous pattern (eight cell types) or a regional pattern (41 cell types). Regional patterns are represented by 12 tissue zones defined by co-located cell types that mimic the organization of real tissues. The assignment of 41 regional cell types to the 12 tissue zones is shown in Supplementary Fig. 2, with each cell type belonging to 1–3 tissue zones and each tissue zone containing 2–8 cell types. The number of cell types present at each location, as well as the absolute abundance (the number of cells per location), were simulated according to either low or high average cell type abundance (Fig. 1b), stratified by ubiquitous and regional location patterns (see below). The mathematical description and the step-by-step procedure for simulating abundance of cell types across locations and generating multi-cell mRNA counts is described in detail in the Supplementary Methods (section 5).

Validation of cell2location using synthetic data. Synthetic gene expression data were constructed as described above to assess the performance of cell2location and alternative methods. The synthetic data provide ground truth values for the relative cell abundance of each cell type (cell type proportions), absolute cell abundance (including partial cells) and total mRNA count of each cell type for each location, which were used to benchmark estimates obtained from cell2location and alternative methods. All considered models were provided with the ground truth snRNA-seq reference data (evaluation subset, see above; Table 1) as well as the synthetic mRNA count matrices for all locations.

Cell2location. Signatures of reference cell types were derived using a negative binomial regression (Supplementary Methods) with default parameters, except training mini-batch size of 2,500 and training duration of 600 epochs. The hyper-parameter values used for the synthetic data analysis are shown in Table 1. In addition, the sensitivity of cell abundance estimates to the choices of these hyper-parameters was assessed (Supplementary Fig. 5, section on sensitivity and robustness below). The prior cell abundance $\hat{N} = 9$ was set to a value close to, but not fully matching, the ground truth $\left(\hat{N} = 12\right)$, with the aim to obtain as realistic results as possible.

Assessment of alternative methods. RCTD, SPOTlight, Seurat 3 and NNLS (AutoGeneS implementation) were used with default or optimized parameter settings as described below. Stereoscope training hyper-parameters were adapted to each dataset as stated in the respective sections. All benchmark datasets were provided with the identical cell type reference snRNA-seq data and spatial count matrix as used for cell2location. The same ground truth relative cell abundance is used for all cell types.

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

Table 1 | Datasets used in the paper and corresponding cell2location hyper-parameters

	Spatial data	*n* samples	*n* locations	Reference data	*n* cell types	*n* cells	*n* genes	Cell abundance prior $\left(\tilde{N}\right)$	Normalization (y_s) alpha prior	*n* iterations
1	Synthetic	1	2,500	Paired (our) mouse brain snRNA-seq	49	8,111	12,080	9	200	40,000
2	Mouse brain Visium	5	14,968	Paired (our) mouse brain snRNA-seq	59	40,532	12,809	8	200	20,000
3	Mouse brain Visium (SSp cortex locations)	5	1,393	Yao et al., SSp cortex and hippocampus	121	6,948	13,565	8	200	40,000
4	Slide-seq V2 (section 2, hippocampus)	1	33,108	Our mouse brain snRNA-seq	59	40,532	10,670	1	20	30,000
6	Human lymph node (10x)	1	4,035	Combined lymph tissue reference	34	73,260	10,217	30	200	30,000
7	Adult gut (Fawkner-Corbett et al.)	2	3,842	Adult and pediatric scRNA-seq (Elmentaite et al.)	76	153,901	14,388	12	20	20,000

All other parameters were used at their default values (Supplementary Methods). *n* genes denote the number of genes shared between scRNA and spatial data.

Stereoscope[3]. This method uses a model-based approach to estimating both the cell type signatures and the relative cell abundance across locations (see the comparison in the Supplementary Methods). A negative binomial observation model is parameterized using a gene- and cell-type-specific log-odds parameter and a gene-specific total count parameter. This custom signature estimation method is hard-coded into the workflow, and both estimated parameters are used as input to the model for estimating relative cell abundance. Unless stated otherwise, default values of hyper-parameters are used. The scvi-tools[19] implementation of the Stereoscope model was used with the following hyper-parameters (default scvi-tools values were modified to achieve improved accuracy):

- No highly variable gene (HVG) selection was used (12,054 genes shared between single-cell and spatial data)
- Estimation of expression signatures of cell types: 0.01 learning rate; 1,500 epochs (increased compared to default to observe convergence); 1,024 mini-batch training size (similar to original Stereoscope default values).
- Estimation of relative cell abundance: 0.01 learning rate; 30,000 epochs (also increased compared to default); 1,250 mini-batch training size (half of the total number of locations, similar to original Stereoscope default values).

RCTD[5]. This method uses a similar model-based approach to the estimation of relative cell abundance across locations (see the comparison in the Supplementary Methods). It estimates reference cell type signatures by computing average mRNA count per cluster (hard-coded). RCTD was used in default mode (not 'doublet' mode) with hyper-parameters selected as suggested in the GitHub tutorial, with a reduced threshold of the number of cells per cell type to ensure that all 49 cell types were considered and including gene filtering based on log fold change (default settings of all other parameters; https://github.com/dmcable/RCTD/tree/7a895ed1666269784cc472fa0a6e4abbd3d2ccb2).

Seurat V3 (principal component analysis, anchor method and mutual nearest neighbor classifier)[1]. Seurat V3 workflow was used to project synthetic locations into the principal component analysis (PCA) space constructed using snRNA-seq reference data (Seurat V3 with anchor selection, Seurat::FindTransferAnchors), with subsequent transfer of cell type labels onto synthetic locations using a mutual nearest neighbr (MNN) classifier (Seurat::TransferData). For the purpose of this comparison, the label transfer scores were interpreted as relative cell type abundance estimates of each cell type for each location. Before integration, both snRNA-seq reference and synthetic spatial datasets were subjected to SCTransform normalization (Seurat::SCTransform)[56]. The PCA method (with default 30 components) was used for integration (Seurat::FindTransferAnchors) as advised in Seurat package documentation and spatial mapping tutorial. All functions were used with default parameters.

SPOTlight[6]. SPOTlight uses a seeded topic model to find expression signatures of reference cell types and a modified non-negative least squares (NNLS) method (denoted NMFreg) to estimate relative cell abundance. The topic model is initialized in a way that guides the topics to resemble expression signatures of cell types. The default SPOTlight workflow was implemented with recommended parameters (https://github.com/MarcElosua/SPOTlight/tree/fb5e7c7de3e6d0ac7618c4b1a55aea06e5f472ea).

In brief, to improve computational efficiency, the snRNA-seq data size was reduced by randomly selecting, at most, 100 cells per cell type (when a cell type is represented by fewer than 100 cells, all cells are kept), and highly variable genes (5,000 genes) were selected using Seurat::FindAllMarkers tool. Untransformed and unnormalized expression counts for both snRNA-seq reference and synthetic locations were provided as input to the 'spotlight_deconvolution' function in the SPOTlight package with the following hyper-parameters: transf = 'uv', method = 'nsNMF', and, crucially, 'minimum allowed proportions of cell types' min_cont = 0.01.

NNLS (AutoGeneS)[18,20]. NNLS was used to estimate the absolute cell abundance of each cell type in each synthetic location given the reference signatures of cell types, which were derived by computing average expression of each gene across cells of each type. Untransformed and unnormalized expression counts of the synthetic locations were used as input to NNLS implementation as part of the AutoGeneS package. Similarly to cell2location, the output was normalized to produce relative cell abundance.

Comparison of cell2location and alternative methods. The estimates of relative cell abundance from all methods were compared to the ground truth using two-dimensional (2D) histogram plots, Pearson *R* (numpy.corrcoef) and Jensen–Shannon divergence (scipy.spatial.distance.jensenshannon) and reported in Fig. 1b and Supplementary Fig. 3a. In addition, the accuracy was assessed using precision recall quantified on all 49 cell types jointly (macro-average). Relative cell abundance was used as a predictor, and cell abundance >0.1 was used as a gold standard label. Precision recall results are reported in Supplementary Fig. 3b. To simplify presentation, results across individual cell types were summarized as an average Pearson *R* (Fig. 1c and Supplementary Fig. 4b) or precision recall score (S3B and S4B) and shown as bar plots, with bar height denoting average values across cell types and error bars denoting standard errors of the mean. To assess absolute cell abundance estimation, estimated cell abundance was compared to simulated cell abundance using 2D histogram plots, Pearson *R* (numpy.corrcoef) and Jensen–Shannon divergence (scipy.spatial.distance.jensenshannon) and reported in Supplementary Fig. 4a.

Sensitivity of cell2location modeling choices and robustness. *Assessment of the effect of modeling choices.* The effect of cell2location modeling choices on the performance of the model was evaluated using both synthetic (Supplementary Fig. 4b) human lymph node data (Supplementary Fig. 24) and human gut data (Supplementary Fig. 25e). In brief, the relevance of the following modeling choices (for details, see Supplementary Methods, section 3) was evaluated using the benchmarking criteria as defined in the main analysis of the respective datasets (Pearson *R* of cell type proportions and precision recall curves of cell type presence/absence):

1. Prior factorization of w_{sf}
2. Gene-specific technology sensitivity scaling m_g
3. Negative binomial regression for estimating reference signatures of cell types versus a hard-coded method

Assessment of the robustness of the cell abundance estimates when removing increasing fractions of cell types from the reference data. Cell2location was assessed using synthetic data combined from 49 cell types (Supplementary Fig. 4c, see above). The accuracy was assessed using precision recall (macro-average) for 17 cell types that were retained in the reference when considering alternative removal schemes to subsample the full set of cell type references. Relative cell abundance was used as a predictor, and cell abundance >0.1 was used as a gold standard label.

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

Assessment of the robustness to hyper-parameter settings. To assess the sensitivity of cell2location with respect to selected parameters (Supplementary Fig. 5 and Supplementary Methods), the benchmark analysis on both synthetic and gut data was performed as described in the corresponding sections. Evaluated were the model performance when varying the two user-defined tissue hyper-parameters (expected cell abundance per location $\hat{N}$ and regularization of within-experiment variation in RNA detection sensitivity α^y) as well as hyper-parameters that govern the hierarchical prior of the decomposition across locations (Supplementary Methods). Hyper-parameters were varied one at a time while retaining the values of the three remaining hyper-parameters constant. Additionally, we assessed the effect of α^y hyper-parameter on the consistency of estimated total cell abundance by cell2location and the nuclei counts estimated by segmenting histology images in the mouse brain (Supplementary Fig. 8f and Supplementary Methods), as quantified by Pearson correlation and root mean squared error.

Mouse brain experiments and analysis. *Mouse brain tissue processing.* Tissue processing. Mouse work was carried out at the Wellcome Sanger Institute in accordance with UK Home Office regulations and the UK Animals (Scientific Procedures) Act of 1986 under a UK Home Office license, which were regularly reviewed by the institutional Animal Welfare and Ethical Review Body.

Brains of wild-type adult C57BL/6 mice (postnatal day 56, one female and one male) were dissected, snap-frozen, embedded in optimal cutting temperature (OCT) compound (Tissue-Tek) and stored at −80 °C. Brain hemispheres were cryosectioned at −20 °C using a cryostat (Leica, CM3050S). To assess tissue quality, RNA was extracted from test tissue sections using the RNeasy Pico Kit (Qiagen) and yielded high RNA integrity number values (9.6 and 9.7) on an Agilent Bioanalyzer, indicating high RNA quality.

For matched single-nuclei and Visium RNA-seq experiments, brain hemispheres were cryosectioned to adjacent thick (200-µm) and thin (10-µm) coronal sections, respectively, and processed the same day. In total, four consecutive sets of thick and thin tissue sections were collected from each brain. Five sets of tissue sections yielded good-quality single-nuclei data as well as good-quality Visium data (three adjacent sections from mouse 1 and two sections from mouse 2), and one additional section from mouse 2 yielded good-quality single-nuclei data; these were considered for analysis in this study.

snRNA-seq. Thick (200-µm) mouse brain sections were cryosectioned, dissected from OCT and kept in a tube on dry ice until subsequent processing. Nuclei were extracted from each section as described previously[57]. In brief, nuclei were released from sections via Dounce homogenization, Hoechst-stained and isolated via fluorescence-activated cell sorting. Nuclei were then loaded into the 10x Chromium Single Cell 3′ Kit (v3) to obtain 3,000–7,000 nuclei per well, and library preparation was done per the manufacturer's protocol. Libraries were sequenced on an Illumina NovaSeq S4 system.

Visium spatial transcriptomics. Thin (10-µm) mouse brain sections were cryosectioned and mounted directly onto separate capture areas on 10x Visium Spatial Gene Expression slides (beta product version). Processing was done per the manufacturer's protocols. In brief, sections were methanol-fixed, H&E-stained and imaged on a NanoZoomer 2.0 slide scanner (Hamamatsu). Sections were then permeabilized and further processed to obtain cDNA libraries that were quality controlled using the Agilent Bioanalyzer. The cDNA libraries were sequenced on the Illumina HiSeq 4000 system, aiming at 300 million raw reads per section with read lengths 28cy R1, 8cy i7 index, 0cy i5 index and 91cy read 2.

Single-nucleus data processing and cell type annotation. Sequencing data were processed using 10x Cell Ranger version 3.0.2, aligned to mouse pre-mRNA genome reference version mm10 (downloaded from the 10x website), and mRNA count matrices were generated by adding intronic and exonic UMI counts for each gene in each cell. Initially, snRNA-seq counts were processed using standard Seurat V3 workflow without correcting batch effects among six individual samples. Specifically, the expression matrices filtered to include droplets containing cells (10x Cell Ranger output), were imported into the Seurat object. Cells were filtered using the following quality control criteria: number of detected genes per cell <5,000, fraction of nuclear-encoded mitochondrial genes <0.10 and doublet scores (scrublet[58]) <0.2. The data were then subjected to normalization by a scale factor of 10,000 and log1p-transformed (Seurat::NormalizeData), scaled gene-wise by subtracting the mean and divided by the standard deviation (Seurat::ScaleData). After that, 50 principal components were found (Seurat::RunPCA) and used as input to the uniform manifold approximation and projection (UMAP) algorithm (Seurat::RunUMAP), which was also used for *k*-nearest neighbor (KNN) graph construction with $k = 100$ (Seurat::FindNeighbors). To generate a comprehensive cell annotation, joint Louvain clustering with resolution 7 was performed (Seurat::FindClusters) on all samples, which yielded 51 cell clusters. Clusters were subsequently annotated using marker genes from the literature and existing mouse brain single-cell reference datasets[10,16,21]. To annotate the regional identity of cell types, we used (1) known regional marker genes[10,16,21], (2) the spatial expression patterns of these genes on in situ hybridization data from the Allen Brain Atlas (https://mouse.brain-map.org/) and (3) the regional localization of cell types estimated by cell2location (Fig. 2 and Supplementray Files 1–5). The marker genes of regional cell types reported in Supplementary Fig. 7 were obtained from Zeisel et al.[10] (http://mousebrain.org/downloads.html, L5_All.agg.loom file). This step yielded an annotation for 44 of 51 clusters, including known regional neuronal and glial subtypes as described above. The remaining seven clusters were composed of low-quality cells or unknown cell types and not analyzed further. Next, to resolve fine-grained astrocyte subtypes, a second targeted analysis was performed using cells from the two astrocyte clusters (2/44) as described below. Collectively, these two steps resulted in reference gene expression signatures for 59 distinct cell subtypes. This integrated reference was used throughout the manuscript to avoid presenting spatial mapping results with varying granularity of astrocyte subtypes. The resulting clusters and cell annotations are reported in Fig. 2b and Supplementary Figs. 6 and 7 and were used as input to cell2location and alternative methods.

Identification of astrocyte subtypes using initial spatial mapping in the mouse brain. To identify regionally enriched astrocytes, we annotated and spatially mapped reference signatures of astrocyte subpopulations. Cell clusters (2/44) annotated as astrocytes were selected from the initial analysis (Fig. 2b), as identified based on the expression of canonical markers (for example, *Aldoc* and *Slc1a3*). These cells were subjected to analysis workflow in scanpy[59] as follows. First, an untransformed and unnormalized count matrix was filtered to select expressed genes at two cutoffs:

1. Selecting genes detected at mRNA count >0 in >5% of cells
2. Selecting genes detected at mRNA count >0 in a few cells (5% > cell count > 0.4%) but with a mean expression >1.122018 across cells with count >0

The filtered data were then log1p-transformed and scaled. PCA was performed (30 components), and the first component, which was strongly associated with total mRNA count per cell, was removed. Next, BBKNN[33] was used to refine the integration across all six snRNA-seq datasets with the default parameters, except neighbors_within_batch = 3. The corrected KNN graph was used to perform UMAP dimensionality reduction and Leiden clustering with resolution 2.4, identifying 17 astrocyte subclusters (Supplementary Fig. 13). Marker genes of these subtypes were identified using differential expression tools in the scanpy package and markers known from the literature.

Next, an initial mapping with cell2location was performed using hyper-parameters described below in the mouse brain Visium section. Reference cell type signatures for initial mapping included 17 astrocyte subclusters as well as all other non-astrocyte cell types shown in Fig. 2b. Clusters that were not sufficiently distinct in their location (Supplementary Fig. 14) and marker genes (Supplementary Fig. 13) were merged. This resulted in ten molecularly and spatially distinct astrocyte subtypes (Fig. 3 and Supplementary Fig. 15b,c). These subtypes were incorporated into cell annotation used throughout the paper.

Visium data processing. 10x Visium spatial sequencing data were aligned to mouse pre-mRNA genome reference version mm10 (downloaded from the 10x website) using 10x Space Ranger 1.0, and mRNA count matrices were generated by adding intronic and exonic reads for each gene in each location. The paired histology H&E images were processed using 10x Space Ranger to select locations covered by tissue by aligning pre-recorded spot locations with fiducial border spots in the histology image. This allows evaluating the correspondence between cell maps produced using our method and the known brain anatomy. This also allows identifying the number of nuclei in each spot using nuclear segmentation, as described in the Supplementary Methods and reported in Supplementary Fig. 8a–d. The histology image was used to manually annotate cortical layers in the primary SSp cortex region using the lasso tool in the 10x Loupe browser.

Spatial mapping of mouse brain cell types to 10x Visium data. Spatial mapping using cell2location. Expression signatures of 59 mouse brain reference cell types (Fig. 2b, see above, and Table 1) were derived using a negative binomial regression model (Supplementary Methods) with training mini-batch size of 2,500 and training duration of 250 epochs. Mouse brain 10x Visium data were processed as described above to untransformed and unnormalized mRNA counts and filtered to genes shared with scRNA-seq (see Table 1 for gene and location count), which were used as input to cell2location with hyper-parameters as stated in Table 1.

Cell2location estimates described in this section are reported in Figs. 2d–g and 3b–e, Supplementary Figs. 8c–e, 9, 10,12b,c, 14, 15 and 18 and Supplementary Files 1–6.

Region identification. Estimated absolute cell abundance was used to calculate a KNN graph (n neighbors = 38) followed by Leiden clustering of locations (resolution = 1.3), yielding 31 clusters. The clusters were annotated by comparing the paired histology images, and clusters marking anatomically similar regions were merged, yielding 29 distinct clusters annotated as tissue regions and reported in Fig. 3b and Supplementary Figs. 10 and 15c.

Assessment of cell2location and alternative methods. RCTD, SPOTlight, Seurat 3 and NNLS (AutoGeneS) were used with default parameters as described in

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

the section on synthetic data (Methods). Stereoscope (implementation using scvi-tools; see Methods, synthetic data section) was used with the following hyper-parameters:

1. No HVG selection was used (12,809 genes shared between single-cell and spatial data)
2. Estimating expression signatures of cell types: learning rate of 0.01 (default); 1,000 epochs (increased compared to default to achieve convergence); batch size of 256 (default)
3. Estimating relative cell abundance in spatial data: learning rate of 0.01 (default); 20,000 epochs (increased compared to default to achieve convergence); batch size of 256 (default)

Assessment of cell2location and alternative methods on the mouse brain was qualitative and is presented in Supplementary Fig. 18.

Analysis of Slide-seq V2 data. Expression signatures of 59 mouse brain reference cell types (Fig. 2b) were derived using the negative binomial regression model as described above. Slide-seq V2 data of the mouse brain originating from the region containing the hippocampus, partial cortex and thalamus were downloaded from the data portal for Stickels et al.[25]. Genes with mRNA count >0 in at least 500 locations and locations with at least 200 UMI counts were retained to reduce data size. One Slide-seq V2 section with the largest number of spatial locations was used as input to cell2location using hyper-parameters stated in Table 1. Cell2location estimates described in this section are reported in Fig. 2h, Supplementary Fig. 12 and Supplementary Files 8 and 9.

smFISH. A wild-type adult C57BL/6 mouse (postnatal day 63, male) was transcardially perfused with ice-cold PBS and 4% paraformaldehyde (PFA) in 1× PBS. The brain was dissected and post-fixed in 4% PFA at 4 °C overnight and cryoprotected in 30% sucrose for 48 h at 4 °C. The brain was then embedded in OCT and frozen on isopentane dry ice slurry and stored at −80 °C. Coronal cryosections (12-µm) were collected on a cryostat (Leica), mounted onto superfrost glass slides (VWR) and stored at −80 °C.

RNAScope smFISH was performed as previously described[60]. Tissue slides were thawed at room temperature for 15 min and baked at 60 °C for 30 min. Afterwards, slides were post-fixed for 15 min in chilled 4% PFA and serially dehydrated through 50%, 70%, 100% and 100% ethanol for 5 min each. Four-plex RNAScope smFISH was automated using a Leica BOND RX (LS Multiplex Assay, Advanced Cell Diagnostics, Bio-Techne). Tissue sections were permeabilized using heat (5 min in Leica BOND ER2 buffer at 95 °C) and protease digestion (Advanced Cell Diagnostics Protease III for 20 min). The probes in C1, C2 and C3 channels were labeled using Opal 520, 570 and 650 fluorophores (PerkinElmer, diluted 1:1,500), respectively. The C4 probe was developed using TSA biotin (TSA Plus Biotin Kit, PerkinElmer, 1:500) and streptavidin-conjugated Atto 425 (Sigma-Aldrich, 1:400). Nuclei staining was performed with DAPI (Life Technologies, 1:50,000).

Imaging was performed on an Opera Phenix High-Content Screening System (PerkinElmer) in spinning disk confocal mode using a ×20 water immersion objective (NA 0.16, 298.99 nm per pixel) with a 1-µm *z*-step size. The excitation (Ex) laser and emission (Em) filter wavelengths were as follows: DAPI (375 nm; 435–480 nm), Atto 425 (425 nm; 463–501 nm), Opal 520 (488 nm; 500–550 nm), Opal 570 (561 nm; 570–630 nm) and Opal 650 (640 nm; 650–760 nm). Illumination correction and stitching were performed on maximum *z*-projection images using Acapella scripts provided by PerkinElmer.

Astrocyte segmentation and smFISH quantification. The segmentation of astrocytes was performed on the *Agt* image channel with pixel classification of ilastik[61]. The Agt channel was Gaussian blurred with a sigma of 2 pixels, and the astrocyte mask was subsequently generated by the trained pixel classifier. The performance of this classifier was evaluated by using visual inspection and a metric called out-of-bag (oob) error, which measures prediction error by using bootstrap aggregating. The oob error of the classifier was 2.857%, measured on a training image of dimension 402 × 2,010 pixels. As Agt is expressed at low levels in astrocytes outside the diencephalon[27], we manually traced the diencephalon to filter all astrocyte signals outside.

Two rounds of 'closing' operations were then executed to connect as many major astrocyte cellular processes to the cell bodies as possible. As a consequence, some cells were erroneously fused together. To split these fused cells, we performed six rounds of opening on large cells (>10,000 pixel^2). Finally, all small partially segmented cells (<250 pixel^2) were filtered out, and the mean pixel intensities (sum intensity/area) of target gene channels were quantified per cell.

Assessment of alternative methods at mouse brain cell type mapping with region-specific scRNA-seq reference. *Construction of single-cell reference data.* For quantitatively validating the cell type maps produced by cell2location, we considered a publicly available[11] gold standard reference of cortical and hippocampal cell locations. In that data, individual layers of the cortex and the hippocampus are dissected and subjected to scRNA-seq using the SMART-Seq 4 protocol (Fig. 2i). Data from this study were obtained from https://brain-map-portal-stage.us.aldryn.io/atlases-and-data/rnaseq. Cells originating from the primary SSp cortex (number of positive cell types, $n = 98$) as well as exclusive to the hippocampus (the subiculum areas indicated as par-post-pre, number of negative cell types, $n = 23$) were selected, discarding cell types that contribute fewer than three cells. The count of cells in this scRNA-seq data was used to compute the proportion of each reference cell type in each cortical layer (Supplementary Fig. 11b), which can serve as a complementary approach to estimate relative cell type abundance (see below). The corresponding reference signatures of 121 cell types were derived as per-cluster average expression (hard-coded method), as the negative binomial regression model is not suitable for SMART-Seq data (Supplementary Methods).

Spatial mapping using cell2location. We mapped cell types from this reference to a subset of the mouse brain 10x Visium data, restricted to the primary SSp cortex (which was annotated based on histology; Supplementary Fig. 11). Data were processed as described in the section on the mouse brain, and mRNA counts were used as input to cell2location using the hyper-parameters stated in Table 1.

Assessment of cell2location and alternative methods. RCTD, SPOTlight, Seurat 3 and NNLS (AutoGeneS) were used with default parameters as described in the section on synthetic data (Methods). Stereoscope (implementation using scvi-tools; see Methods, synthetic data section) was used with the following hyper-parameters:

4. No HVG selection was used (13,565 genes shared between single-cell and spatial data).
5. Estimating expression signatures of cell types: learning rate of 0.01 (default), 2,000 epochs and batch size of 1,024 were used to achieve convergence and acceptable accuracy of relative cell abundance estimation.
6. Estimating relative cell abundance in spatial data: learning rate of 0.01 (default), 20,000 epochs and batch size of 1,393 (full data) were used to achieve acceptable accuracy of relative cell abundance estimation.

Quantitative assessment of the specificity and sensitivity of cell2location at excluding cell types that should be absent in the spatial data. For all methods, precision recall was quantified for all 121 cell types and five Visium tissue sections jointly. All analyses and visualizations are based on the 99.8% quantile of relative cell abundance of each cell type in each Visium sample from the primary SSp cortex region as a predictor (Supplementary Fig. 11e), representing the presence of a given cell type at least at some locations within the sample. The cell count >3 in reference data from the SSp cortex region was used as a gold standard label, yielding $n = 98$ positive cell types and $n = 23$ negative cell types (Supplementary Fig. 11e). Identical criteria were used for alternative methods (Fig. 2i,j and Supplementary Fig. 11f), after the estimation of relative cell abundances for individual locations using those methods (see above).

Quantitative assessment of the consistency of the estimated relative cell abundance to reference data from manually dissected cortical layers. Cell2location cell type abundance estimates were averaged and normalized per cortical layer to obtain the proportion of cells per cell type within each cortical layer. These estimates were then compared to the cell proportions in the reference scRNA data (Supplementary Fig. 11a–d), considering 23 most prevalent cell types exclusive to the SSp cortex (cell count in SSp cortex >15 and cell count = 0 in the hippocampal region in the scRNA data). The same criteria were used to assess alternative methods (Supplementary Fig. 11d; RCTD, Stereoscope, SPOTlight, Seurat V3 and NNLS) after the estimation of relative cell abundance as described above.

Analysis of human lymph node Visium data. *Construction of single-cell reference data.* Published scRNA-seq datasets of lymph nodes have typically lacked an adequate representation of GC-associated immune cell populations due to the age of patient donors[62,63]. We, therefore, included scRNA-seq datasets spanning lymph nodes, spleen and tonsils in our single-cell reference to ensure that we captured the full diversity of immune cell states likely to exist in the spatial transcriptomic dataset (Fig. 4a,b). The scRNA-seq reference was constructed by combining data from three studies[35–37]. Integration was performed using scanpy[59], including removal of doublets with scrublet[58], regressing out batch from principal component space and performing BBKNN integration. The integrated reference was used to harmonize cell type and subtype labels among datasets, yielding 34 annotated cell populations represented across 73,260 cells.

Spatial data. 10x Visium spatial transcriptomics data of human lymph nodes were downloaded from the 10x Genomics website. A paired histology image was used to annotate GC locations.

Spatial mapping using cell2location. Expression signatures of 34 cell populations (Fig. 4b) were estimated using the negative binomial regression model (Supplementary Methods), accounting for the effect of different batches as well as technologies (Supplementary Fig. 24). Before model-based estimation, an untransformed and unnormalized count matrix was filtered to select expressed genes at two cutoffs:

1. Selecting genes detected at mRNA count >0 in >3% of cells
2. Selecting genes detected at mRNA count >0 in a few cells (3% > cell count > 0.1%) but with a mean expression across non-zero cells >1.122018

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

The untransformed and unnormalized mRNA counts from a single Visium experiment were used as input to cell2location using the hyper-parameters stated in Table 1. The prior cell abundance $\hat{N} = 30$ was derived by examining histology images of lymph node tissue (paired images had insufficient image quality).

NMF. Absolute cell type abundance estimates obtained from cell2location were used as input for NMF to identify spatially interlaced tissue zones/compartments of lymph nodes (Fig. 4d). In brief, NMF decomposes cell abundance estimates $w_{s,f}$ into additive components:

$$w_{s,f} = \sum_r z_{s,r} x_{r,f},$$

where $x_{r,f}$ represents NMF weight describing the contribution of each cell type group r to the abundance of cell types f, and $z_{s,r}$ represents NMF weight describing the abundances of each cell type group r across locations s.

NMF was trained for a range of $R = \{8, .., 28\}$, and the decomposition into 14 factors was chosen as a balance between capturing fine tissue zones and splitting known compartments such as light zone of the GC into several distinct factors (Fig. 4e, panel 3) into separate zones. NMF weight $x_{r,f}$ is shown in Fig. 4d and Supplementary Fig. 20c.

Cell2location estimates from this section are reported in Fig. 4, Supplementary Figs. 12b and 20–24 and Supplementary Files 10 and 11.

Assessment of cell2location and alternative methods. RCTD, SPOTlight, Seurat 3 and NNLS (AutoGeneS) were used with default parameters as described in the section on synthetic data (Methods). Stereoscope (implementation using scvi-tools; see Methods, synthetic data section) was used with the following hyper-parameters:

7. No HVG selection was used (10,230 genes shared between single-cell and spatial data).
8. Estimating expression signatures of cell types: learning rate of 0.01 (default); 1,000 epochs (increased compared to default to achieve convergence); batch size of 256 (default)
9. Estimating relative cell abundance in spatial data: learning rate of 0.01 (default); 20,000 epochs (increased compared to default to achieve convergence); batch size of 256 (default)

Assessment criteria of cell2location and alternative methods. The accuracy of cell2location and alternative methods was assessed by mapping six GC cell populations (B_Cycling, B_GC_DZ, B_GC_LZ, B_GC_prePB, FDC and T_CD4$^+$_TfH_GC)[37] to the GC locations, which were anatomically annotated using histology images paired to the Visium data (Fig. 4f). Precision recall, false-positive rate and true-positive rate were evaluated as a macro-average and separately for each cell type (scikit-learn) across six GC cell types expected to be present in annotated GC locations (positive set of locations) and absent from other locations (negative set) (Fig. 4f–h and Supplementary Fig. 23). Precision recall curves are shown for every cell type (Fig. 4g and Supplementary Fig. 23). To simplify the presentation, we additionally used bar plots (Fig. 4h and Supplementary Fig. 22), where bar height represents average precision recall score across cell types, and error bars show standard error of the mean.

Six known GC cell types were classified as either transcriptionally fine or transcriptionally distinct, based on the Euclidean distance of their expression signatures (linear mRNA count space rather than log-transformed) to the closest cell type from the full set of 34 cell types. Euclidean distance represents both the number of genes marking each cell type and the absolute expression levels of those genes, both of which are important for mapping. Cell types were classified as transcriptionally distinct if the Euclidean distance was greater than 100 and as transcriptionally fine if it was less than 100 (Supplementary Fig. 22a).

Additionally, we performed the comparison of model-based and hard-coded methods (Supplementary Methods, sections 2 and 3) for estimating the reference signatures of cell types and other cell2location modeling choices (see Supplementary Fig. 24 and the modeling choices section above), keeping all other parameters constant.

Analysis of human gut Visium data. *Construction of single-cell reference data.* We used a published gut scRNA dataset that covers a range of time points and gut regions from Elmentaite et al.[42]. Crucially, the dataset contains the cells from lymphoid structures within the gut (pediatric subset in particular) and covers the age range of the considered spatial data (adult gut). Literature evidence was used to annotate 19 of 76 cell types as expected in the lymphoid follicle.

Spatial data. 10x Visium spatial data consisting of two replicates of the biopsy of colon tissue were obtained from Fawkner-Corbett et al.[43]. Crucially, the dataset contains mature lymphoid follicles. Both the paired histology image and expression of marker genes (Supplementary Fig. 25b) were considered to annotate lymphoid follicles, prioritizing histology when marker genes were detected outside expected histology areas; the annotation was performed using the 10x Loupe browser. Replicates also contain distinct sequencing depth/RNA detection sensitivity, requiring normalization across batches.

Spatial mapping using cell2location. Reference signatures for 76 cell types were estimated using the negative binomial regression model, accounting for the effect of different batches and technologies (3′ versus 5′) with training mini-batch size of 2,500 and training duration of 200 epochs. Before model-based estimation, an untransformed and unnormalized count matrix was filtered to select expressed genes at two cutoffs:

1. Selecting genes detected at mRNA count >0 in >3% of cells
2. Selecting genes detected at mRNA count >0 in a few cells (3% > cell count > 0.02%) but with a mean expression across non-zero cells >1.12.

The untransformed and unnormalized mRNA counts from a single Visium experiment were used as input to cell2location using the hyper-parameters stated in Table 1. The prior cell abundance $\hat{N} = 25$ was derived by computing average cell count in the paired histology images for 20 random locations.

Cell2location estimates from this section are reported in Fig. 5, Supplementary Figs. 25 and 26 and Supplementary Files 12 and 13.

Assessment of cell2location and alternative methods. RCTD, SPOTlight, Seurat 3 and NNLS (AutoGeneS) were used with default parameters as described in the section on synthetic data (Methods). Stereoscope (implementation using scvi-tools; see Methods, synthetic data section) was used with the following hyper-parameters:

10. No HVG selection was used (14,388 genes shared between single-cell and spatial data).
11. Estimating expression signatures of cell types: learning rate of 0.01 (default); 8,000 epochs (increased compared to default to achieve convergence); batch size of 1,024 (increased compared to default in an attempt to achieve sufficient accuracy)
12. Estimating relative cell abundance in spatial data: learning rate of 0.01 (default); 50,000 epochs (increased compared to default to achieve convergence); batch size of 1,024 (increased compared to default in an attempt to achieve sufficient accuracy)

Similarly to lymph node analysis described above, precision recall curves and scores were used to compare alternative methods. Cell proportions across locations for each of the 19 cell types expected in the lymphoid follicles were used as a predictor; histology-based annotation of the lymphoid follicles was used as a positive set; and all other locations were used as a negative set (Supplementary Figs. 25d and 26). Again, these 19 cell types were classified into transcriptionally fine and transcriptionally distinct using Euclidean distance to the closest other cell type as described in the lymph node section above (Fig. 5f and Supplementary Fig. 25d).

Importance of model features (Supplementary Fig. 25e) and sensitivity to hyper-parameters (Supplementary Fig. 5b) were assessed using precision recall procedures as described above and in the section about sensitivity of model features and hyper-parameters.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Data generated for this manuscript (snRNA-seq and Visium from adjacent sections in the mouse brain (Fig. 2a), sequencing reads as well as Cell Ranger and Space Ranger output) were submitted to ArrayExpress under accession numbers E-MTAB-11114 (Visium) and E-MTAB-11115 (snRNA-seq). Annotated snRNA-seq data are publicly available via cellxgene portals[64]: full dataset (annotation_1_print column denotes cell types) and astrocyte subclusters. The integrated secondary lymphoid organ scRNA-seq data are publicly available for download through S3 bucket. Image data generated in this manuscript were deposited to the BioImage Archive (accession no. S-BIAD207).
Ground truth annotations of germinal center zones in lymph node Visium data can be found on GitHub. Ground truth annotations of the gut lymphoid follicles can be found on GitHub.
Published datasets. snRNA-seq data from Yao et al. were downloaded from the Allen Brain Institute data portal. Slide-seq V2 data from Stickels et al. were downloaded from the Broad Institute data portal (subject to use agreement). Visium data of human lymph nodes can be downloaded from the 10x website (via a function in the scanpy package).

Code availability

The cell2location package is available at https://github.com/BayraktarLab/cell2location/. Documentation and tutorials are available at https://cell2location.readthedocs.io/.
Code used to generate synthetic data is available at https://github.com/vitkl/cell2location_paper/blob/master/notebooks/benchmarking/synthetic_data_construction_improved_real_mg.ipynb. Code used to segment nuclei in histology images is available at https://github.com/yozhikoff/segmentation. Jupyter notebooks covering the analysis in this paper are available at https://github.com/vitkl/cell2location_paper/ and upon reasonable request.

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

References

56. Hafemeister, C. & Satija, R. Normalization and variance stabilization of single-cell RNA-seq data using regularized negative binomial regression. *Genome Biol.* **20**, 296 (2019).
57. Krishnaswami, S. R. et al. Using single nuclei for RNA-seq to capture the transcriptome of postmortem neurons. *Nat. Protoc.* **11**, 499–524 (2016).
58. Wolock, S. L., Lopez, R. & Klein, A. M. Scrublet: computational identification of cell doublets in single-cell transcriptomic data. *Cell Syst.* **8**, 281–291 (2019).
59. Wolf, F. A., Angerer, P. & Theis, F. J. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol.* **19**, 15 (2018).
60. Roberts, K. & Bayraktar, O. A. Automation of multiplexed RNAscope single-molecule fluorescent in situ hybridization and immunohistochemistry for spatial tissue mapping. *Methods Mol. Biol.* **2148**, 229–244 (2020).
61. Berg, S. et al. ilastik: interactive machine learning for (bio)image analysis. *Nat. Methods* **16**, 1226–1232 (2019).
62. Turner, J. S. et al. Human germinal centres engage memory and naive B cells after influenza vaccination. *Nature* **586**, 127–132 (2020).
63. Madissoon, E. et al. scRNA-seq assessment of the human lung, spleen, and esophagus tissue stability after cold preservation. *Genome Biol.* **21**, 1 (2019).
64. Megill, C. et al. chanzuckerberg/cellxgene: release 0.15.0. https://doi.org/10.5281/zenodo.3710410 (2020).

Acknowledgements

We thank B. Velten, Y. Huang and L. Marconato for feedback on the cell2location model; M. Prete and V. Kiselev for dockerizing the cell2location tool and creating the web portals for sharing our data; N. Kumasaka for helpful comments on single-cell analysis; S. Leonard and K. Polanski for help with spatial and single-nucleus data processing; K. James for advice on gut immune cell types; K. Roberts for advice on smFISH; J. E. Kwa for advice on snRNA-seq; J. Eliasova for illustrations and logo design; K. James advice on human gut immune cells; A. Antanaviciute, H. Koohy and A. Simmons for sharing gut Visium data; F. Obermeyer and M. Jankowiak for help with advanced use of pyro code base; and D. Rowitch and S. Teichmann for comments on the manuscript. H.W.K. was funded by a Sir Henry Wellcome Postdoctoral Fellowship (213555/Z/18/Z). This study was supported by Wellcome Trust Core Funding to O.A.B.

Author contributions

V.K., O.S. and O.A.B. conceived the study. V.K. developed and implemented the cell2location model with feedback from O.S. and worked on validation, mouse brain, human lymph node, gut and benchmarking analyses. A.S. performed Slide-seq and hyperparameter sensitivity analysis, performed histology image segmentation, wrote data visualization modules and contributed to implementation of cell2location in pyro and benchmarking analyses. E.D. worked on generating synthetic data, mouse brain cell annotation and validation of cell type mapping and contributed to lymph node analysis. V.KE. and A.S. contributed to the development and interpretation of the model for estimating cell type signatures. A.L., M.G., A.A.I. and M.S.J. contributed to the developing of model architecture, writing the model and downstream analysis code, interpretation of the cell2location results as well as downstream clustering and NMF. A.A.R. performed benchmarking of stereoscope, supervised by R.V.-T. and O.S. A.G. contributed to implementation of cell2location using scvi-tools and pyro. H.W.K. contributed to lymph node data analysis and interpretation, supervised by L.J. R.E. contributed to gut data analysis. L.R. performed snRNA-seq experiments. L.T. performed Visium experiments. J.S.P. performed smFISH experiments and imaging. T.L. performed smFISH image processing and astrocyte segmentation analysis. V.K., O.S. and O.A.B. wrote the manuscript, with feedback from all authors.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41587-021-01139-4.

Correspondence and requests for materials should be addressed to Oliver Stegle or Omer Ali Bayraktar.

Peer review information *Nature Biotechnology* thanks Steven Sloan, Xiaoqun Wang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

nature research

Corresponding author(s): Oliver Stegle, Omer Bayraktar

Last updated by author(s): Oct 26, 2021

Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☐	☒	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	10X CellRanger version 3.0.2, 10X SpaceRanger version 1.0
Data analysis	The cell2location package is available at https://github.com/BayraktarLab/cell2location/. Documentation and tutorials are available at https://cell2location.readthedocs.io/. Code used to generate synthetic data is available in https://github.com/vitkl/cell2location_paper/blob/master/notebooks/benchmarking/synthetic_data_construction_improved_real_mg.ipynb. Code used to segment nuclei in histology images is available in https://github.com/yozhikoff/segmentation. Jupyter notebooks covering the analysis in this paper are available at https://github.com/vitkl/cell2location_paper/ and upon request. Specific package version used for analysis are as follows (python): python 3.7.10 scvi 0.13.0 pyro 1.6.0+823dce2b scanpy 1.7.2 numpy 1.20.2 scipy 1.6.2 sklearn 0.24.1 Other python dependencies: torch 1.8.1+cu102 anndata 0.7.5 pandas 1.2.3 leidenalg 0.8.3 louvain 0.7.0

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

Specific package version used for analysis are as follows (R):
R 4.0.3
Seurat 4.0
SeuratDisk 0.0.0.9019
RCTD (version from github commit 037a5da2c282fab94b1aee5cf489de73c494dc39)

R SPOTlight was used via the docker image provided by the developers (marcelosua/spotlight_env_r:latest):
SPOTlight 0.1.7
R 3.6.3

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Data generated for this manuscript (snRNA-seq and Visium from adjacent sections in the mouse brain, Fig 2A, sequencing reads as well as CellRanger and SpaceRanger output) is submitted to ArrayExpress under accession numbers E-MTAB-11114 (Visium) and E-MTAB-11115 (snRNA-seq). Image data generated in this manuscript was deposited to BioImage Archive (accession number: S-BIAD207).

The integrated secondary lymphoid organ scRNA-seq data are publicly available for download through S3 buckets at https://cell2location.cog.sanger.ac.uk/browser.html (lymph nodes - /paper/integrated_lymphoid_organ_scrna/). snRNAseq data can be publicly viewed without download via cellxgene portals at https://cell2location.cellgeni.sanger.ac.uk/mouse-brain (full dataset, annotation_1_print column denotes cell types) and https://cell2location.cellgeni.sanger.ac.uk/mouse-brain-astrocytes (astrocyte subclusters).
Ground truth annotations of Germinal Center zones in lymph node Visium data can be found at https://github.com/vitkl/cell2location_paper/blob/master/notebooks/selected_results/lymph_nodes_analysis/manual_GC_annot.csv . Ground truth annotations of the gut lymphoid follicles can be found at https://github.com/vitkl/cell2location_paper/tree/master/notebooks/selected_results/tissue_annotation/oxford .

Published datasets. snRNA-seq data from Yao et al were downloaded from Allen Brain Institute data portal (https://portal.brain-map.org/atlases-and-data/rnaseq/mouse-whole-cortex-and-hippocampus-smart-seq). Slide-Seq V2 data from Stickels, Murray et al were downloaded from the Broad Institute data portal (https://singlecell.broadinstitute.org/single_cell/study/SCP815/sensitive-spatial-genome-wide-expression-profiling-at-cellular-resolution#study-download, subject to use agreement). Visium data of human lymph nodes can be downloaded from 10X website (via a function in Scanpy package).

Genome references used for mapping mouse brain data were downloaded from 10X website.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

We aimed to profile 2 mice at 4 adjacent tissue positions (i.e. sections spaced by 200 microns). In total, four consecutive sets of thick and thin tissue sections were collected from each brain. Five sets of tissue sections yielded both good quality single nuclei and Visium data (three adjacent sections from mouse 1 and two sections from mouse 2) while one additional section from mouse 2 yielded good single nuclei; these were sequenced and considered for analysis in this study (Fig 2-3).

For analysis of other tissues and Slide-Seq V2 published spatial transcriptomics datasets were used (human lymph node, n=1; human gut, n=2; Slide-Seq V2, n=1) with the total number of samples limited by availability.

No statistical methods were used to pre-determine sample sizes but our sample sizes are similar to those reported previously (references 3, 25, 42-43).

Data exclusions

Visium spots located outside tissue areas were excluded from all analyses in the paper based on the histological examination. Specifically this was done in 2 steps:
1. Automatic tissue detection using SpaceRanger workflow with subsequence location filtering. No further data exclusion for mouse brain and human lymph node data were performed.
2. When the automatic tissue detection failed (adult human gut samples) - manual annotation was performed, with subsequent filtering of location that do not overlap tissue.
The rationale behind this filtering is as follows: 10X Visium locations are manufactured in a hexagonal grid within square capture areas - as a

iranpaper Downloaded from https://iranpaper.ir
https://www.tarjomano.com ترجمه تخصصی این مقاله ترجمانو

result not all locations are covered by tissue and the locations that are not covered contain background RNA rather t
tissue.

Replication	For mouse experiments, multiple animals were processed independently on the same day. Fig 2G and Fig S10 report on consistency of cell2location cell abundance estimates between adjacent sections and replicate animals. Differences cell type abundances between sections can be attributed to change in the regional composition (by histology examination). Cell type locations of novel astrocyte subtypes were validated using single molecule FISH of mRNA for gene marking the novel subtypes at corresponding anatomical location in different mice (Fig 3D) - thus replicating the detection of novel subtypes using alternative technology and an independently processed animal.
Randomization	Randomization is not relevant to our study because only wild-type mice were used for mouse brain mapping experiments. For lymph node and gut mapping experiments, the data were derived from published observational studies.
Blinding	Blinding is not relevant to our study because the objective was not to compare the effect of 2 conditions. Real world data were generated from the mouse brain or derived from published observational studies that included wild-type / healthy tissues only and the simualtions were generated by the research team.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Laboratory animals	Two wild-type adult C57BL/6 mice (postnatal day 56, 1 female and 1 male) were used for 10X Visium experiments. One additional wild-type adult C57BL/6 mouse (postnatal day 63, male) was used for single-molecule fluorescent in situ hybridization.
Wild animals	Study did not involve wild animals.
Field-collected samples	Study did not involve field-collected samples.
Ethics oversight	Mouse work was carried out at the Wellcome Sanger Institute in accordance with UK Home Office regulations and UK Animals (Scientific Procedures) Act of 1986 under a UK Home Office license, which were regularly reviewed by the institutional Animal Welfare and Ethical Review Body.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

nature portfolio

https://doi.org/10.1038/s41587-021-01139-4

Supplementary information

Cell2location maps fine-grained cell types in spatial transcriptomics

In the format provided by the authors and unedited

Figure S1

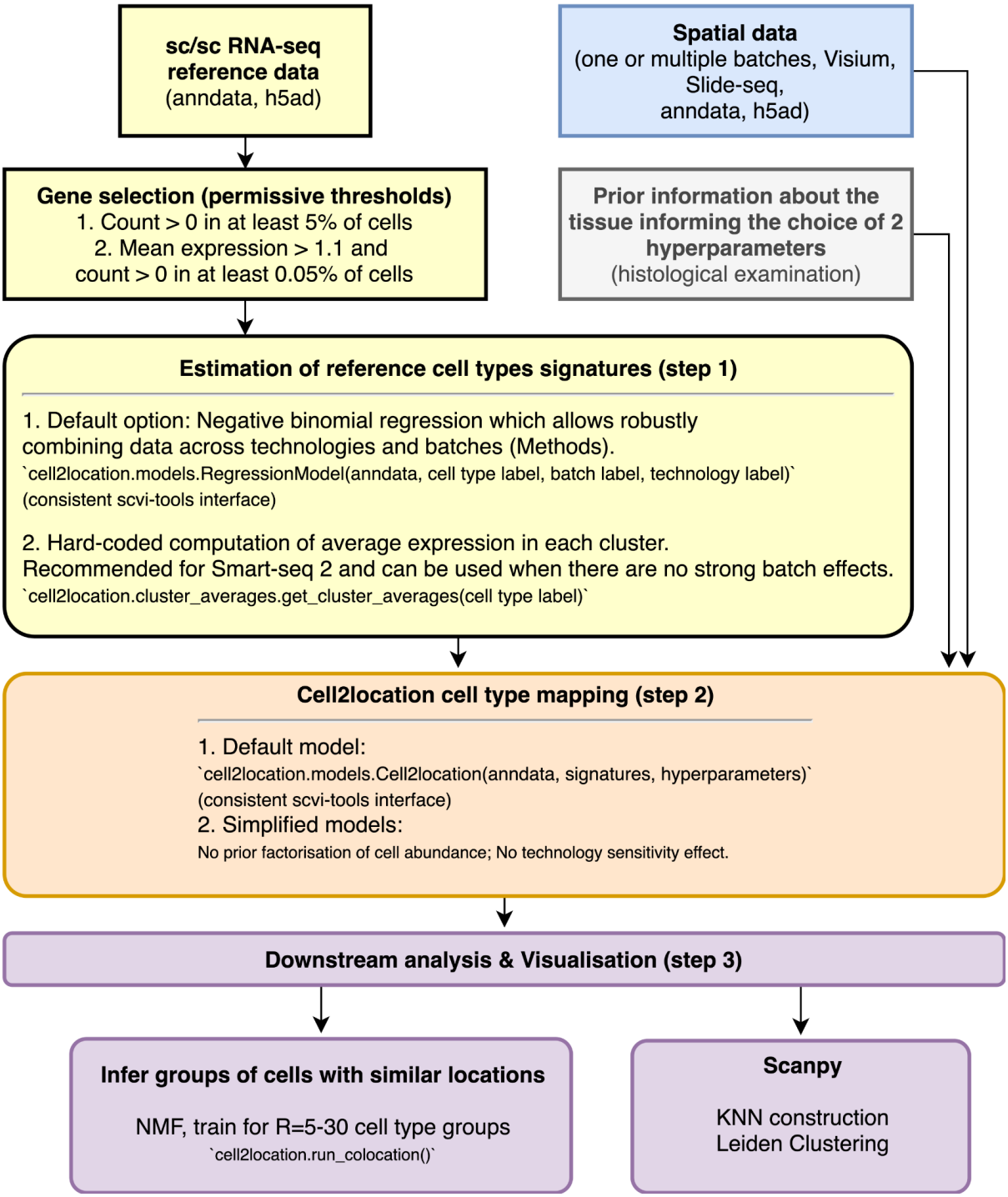


Figure S1. Details of cell2location workflow. Diagram of the cell2location workflow, which consists of estimation of reference cell type signatures estimation from scRNA-seq, cell2location cell type mapping and downstream analyses. Each block highlights relevant models, tutorial notebooks show the 3 steps outlined.

Figure S2

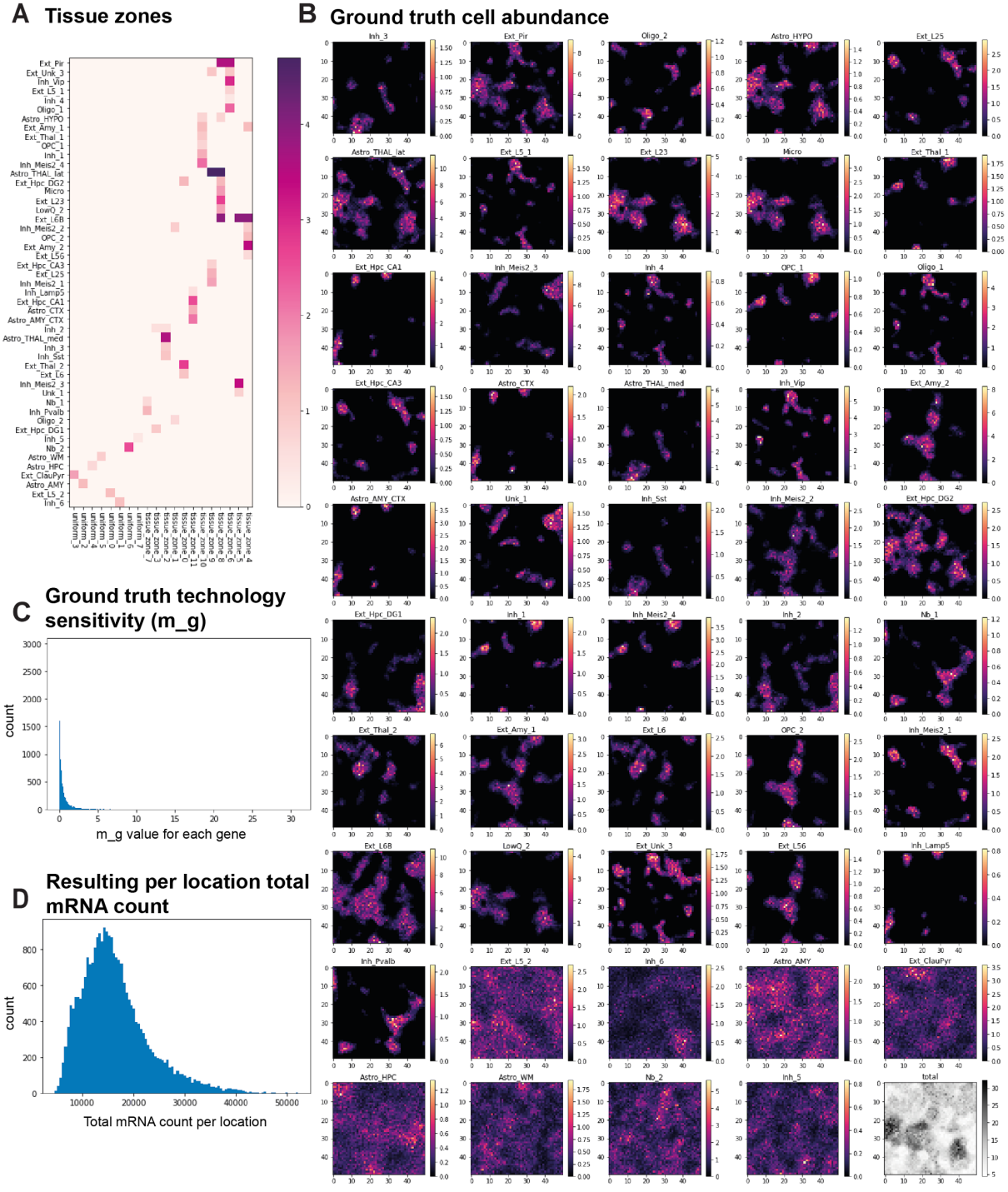


Figure S2. Spatial patterns and tissue zones in the synthetic data.

A. Assignment of cell types (rows) to alternative distribution patterns (columns) for generating a synthetic visium dataset. A subset of cell types are assigned to ubiquitous distribution patterns (“uniform”, left), whereas the majority of cell types are assigned to one of 11 tissue zones (“tissue_zone”, right). Colour denotes a scaling factor per cell type to simulate variable abundance across cell types.

B. Simulated cell type abundance patterns for all 49 reference cell types (facets) and the total cell abundance. The first 41 cell types were simulated to exhibit smoothly varying regional and sparse location patterns (most locations have 0 abundance); the last 8

cell types are drawn from a ubiquitous abundance pattern. Bottom right panel denotes the total cell abundance generated from all cell types. Colour denotes absolute cell abundance.

C. Distribution of the simulated technology-specific sensitivity coefficient between snRNA-seq and synthetic spatial data. Shown is a histogram of the sensitivity parameter (X-axis, corresponding to m_g in the cell2location model). The distribution mimics parameters observed in the mouse brain; Methods.
D. Distribution of the total mRNA count per location (X-axis) in the synthetic data across 2,500 simulated locations. The distribution resembles empirical observations in the mouse brain.

Figure S3

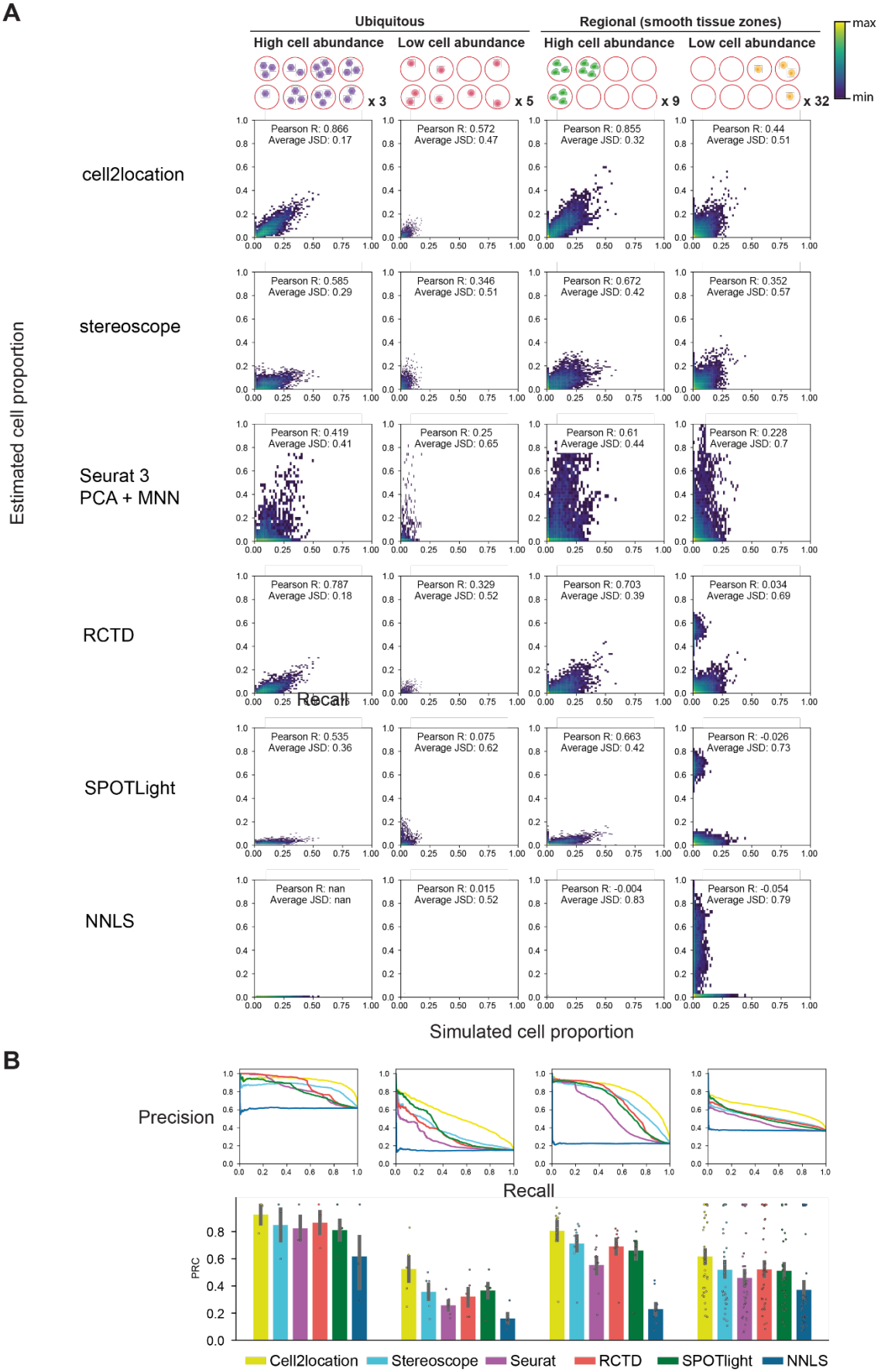

A
Ubiquitous
Regional (smooth tissue zones)
High cell abundance
Low cell abundance
High cell abundance
Low cell abundance
x 3
x 5
x 9
x 32
max
min
cell2location
stereoscope
Seurat 3
PCA + MNN
RCTD
SPOTLight
NNLS
Estimated cell proportion
Recall
Simulated cell proportion
Pearson R: 0.866
Average JSD: 0.17
Pearson R: 0.572
Average JSD: 0.47
Pearson R: 0.855
Average JSD: 0.32
Pearson R: 0.44
Average JSD: 0.51
Pearson R: 0.585
Average JSD: 0.29
Pearson R: 0.346
Average JSD: 0.51
Pearson R: 0.672
Average JSD: 0.42
Pearson R: 0.352
Average JSD: 0.57
Pearson R: 0.419
Average JSD: 0.41
Pearson R: 0.25
Average JSD: 0.65
Pearson R: 0.61
Average JSD: 0.44
Pearson R: 0.228
Average JSD: 0.7
Pearson R: 0.787
Average JSD: 0.18
Pearson R: 0.329
Average JSD: 0.52
Pearson R: 0.703
Average JSD: 0.39
Pearson R: 0.034
Average JSD: 0.69
Pearson R: 0.535
Average JSD: 0.36
Pearson R: 0.075
Average JSD: 0.62
Pearson R: 0.663
Average JSD: 0.42
Pearson R: -0.026
Average JSD: 0.73
Pearson R: nan
Average JSD: nan
Pearson R: 0.015
Average JSD: 0.52
Pearson R: -0.004
Average JSD: 0.83
Pearson R: -0.054
Average JSD: 0.79
B
Precision
Recall
PRC
Cell2location
Stereoscope
Seurat
RCTD
SPOTlight
NNLS

Figure S3. Assessment of cell2location and alternative methods using simulated data. Shown are supplementary results that complement main text Fig 1, based on simulated data that comprise 49 reference cell types distributed across 2,500 locations.

A. Comparison of simulated and estimated cell type proportions for cell2location and alternative methods (rows), considering cell types assigned to one of four abundance patterns. Top panel: Simulation scheme according to which alternative cell type abundance patterns are generated, and the number of cell types assigned to each pattern (c.f. Fig S2, Methods). Lower panels: 2D histogram counts (log10, colour), displaying the concordance between simulated (X-axis) and estimated (Y-axis) cell type proportions across 2,500 locations and cell types. R denotes Pearson correlation and JSD denotes Jensen–Shannon divergence.
B. Assessment of cell2location and alternative methods for detecting locations with non-zero cell abundance across the four cell type abundance patterns (2,500 locations, 49 cell types). Top: macro-average precision-recall curves across all cell types within the respective abundance pattern for alternative methods. Bottom: bar plots depicting precision-recall scores (Y-axis) for alternative methods (colour, X-axis). Bar height denotes the average precision-recall score across cell types; error bars denote standard errors across cell types.

Figure S4

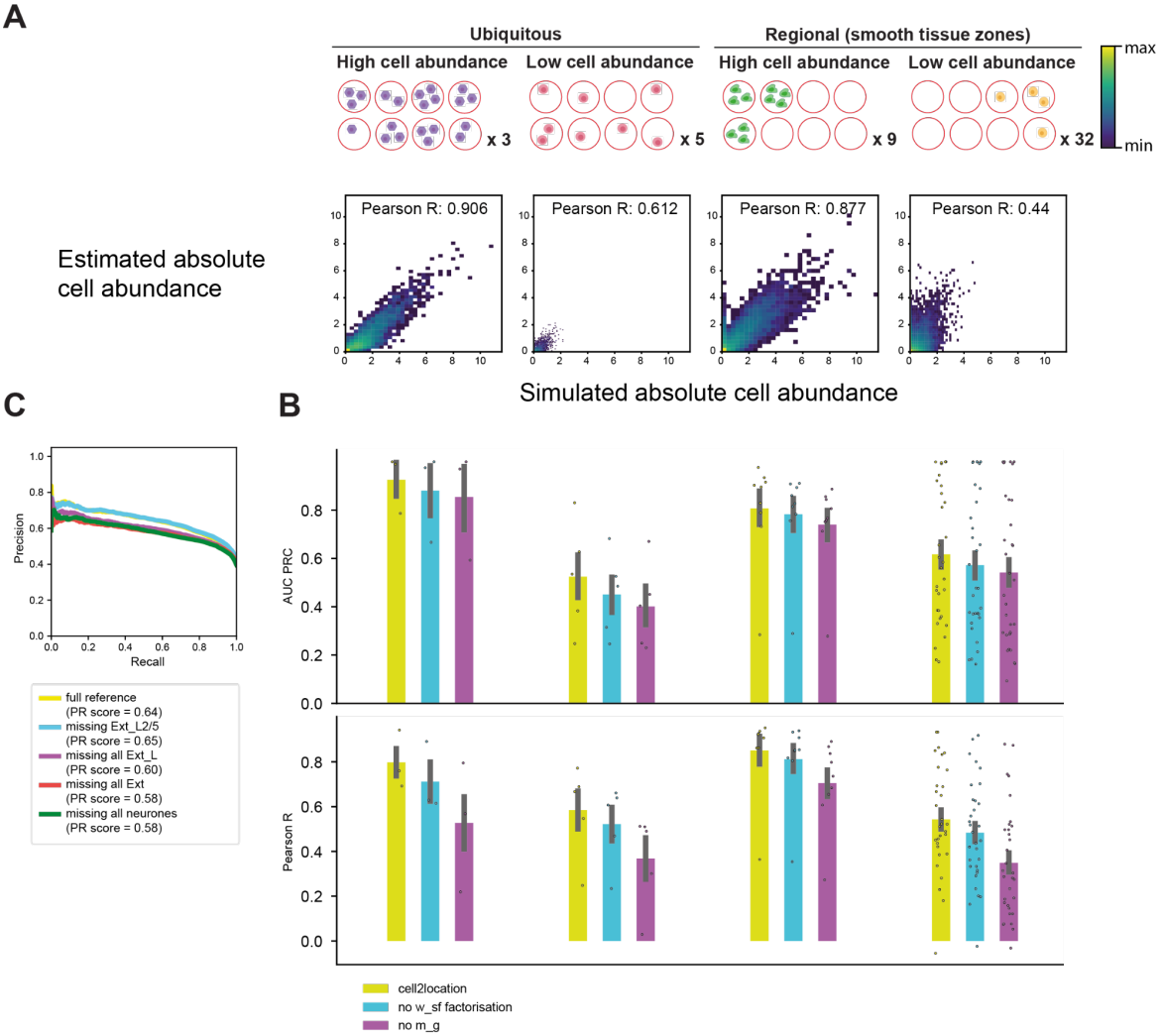


Figure S4. Assessment of cell2location and alternative methods using simulated data. Shown are supplementary results that complement main text Fig 1, based on simulated data that comprise 49 reference cell types distributed across 2,500 locations.

A. Top panel: 4 alternative cell type abundance patterns considered for data simulation (see also Fig S2) and the number of cell types assigned to each pattern (Methods). Bottom panel: 2D histogram plots, displaying the concordance between simulated (X-axis) and estimated (Y-Axis) absolute cell-type abundance across 2,500 locations, aggregated for cell types assigned to the respective pattern. Colour denotes 2D histogram counts (50 bins along both X- and Y-axis). R denotes Pearson correlation.

B. Assessment of cell2location and two reduced variants of the model without specific features: prior factorisation of cell abundance w_sf and gene-specific technology scaling m_g (used in conjunction with the NB regression model, Suppl. Methods). Shown are precision recall curves for cell type detection (top panel) and correlation coefficients between simulated and estimated cell type proportions (bottom panel; cf. FIg S3B), aggregated for cell types assigned to the respective cell type abundance pattern (as in A). Bar height denotes the average precision-recall score across cell types; error bars denote standard errors across cell types. The number of cell types assigned to each pattern is as stated in panel A.

C. Assessment of cell2location when removing reference expression signatures of increasing fractions of cell types. Shown are macro-average precision-recall curves across 17 retained cell types, when subsampling the remaining cell types in the reference: one Ext_L25 subtype removed, 48/49 retained; all excitatory layer Ext_L neurones removed, 42/49 retained; all excitatory Ext neurones removed, 31/49 retained; all neurones removed, 17/49 retained. The corresponding precision-recall scores are stated in the legend.

Figure S5

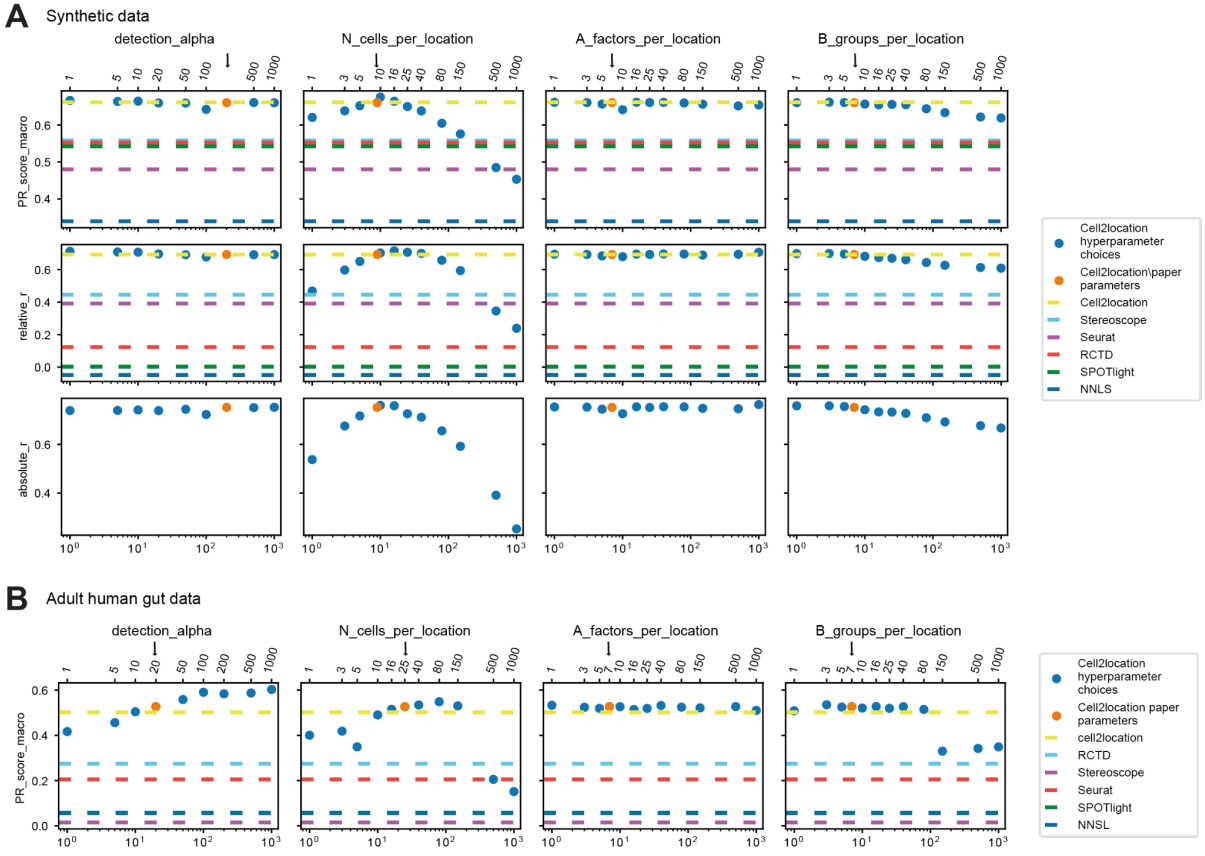


Figure S5. Stability of cell2location cell abundance estimates with respect to model hyperparameter settings. Assessment of the accuracy of cell2location when varying selected hyperparameters using alternative benchmarks. Rows: alternative datasets and assessment metrics.

A. Simulated data. From top to bottom: Precision recall curve for identifying cell types with non-zero abundance using cell2location or alternative methods (as in Fig S2B). Correlation coefficients (Pearson R) between simulated and estimated cell type proportions (as in Fig 1C). Correlation coefficient (Pearson R) between stimulated and estimated absolute cell type proportions (as in Fig S4A). Arrows denote hyperparameter values used in the paper (from left to right 200, 9, 7, 7).

B. Gut benchmark. Precision recall curve for identifying cell types with non-zero abundance in annotated GC locations (positive set of locations) (as in Fig 5F). Arrows denote hyperparameter values used in the paper (from left to right 20, 25, 7, 7).

Columns denote result when varying selected hyperparameters of cell2location, from left to right: varying the hyperparameter α^y for regularising within-experiment variation in RNA detection sensitivity (detection_apha), the expected cell abundance $\widehat{N}$ per location (N_cells_per_location), the number of cell types per location $\hat{A}$ (A_factors_per_location) and the number of co-located cell type groups $\hat{B}$ per location (B_groups_per_location). Note that only the first two hyperparameters are advised to be adjusted by the user (Methods, Fig S27). In each panel, horizontal dashed lines denote results from the cell2location and alternative methods as presented for the main analysis presented in the main text. Blue dots show results obtained from cell2location when varying each of the considered hyperparameters. The orange dot denotes the hyperparameter settings used for the main analysis presented in the main text.

Figure S6

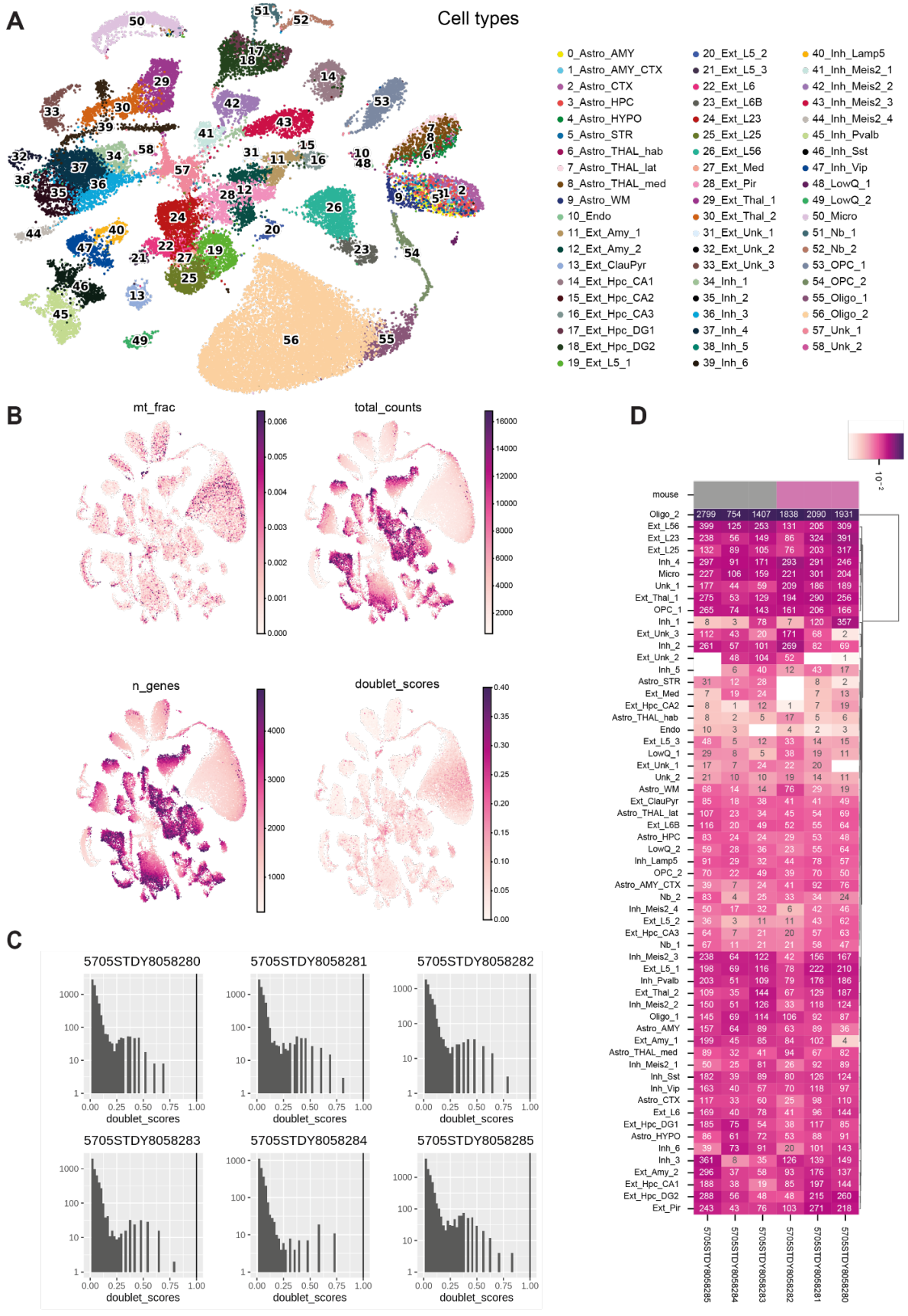

A
Cell types
0_Astro_AMY
1_Astro_AMY_CTX
2_Astro_CTX
3_Astro_HPC
4_Astro_HYPO
5_Astro_STR
6_Astro_THAL_hab
7_Astro_THAL_lat
8_Astro_THAL_med
9_Astro_WM
10_Endo
11_Ext_Amy_1
12_Ext_Amy_2
13_Ext_ClauPyr
14_Ext_Hpc_CA1
15_Ext_Hpc_CA2
16_Ext_Hpc_CA3
17_Ext_Hpc_DG1
18_Ext_Hpc_DG2
19_Ext_L5_1
20_Ext_L5_2
21_Ext_L5_3
22_Ext_L6
23_Ext_L6B
24_Ext_L23
25_Ext_L25
26_Ext_L56
27_Ext_Med
28_Ext_Pir
29_Ext_Thal_1
30_Ext_Thal_2
31_Ext_Unk_1
32_Ext_Unk_2
33_Ext_Unk_3
34_Inh_1
35_Inh_2
36_Inh_3
37_Inh_4
38_Inh_5
39_Inh_6
40_Inh_Lamp5
41_Inh_Meis2_1
42_Inh_Meis2_2
43_Inh_Meis2_3
44_Inh_Meis2_4
45_Inh_Pvalb
46_Inh_Sst
47_Inh_Vip
48_LowQ_1
49_LowQ_2
50_Micro
51_Nb_1
52_Nb_2
53_OPC_1
54_OPC_2
55_Oligo_1
56_Oligo_2
57_Unk_1
58_Unk_2
B
mt_frac
total_counts
n_genes
doublet_scores
C
5705STDY8058280
5705STDY8058281
5705STDY8058282
5705STDY8058283
5705STDY8058284
5705STDY8058285
doublet_scores
D
mouse
Oligo_2 2799 754 1407 1838 2090 1931
Ext_L56 399 125 253 131 205 309
Ext_L23 238 56 149 86 324 391
Ext_L25 132 89 105 76 203 317
Inh_4 297 91 171 293 291 246
Micro 227 106 159 221 301 204
Unk_1 177 44 59 209 186 189
Ext_Thal_1 275 53 129 194 290 256
OPC_1 265 74 143 161 206 166
Inh_1 8 3 78 7 120 357
Ext_Unk_3 112 43 20 171 68 2
Inh_2 261 57 101 269 82 69
Ext_Unk_2 48 104 52 1
Inh_5 6 40 12 43 17
Astro_STR 31 12 28 8 2
Ext_Med 7 19 24 7 13
Ext_Hpc_CA2 8 1 12 1 7 19
Astro_THAL_hab 8 2 5 17 5 6
Endo 10 3 4 2 3
Ext_L5_3 48 5 12 33 14 15
LowQ_1 29 8 5 38 19 11
Ext_Unk_1 17 7 24 22 20
Unk_2 21 10 10 19 14 11
Astro_WM 68 14 14 76 29 19
Ext_ClauPyr 85 18 38 41 41 49
Astro_THAL_lat 107 23 34 45 54 69
Ext_L6B 116 20 49 52 55 64
Astro_HPC 83 24 24 29 53 48
LowQ_2 59 28 36 23 55 64
Inh_Lamp5 91 29 32 44 78 57
OPC_2 70 22 49 39 70 50
Astro_AMY_CTX 39 7 24 41 92 76
Nb_2 83 4 25 33 34 24
Inh_Meis2_4 50 17 32 6 42 46
Ext_L5_2 36 3 11 11 43 62
Ext_Hpc_CA3 64 7 21 20 57 63
Nb_1 67 11 21 21 58 47
Inh_Meis2_3 238 64 122 42 156 167
Ext_L5_1 198 69 116 78 222 210
Inh_Pvalb 203 51 109 79 176 186
Ext_Thal_2 109 35 144 67 129 187
Inh_Meis2_2 150 51 126 33 118 124
Oligo_1 145 69 114 106 92 87
Astro_AMY 157 64 89 63 89 36
Ext_Amy_1 199 45 85 84 102 4
Astro_THAL_med 89 32 41 94 67 82
Inh_Meis2_1 50 25 81 26 92 89
Inh_Sst 182 39 89 80 126 124
Inh_Vip 163 40 57 70 118 97
Astro_CTX 117 33 60 25 98 110
Ext_L6 169 40 78 41 96 144
Ext_Hpc_DG1 185 75 54 38 117 85
Astro_HYPO 86 61 72 53 88 91
Inh_6 39 73 91 20 101 143
Inh_3 361 8 35 126 139 149
Ext_Amy_2 296 37 58 93 176 137
Ext_Hpc_CA1 188 38 19 85 197 144
Ext_Hpc_DG2 288 56 48 48 215 260
Ext_Pir 243 43 76 103 271 218
5705STDY8058285
5705STDY8058284
5705STDY8058283
5705STDY8058282
5705STDY8058281
5705STDY8058280

Figure S6. Quality control metrics of the mouse brain snRNA-seq data and cell type clusters.

A. UMAP representation (X- and Y-axis) of 59 cell subtypes identified by Louvain clustering on the integrated snRNA-seq dataset across sections (extending Fig 2B).
B. Different quality control metrics shown on the UMAP representation as in A (X- and Y-axis, axis rotated 90 degrees compared to A): the proportion of mRNA counts per cell coming from mitochondrial-encoded genes, total mRNA counts per cell, the total number of genes with mRNA counts > 0, doublet cell scores as identified by scrublet tool.
C. Histogram of doublet scores (X-axis, obtained using scrublet) across the 6 profiled mouse brain tissue sections (panels), with high values corresponding to a higher likelihood that a given cell barcode corresponds to a doublet (a droplet containing two cells and one bead with a single cell barcode). The Y-axis of histogram counts is shown on a log scale. Cells with doublet scores > 0.2 were discarded from the analysis in panel A and the rest of the paper.
D. Heatmap denoting the proportion of cells (colour, log scale, normalised per tissue section) of each of the 59 cell subtypes (columns) coming from each tissue section (rows). Colour bar denotes mouse 1 (purple) and mouse 2 (grey). Text denotes the count of cells.

Figure S7

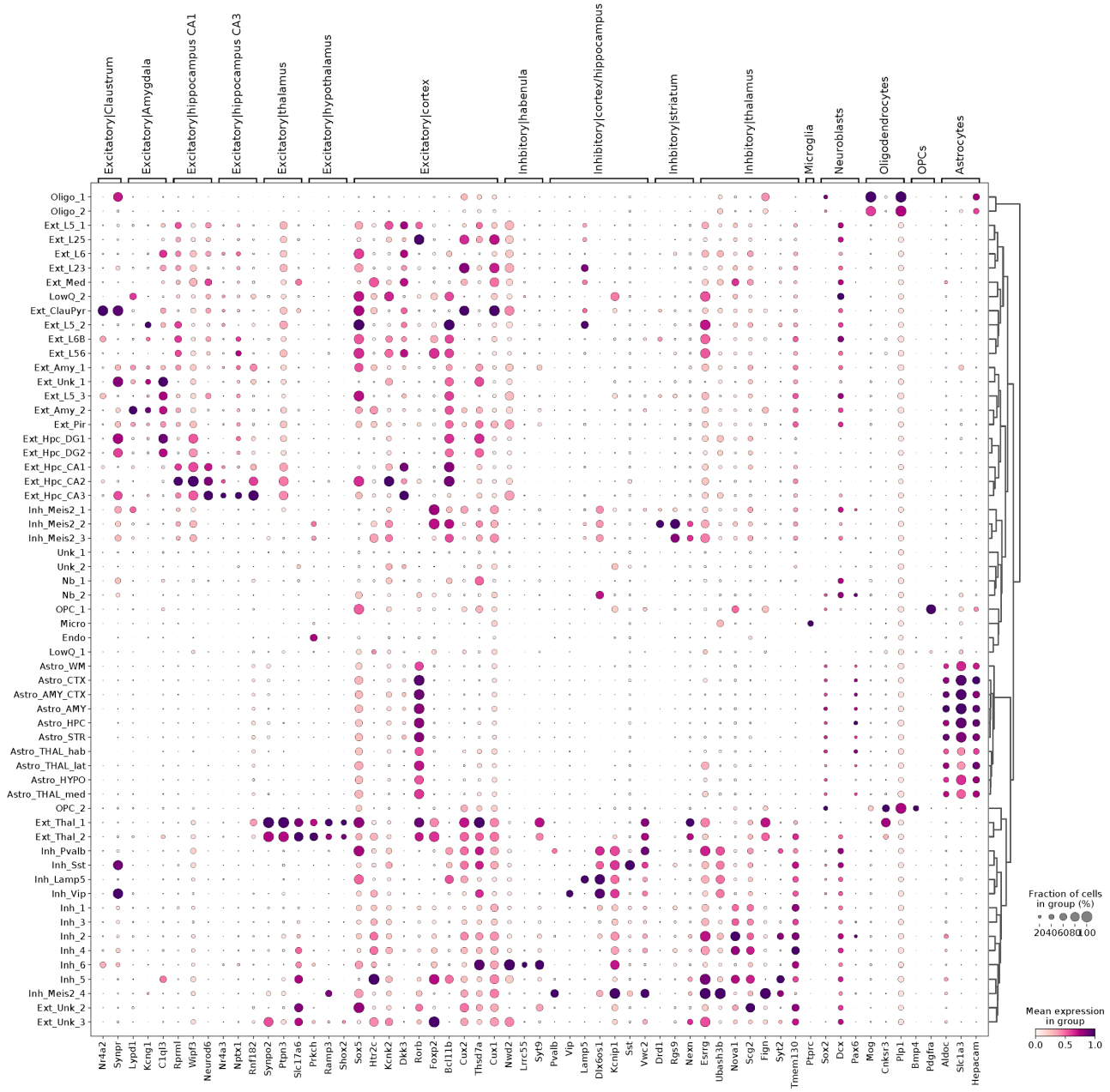


Figure S7. Regional marker genes of the 59 cell type clusters in the mouse brain snRNA-seq data. Shown are expression levels of regional marker genes across 59 mouse brain reference cell types. Dot size denotes the fraction of cells per cluster (rows) that express a given marker gene (count > 0; columns). Dot colour denotes the relative expression of each gene in each cluster, normalised by the maximum values for each gene (scaled between 0 and 1). Marker genes are grouped by region and broad cell type (regional excitatory and inhibitory neurons from Zeisel et al, 2015, 2018 and Tasic et al 2018, as well as glial cell types). Briefly, the 59 cell types were identified using two consecutive analyses (see Methods). First, Louvain clustering of mouse brain snRNAseq data identified 51 clusters. Out of these, 44 clusters could be annotated as known regional neuronal and glial subtypes as above; the remaining 7 clusters were composed of low quality cells or unknown cell types. Second, we selected two astrocyte clusters (2/44) and performed a targeted analysis to identify regionally

and molecularly distinct astrocyte subtypes. Briefly, an initial clustering analysis using Leiden clustering yielded 17 putative astrocyte subtypes. Next, we used distinct locations in the initial spatial mapping of these subtypes (using cell2location, Fig S14) and marker gene expression (Fig S13) to consolidate 17 initial clusters into 10 transcriptionally and regionally distinct subtypes. Collectively, these analyses resulted in a combined set of 59 distinct reference cell types with molecular signatures.

Figure S8

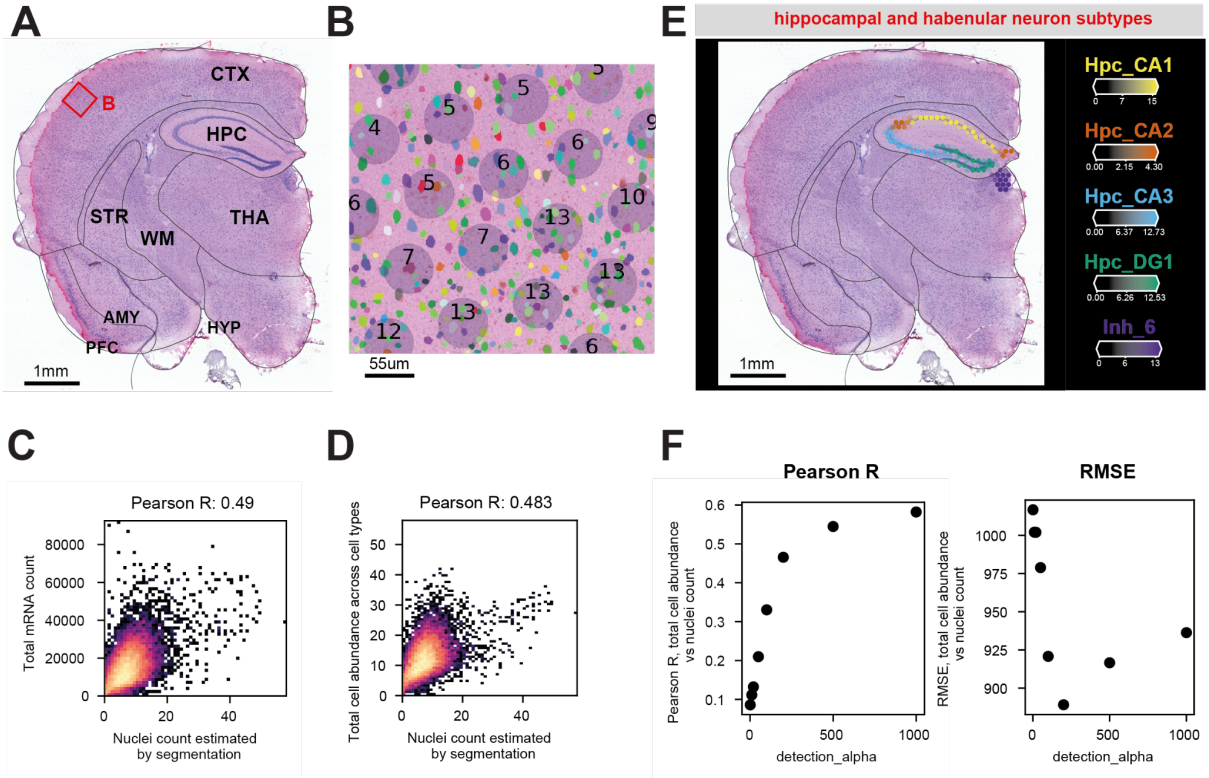


Figure S8. Cell type abundance estimates from histology image segmentation and cell2location.

A. Mouse brain regions present in the assayed section. Shown is a histology (H&E) image of the tissue section and the outlines of major brain regions. The red bounding box corresponds to the region in B. Scale bar: 1 mm.

B. A close-up image of the somatosensory cortex. Nuclei segmentation masks (pseudocolours) are shown over H&E image of the tissue section in the background. Grey circles denote the positions and size of the 10X Visium array “spots” (i.e. mRNA capture locations). The numbers denote the number of nuclei that overlap with the capture locations. Scale bar: 50 micrometers.

C. 2D histogram plots, displaying the correlation between the number of nuclei estimated via segmentation of the H&E image (X-axis) and the total mRNA count per location detected using the Visium technology (Y-Axis). R denotes Pearson correlation.

D. 2D histogram plots, displaying the correlation between the number of nuclei estimated via segmentation of the H&E image (X-axis) and the total absolute cell abundance estimated by cell2location (Y-Axis). R denotes Pearson correlation.

E. Estimated cell abundances (colour intensity) of excitatory neurons across hippocampal divisions (Ext Hpc CA1, Ext Hpc CA2, Ext Hpc CA3, Ext Hpc DG1) and habenular inhibitory subtype Inh 6. Scale bar: 1 mm.

F. The impact of the choices of the hyperparameter α^y for regularising within-experiment variation in RNA detection sensitivity (detection_alpha, X-axis) on the consistency of total cell abundance estimates by cell2location and nuclei counts estimated via segmentation (Y-axis, as in D). Right: Pearson correlation R; Left: root mean squared error (RMSE).

Figure S9

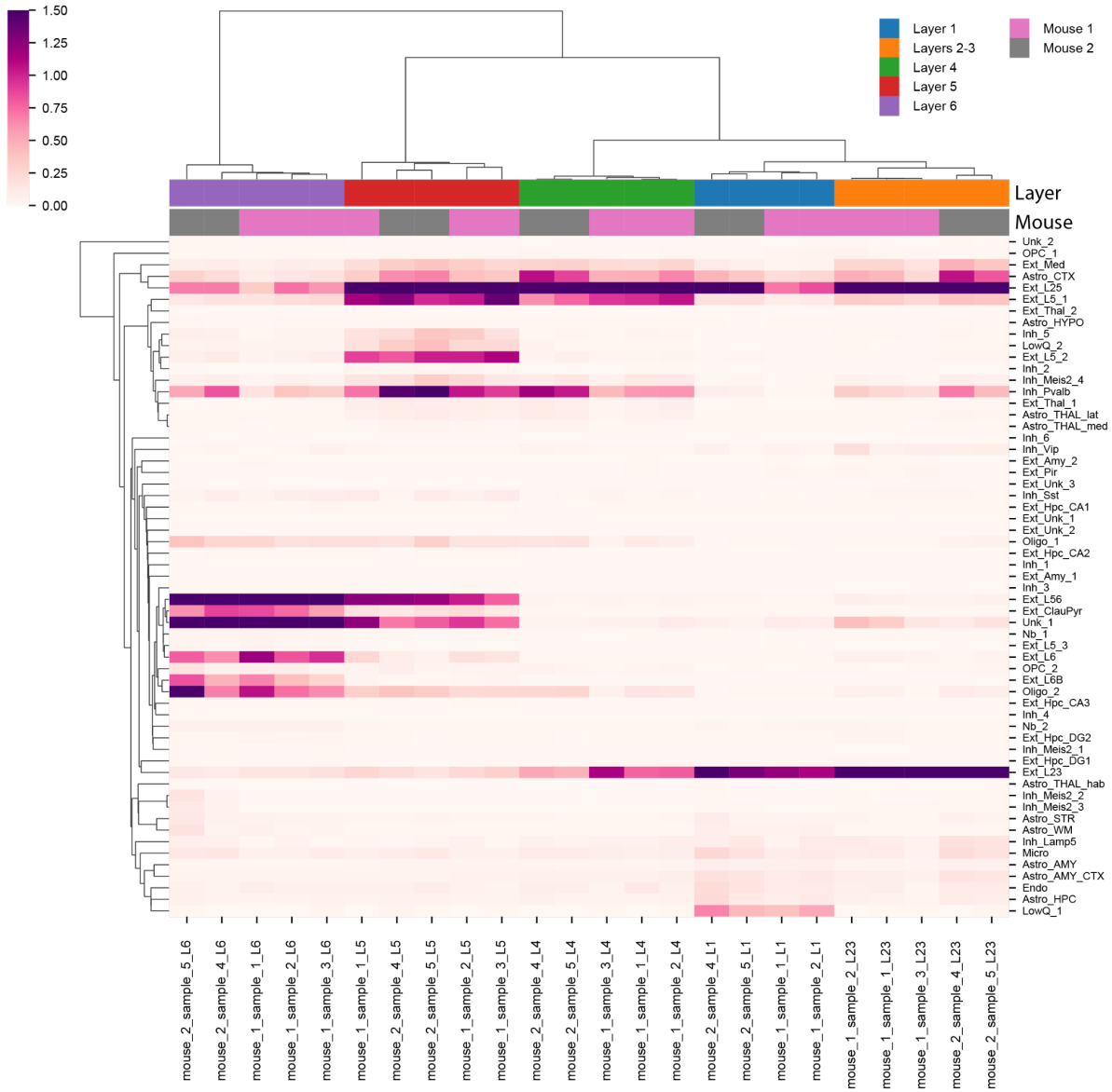


Figure S9. Hierarchical clustering of cell type abundance estimates recapitulate cortical layers, consecutive sections and replicate mice. Estimated cell abundances of 59 cell subtypes across 5 cortical layers, 3 sections from mouse #1 and 2 sections from mouse #2. The heatmap with a colour corresponding to the absolute cell abundance for each cell type (rows) across cortical layers separately in each section (columns). Colour bars denote cortical layers (top) and mice samples were taken from (bottom). The correlation metric and “single” linkage strategy were used for clustering.

Figure S10

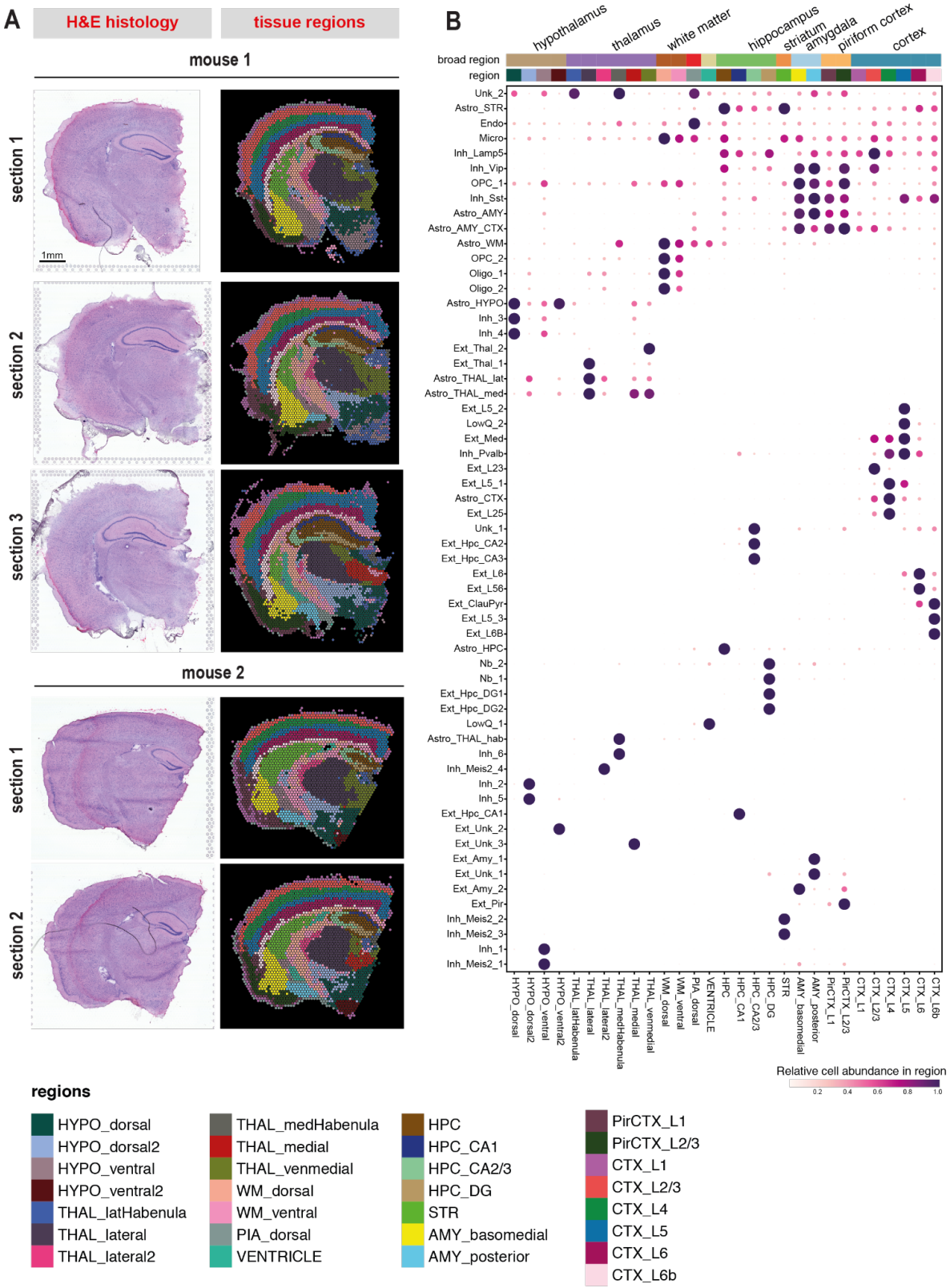


Figure S10. Distribution of mouse brain cell types across tissue regions identified by clustering of cell type abundance.

A. The distribution of tissue regions identified by joint clustering of locations across adjacent tissue sections and replicate mice (n=3 and n=2). Sections (rows) are grouped by mouse and adjacent sections are approximately 250 microns apart. Shown

are histology images (left) and Visium locations and regions identified from clustering (right). The shared legend shows tissue regions identified by clustering (colour). Clustering of locations is based on the cell2location cell type abundance estimates across all slides, followed by anatomical annotation (Methods). Scale bar: 1 mm.

B. Estimated cell abundances of 59 cell subtypes across 29 brain regions. Shown is a dot plot, with dot size and colour corresponding to the relative cell abundance of cell types (rows) across regions (columns), scaled between 0 and 1 by normalising by the maximum values for each cell type. Colour bars (top rows) denote broad brain areas (first row) and subregion (second row, colour code matching panel A).

Figure S11

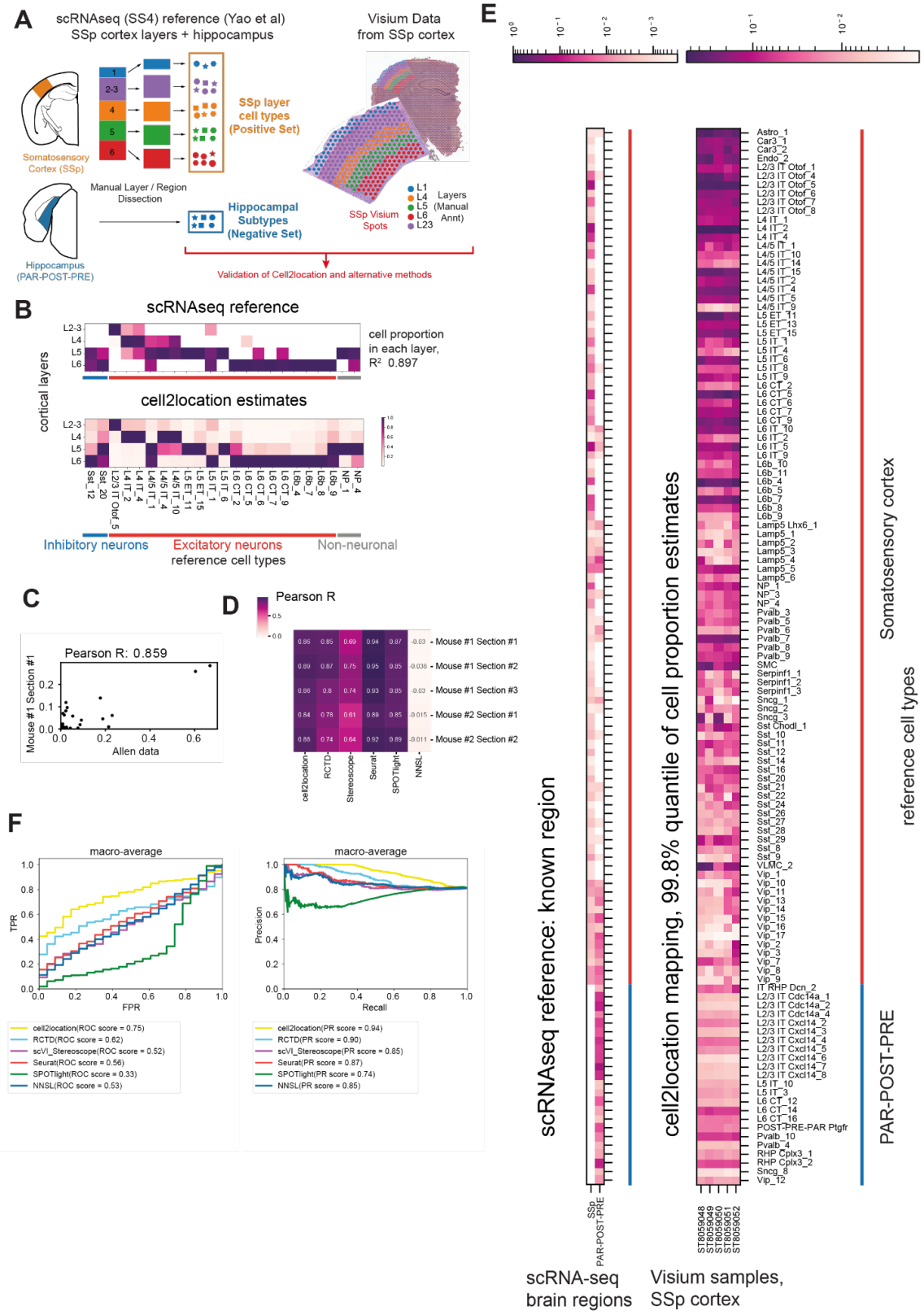


Figure S11. Additional results, assessing quantitative estimates of cell2location and alternative methods for quantitative benchmarks on the mouse brain.

A. Benchmarking workflow for the quantitative assessment of relative cell abundance estimates for cell2location and alternative spatial mapping approaches based on

scRNA-seq from manually dissected cortical layers (SSp) and hippocampus (PAR-POST-PRE). For each subtype in each layer, compared were cell-type proportion estimates from cell2location and alternative methods with estimates based on the scRNA-seq reference of neurons from manually dissected layers (Yao et al, 2020).

B. Heatmap of (relative) proportions of cell types (columns; within each layer; colour) in each cortical layer (rows). Top: estimates derived from scRNA-seq from manually dissected layers; bottom: estimates based on cell2location output.
C. Scatter plot, displaying the concordance between relative cell subtype proportions in each cortical layer in Allen reference (X-axis) versus estimates derived using cell2location based on Visium slides (Y-axis). R denotes Pearson correlation.
D. Assessment of the consistency of cell2location and alternative methods for estimated cell type proportions in each cortical layer in Allen reference. Shown is a heatmap of Pearson R correlation coefficients as in C for each for individual 10X Visium sections (rows) and alternative methods (columns).
E. Comparison of cell proportion in scRNA-seq reference (left) and the cell2location relative cell abundance estimates (right) as in Fig 2I-J, considering all 121 reference cell types. Left: Heatmap of relative proportion (colour, log10 scale) of each cell subtype (rows) within the somatosensory cortex (SSp) and hippocampus (PAR-POST-PRE)(columns) in scRNA-seq data from Yao et al. Right: Heatmap of the 99.8% quantile of relative cell abundance (colour, log10 scale) of each cell type (rows) in each Visium sample from the primary somatosensory cortex region (columns).
F. Assessment of the specificity of cell2location and alternative methods for mapping cell types present (positive set, n=98) and excluding cell types absent (negative set, n=23) in the somatosensory cortex region as in Fig 2I-J. Shown are results based on the 99.8% quantile of the estimated cell type proportions for each cell type across locations within the SSp cortex(panel E right), which represents the presence of a given cell type at least at some locations within the target region. The cell count >= 3 in the SSp region was used as a gold standard label (panel E left). Left: Receiver Operating Characteristic (ROC) curves, displaying false positive rate (FPR, X-axis) versus true positive rate (TPR, Y-axis) for 6 alternative methods (colour). Right: Precision-recall curves, displaying recall (true positive rate, X-axis) versus precision (Y-axis) for 6 alternative methods (colour). The legend includes the macro-average score.

Figure S12

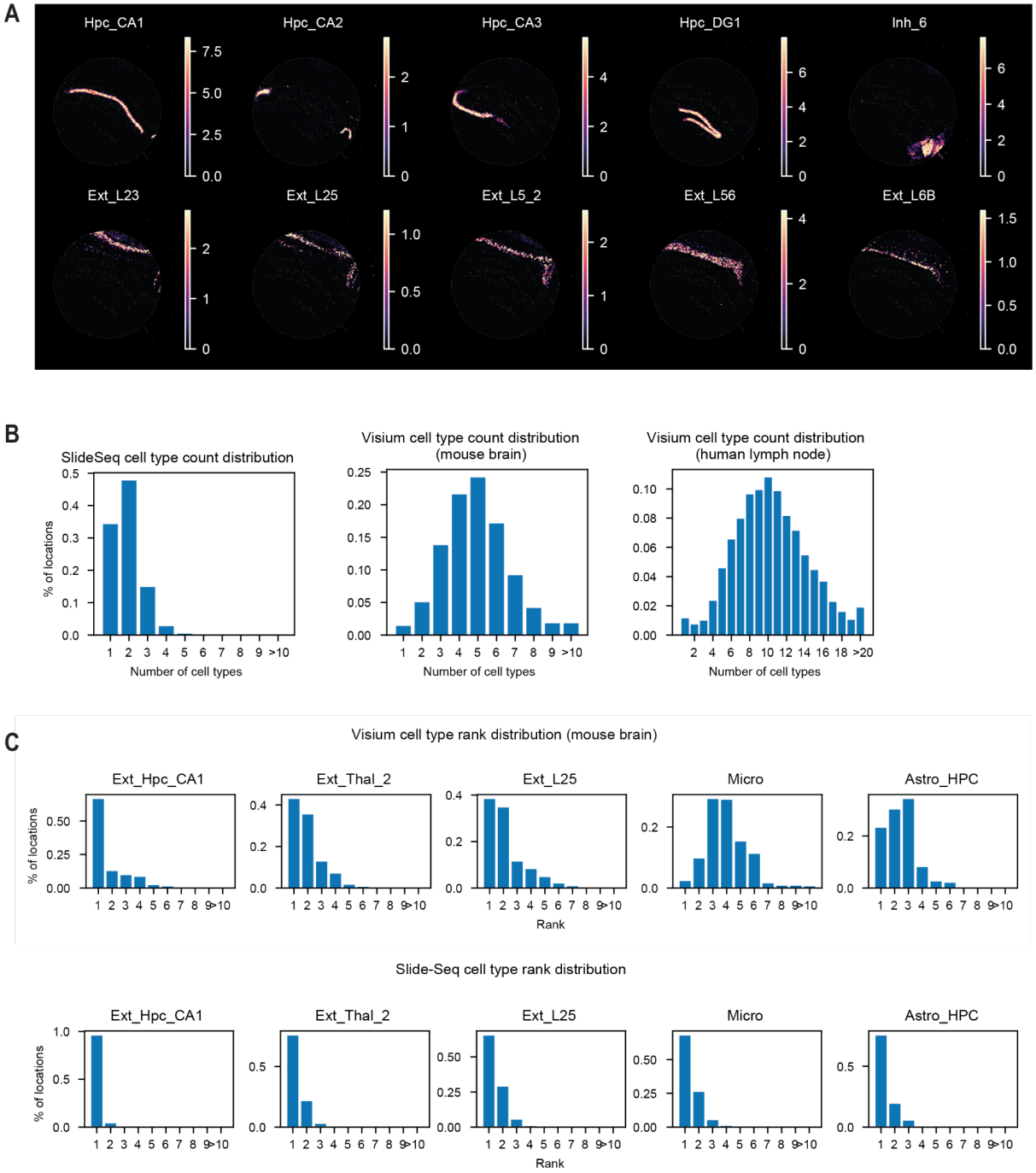


Figure S12. Comparison of the number of cell types per location in Slide-Seq V2 and 10X Visium data.

A. Cell2location estimates of cell abundance for 10 cell types (colour) in the hippocampus region (top row) and across cortical layers (bottom row) in the Slide-Seq V2 data (extending Fig2H, Methods, Analysis of Slide-Seq V2 data). Each point represents a Slide-Seq V2 “bead”. Data was clipped to the 99.5% percentile.

B. The histogram (Y-axis) represents the distribution of cell type count (X-axis) across locations (10X Visium “spots” and Slide-Seq V2 “beads”). Cell type is considered to be present in a location if the estimated value of its mRNA count or cell abundance

exceeds a certain threshold which was selected by examining cell type location maps following filtering by this threshold (See below).

Left: 59 cell subtypes located in Slide-Seq V2 data of the hippocampus region (Stickels et al), partially covering the cortex and the thalamus, shown in panel A. The threshold is mRNA count > 40. Beads with total mRNA count outside 99.5% quantile were excluded from the analysis as outliers.

Middle: 59 cell subtypes located in 10X Visium data of the mouse brain (shown in Fig 2D-F), representing the tissue with intermediate spatial interlacing and 4-5 cell types observed in each location The threshold is cell abundance > 0.4.

Right: 34 immune cell populations located in the 10X Visium data of human lymph node region (shown in Fig 4), representing the tissue with high spatial interlacing and many cell types observed in each location (>10). The threshold is cell abundance > 0.8.

C. The distribution of cell type rank in terms of mRNA count across all locations, where the given cell type is present. Cell type is considered to be present if the estimated value of its mRNA count or cell abundance exceeds a certain threshold.

Top row: 10X Visium data of the mouse brain, including the hippocampus region, the threshold is cell abundance 0.5.

Bottom row: Slide-Seq V2 data of the hippocampus region, the threshold is 60 mRNA. Beads with total mRNA count outside 99.7% quantile were excluded from the analysis as outliers.

Figure S13

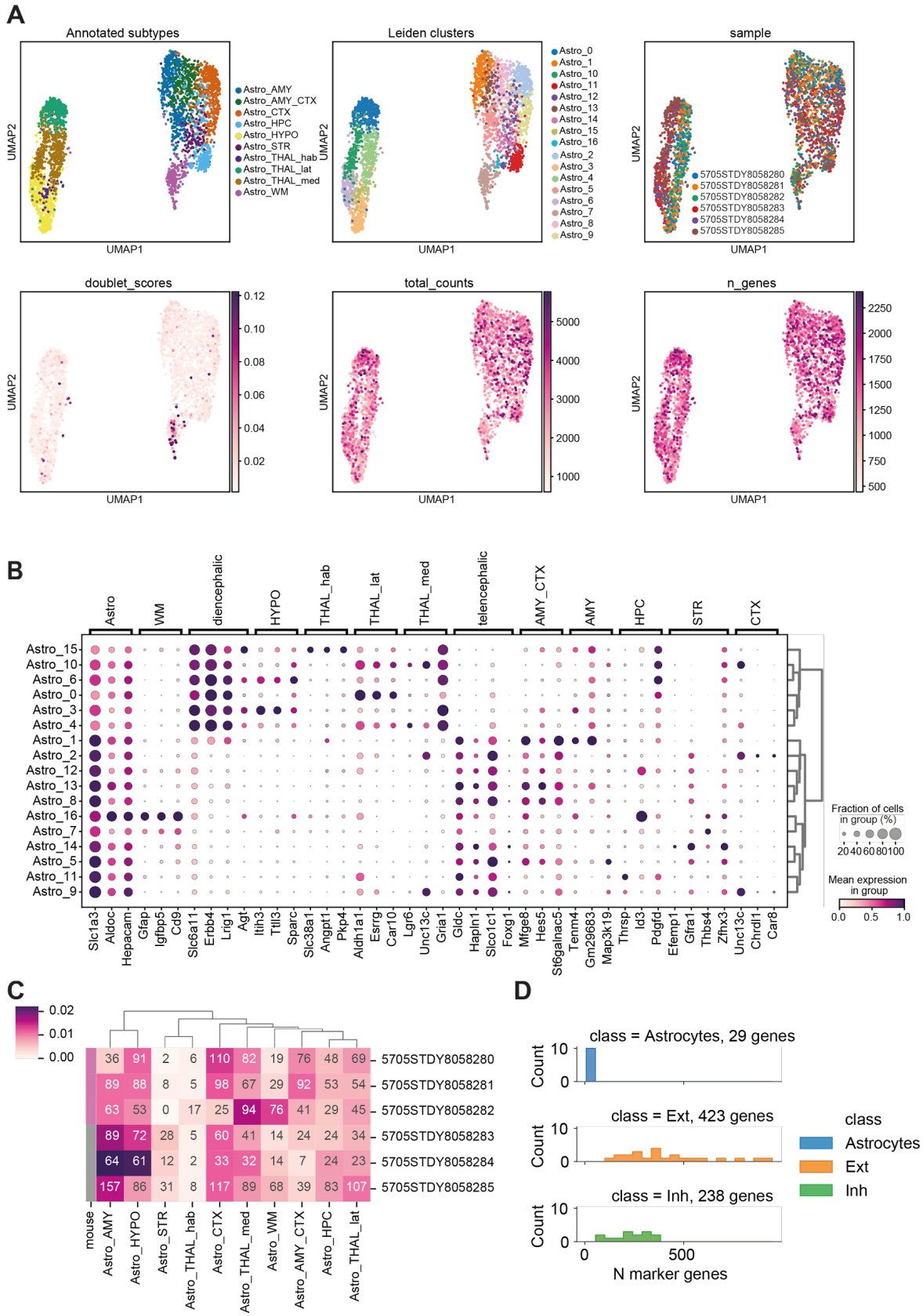


Figure S13. Definition of mouse astrocyte subtypes and quality control.

A. UMAP representation (X- and Y-axis) of 10 astrocyte subtypes, following consolidation using molecular and spatial properties (1st panel) versus the 17 candidate cell types identified based on Leiden clustering (2nd panel). Remaining panels show quality control metrics: doublet cell scores as identified by scrublet tool, total mRNA counts per cell, the total number of genes with mRNA counts > 0.
B. Expression levels of marker genes for the initial 17 candidate astrocyte Leiden subclusters (c.f. Fig S14 for their spatial locations). Dot size corresponds to the fraction of cells in each astrocyte subtype cluster (rows) that express a given marker gene (count > 0; columns). Dot colour denotes the relative expression of each gene in each cluster, normalised by the maximum values for each gene across 17 clusters (scaled between 0 and 1). Marker genes are grouped by subtypes.
C. Heatmap displaying the proportion of cells (colour, log scale, normalised per tissue section) of each of the 10 consolidated astrocyte subtypes (columns) in the scRNA-seq data coming from each dissected tissue section (rows). Colour bar denotes mouse 1 (purple) and mouse 2 (grey). Text denotes the count of cells.
D. Histogram of the number of marker genes per subtype (counts) within each cell class (rows): excitatory (Ext, n=23) and inhibitory (Inh, n=14) neurons and astrocytes (n=10). Average number of marker genes per class is specified in the title. Marker genes were determined by specificity to cell subtypes using normalised expression: reference expression signatures estimated by NB regression and normalised by sum for each gene, $g_{fg} \; / \; \sum_g \quad g_{fg}$. A given gene was assigned as marker to a given subtype when normalised expression > 0.2. To make the distribution of the number of marker genes between classes directly comparable, excitatory and inhibitory neurons were randomly downsampled to 10 subtypes (within each class) and the gene selection was applied. Average number of marker genes per subtype was computed across 1000 random subsamples.

Figure S14

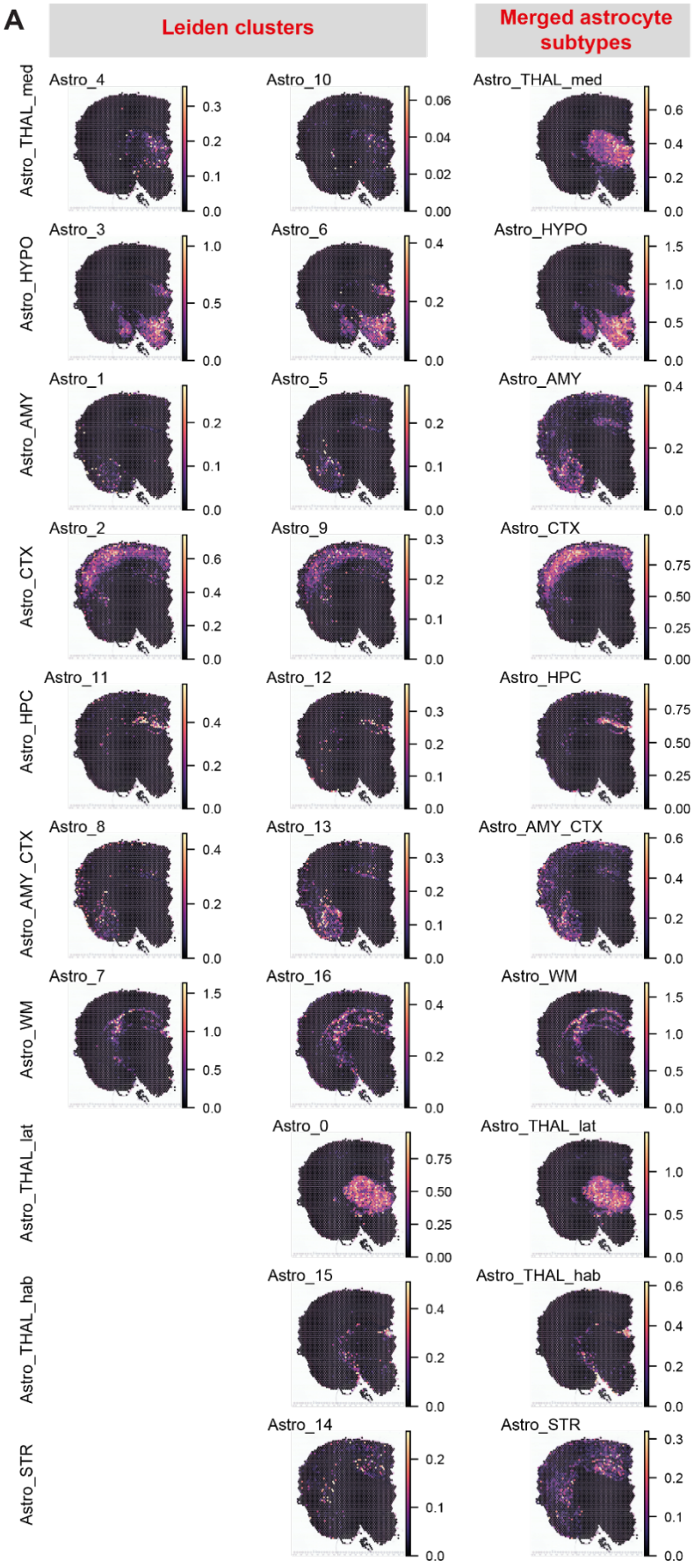

A
Leiden clusters
Merged astrocyte subtypes
Astro_THAL_med
Astro_HYPO
Astro_AMY
Astro_CTX
Astro_HPC
Astro_AMY_CTX
Astro_WM
Astro_THAL_lat
Astro_THAL_hab
Astro_STR
Astro_4
Astro_10
Astro_THAL_med
Astro_3
Astro_6
Astro_HYPO
Astro_1
Astro_5
Astro_AMY
Astro_2
Astro_9
Astro_CTX
Astro_11
Astro_12
Astro_HPC
Astro_8
Astro_13
Astro_AMY_CTX
Astro_7
Astro_16
Astro_WM
Astro_0
Astro_THAL_lat
Astro_15
Astro_THAL_hab
Astro_14
Astro_STR

Figure S14. Refined annotation of astrocyte clusters based on cell2location mapping. Comparison of absolute cell abundance (colour) from each of 10 consolidated astrocyte subtypes (right, one row per subtype, as reported in Fig 3A) to the cell abundance of initial 17 candidate subtype Leiden clusters (left).

Figure S15

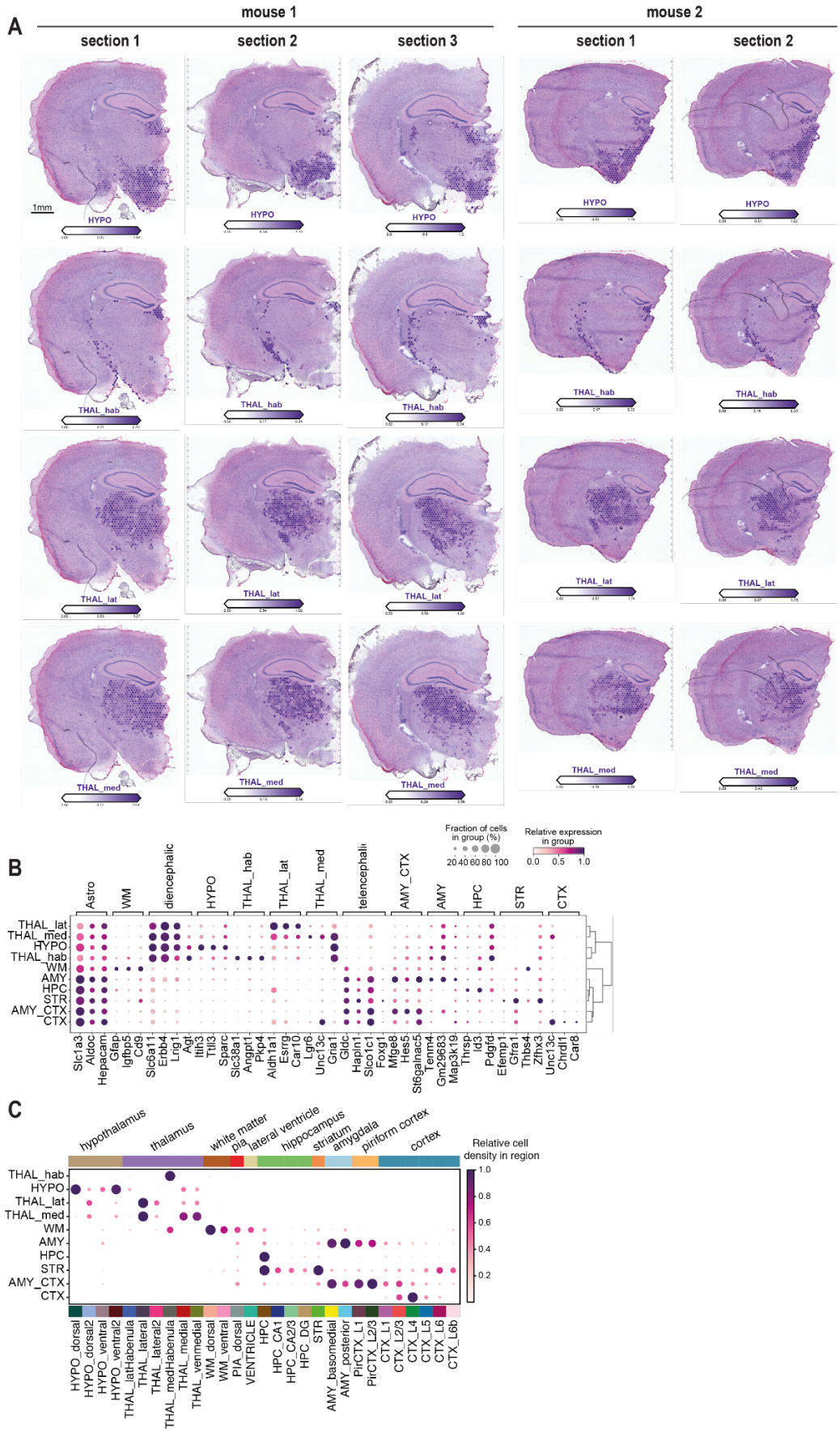


Figure S15. Spatial mapping of diencephalic astrocytes using Visium data.

A. Estimated cell abundances (colour intensity) of 4 diencephalic astrocyte subtypes on tissue sections (approximately 250 microns between consecutive sections) across 2 replicate animals (n=3, n=2). Sections (rows) are grouped by mouse and histology images are shown in the plot background. Rows show HYPO, THAL_hab, THAL_lat and THAL_med astrocyte subtypes, respectively. Scale bar: 1 mm.
B. Astrocyte subtype markers. Dot size corresponds to the fraction of cells in each astrocyte subtype cluster (rows) that express a given marker gene (count > 0; columns). Dot colour denotes the relative expression of each gene in each cluster, normalised by the maximum values for each gene (scaled between 0 and 1). Marker genes are grouped by subtypes.
C. Estimated cell densities of astrocyte subtypes across brain regions (as in Fig 3C). Dot plot, with dot size and colour corresponding to the relative cell density for each astrocyte subtype (rows) across regions (columns), scaled between 0 and 1 by normalising by the maximum values for each subtype. Colour bars denote broad brain areas (top) and subregion (bottom, matching Fig 3C).

Figure S16

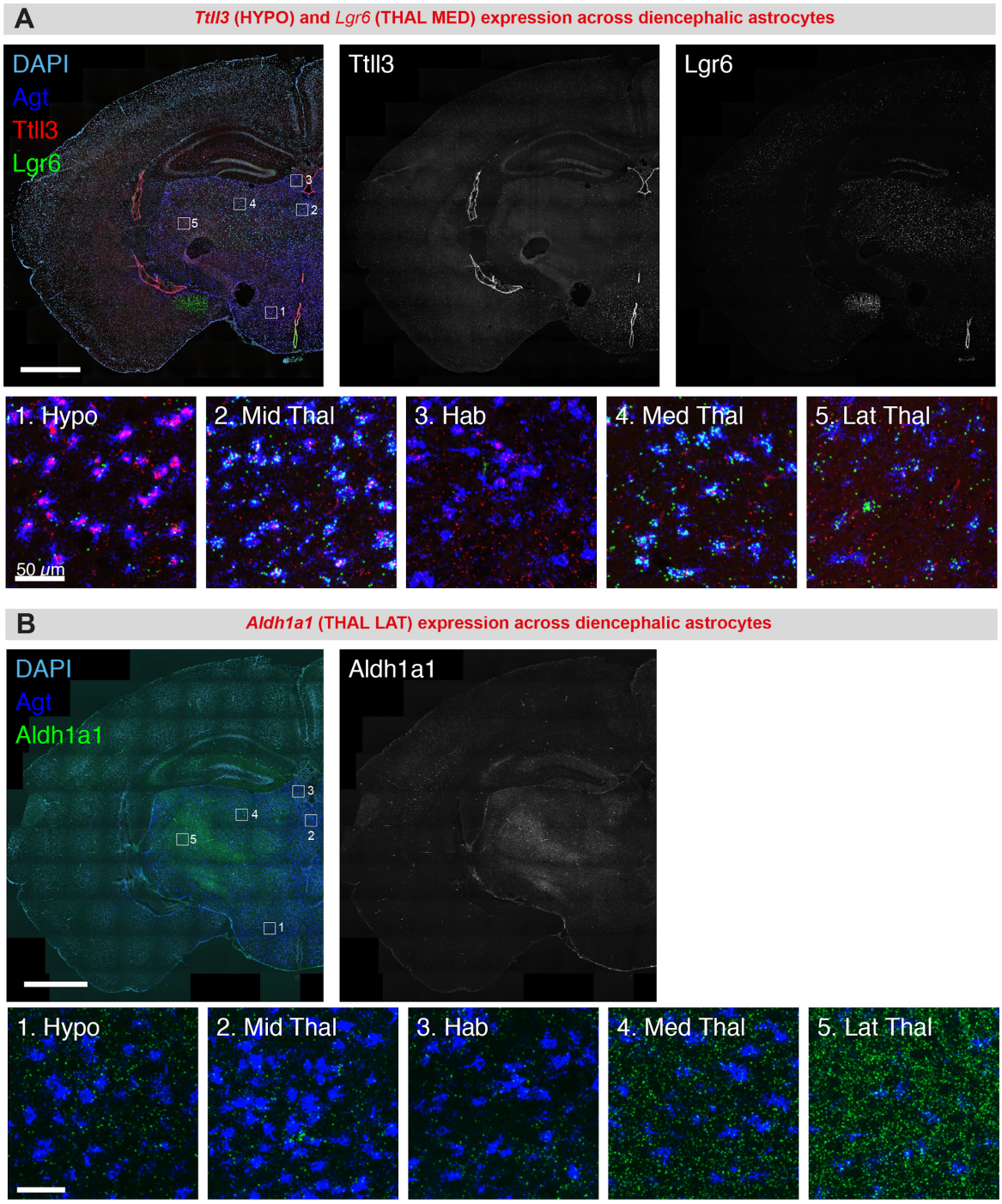


Figure S16. smFISH validation of hypothalamic and thalamic astrocytes. RNAScope smFISH validation of regional astrocyte subtype markers across the thalamus and hypothalamus. Large panels show stitched maximum z-projections of coronal hemi-sections. Small panels show close-up images of boxed areas on the large images in the hypothalamus (Hypo), midline thalamic nuclei (Mid Thal), lateral habenula (Hab), medial thalamic nuclei (Med Thal) and lateral thalamic nuclei (Lat Thal). Agt expression marks the soma of diencephalic astrocytes while DAPI labels all cell nuclei. *Aldh1a1* and *Ttll3* also express in non-astrocyte cell types in the brain. n=1 mouse, 3 technical replicate tissue sections for each marker independently imaged. Scale bars: large panels, 1 mm; small panels, 50 microns.

Figure S17

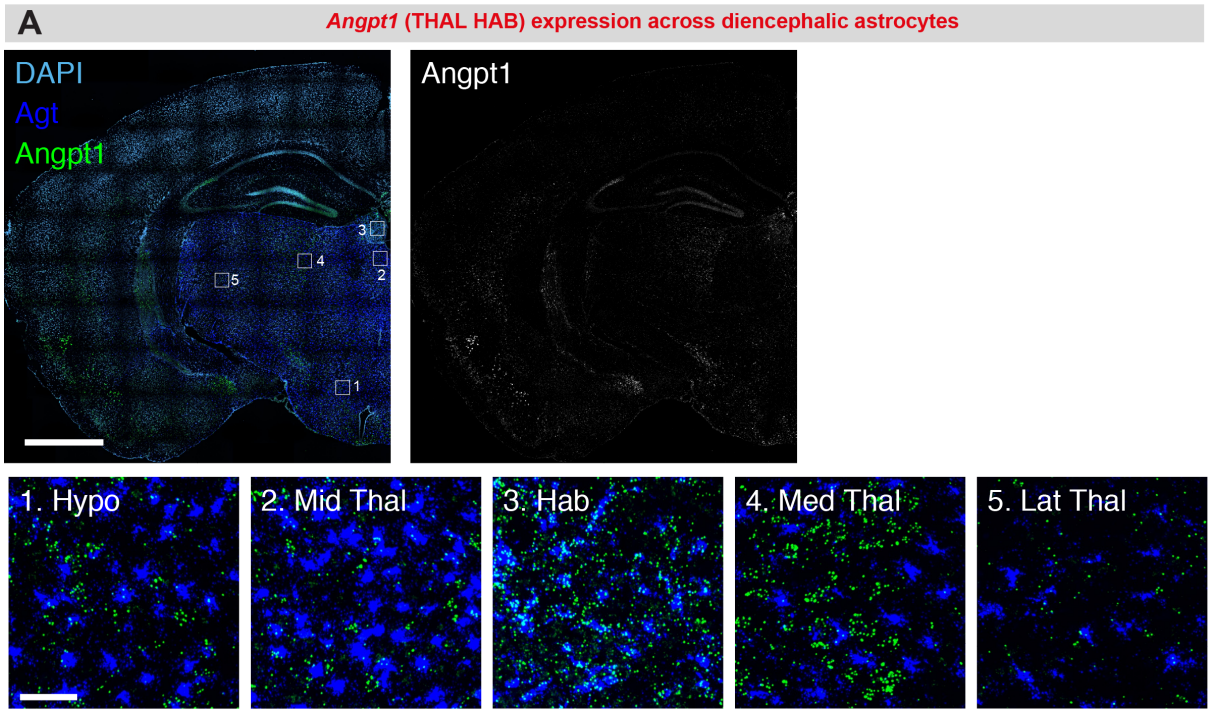


Figure S17. smFISH validation of habenular astrocytes. RNAScope smFISH validation of regional astrocyte subtype markers across the thalamus and hypothalamus. Large panels show stitched maximum z-projections of coronal hemi-sections. Small panels show close-up images of boxed areas on the large images in the hypothalamus (Hypo), midline thalamic nuclei (Mid Thal), lateral habenula (Hab), medial thalamic nuclei (Med Thal) and lateral thalamic nuclei (Lat Thal). Agt expression marks the soma of diencephalic astrocytes while DAPI labels all cell nuclei. *Angpt1* also expresses in non-astrocyte cell types in the brain. Scale bars: large panels, 1 mm; small panels, 50 microns.

Figure S18

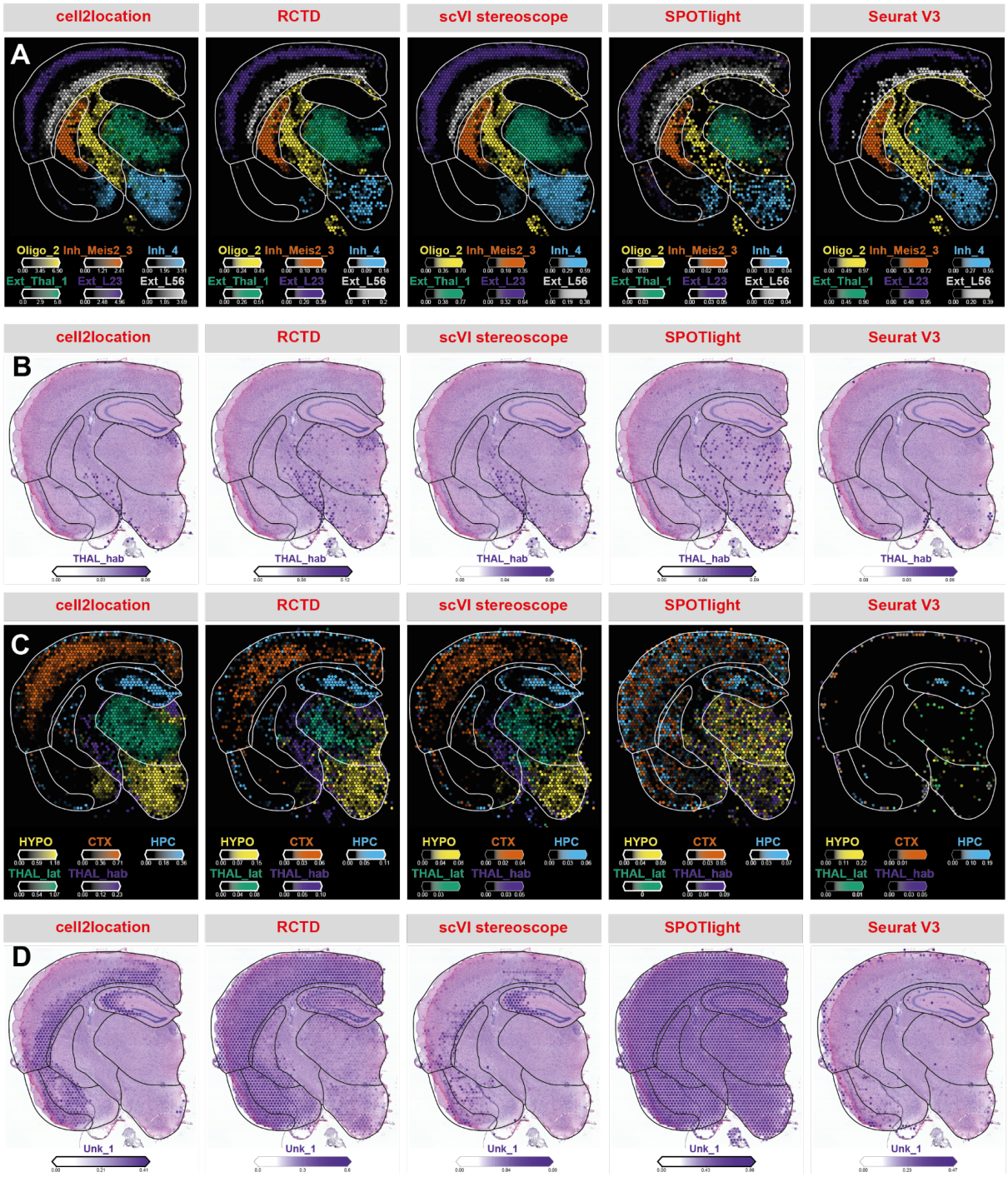


Figure S18. Comparison of spatial mapping methods in the mouse brain.

A. Estimated cell abundances (colour intensity) of major regional cell types (colour) for 5 considered methods (columns, excluding NNLS to preserve readability of visualisation and because it is a generic method not tailored to expression data).

B. Estimated cell abundances (colour intensity) of THAL_hab astrocyte subtype for 5 considered methods (columns).

C. Estimated cell abundances (colour intensity) of 5 astrocyte subtypes show in Fig 3C for 5 considered methods (columns).

D. Estimated cell abundances (colour intensity) of Unk_1 population (that we were not able to annotate) for 5 considered methods (columns). SPOTlight and RCTD assigned

very large relative abundance (> 0.5) to large areas on the tissue, likely indicating artifactual mapping.

Figure S19

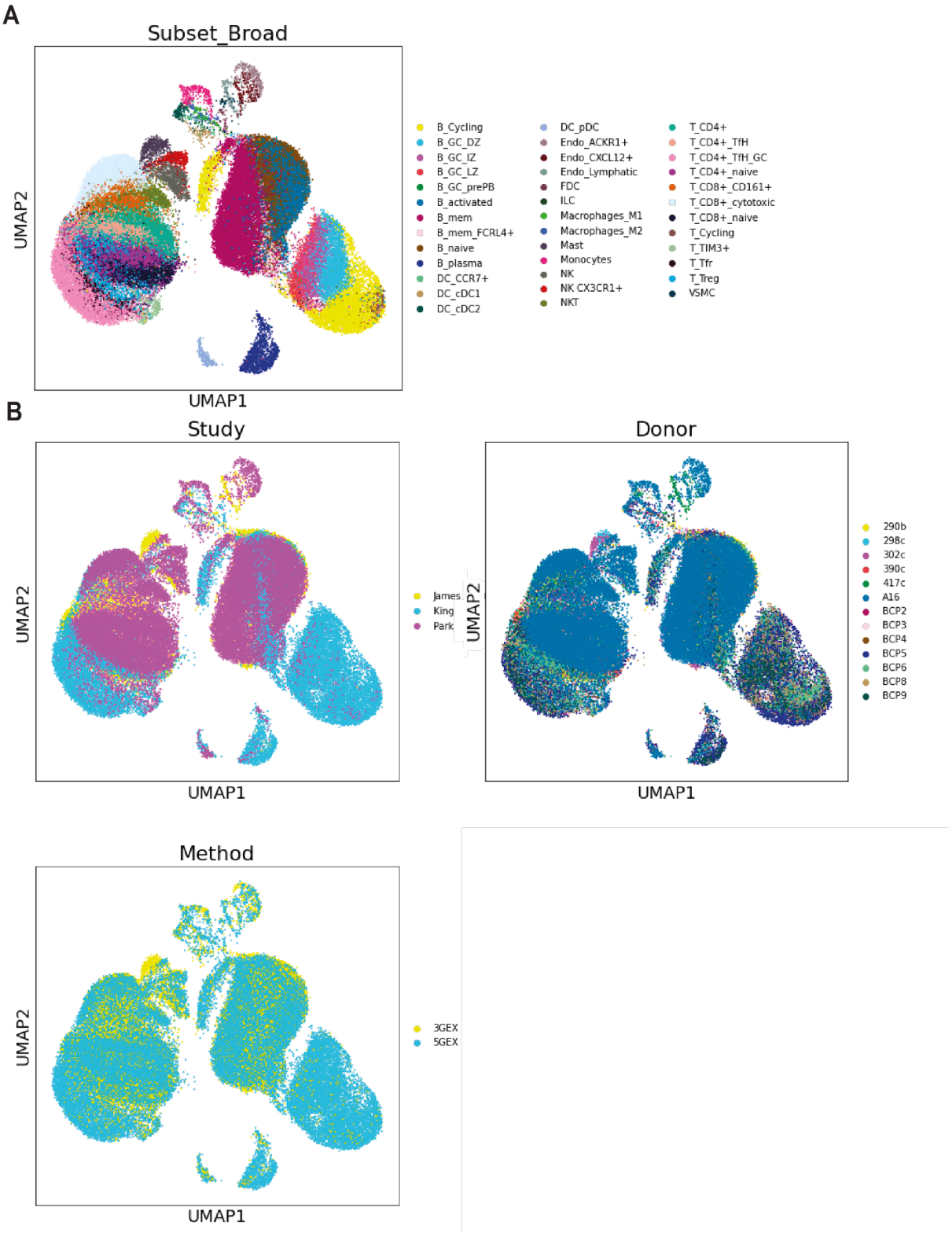


Figure S19. Construction of reference cell type signatures for the lymph node.

A. UMAP representation (X- and Y-axis) of 34 immune and non-immune cell subtypes (colour), combining scRNA-seq data from 3 studies (James et al 2020, Park et al 2020, King et al 2020) used to define reference gene expression signatures of cell types. UMAP was constructed by applying standard Scanpy workflow to scRNA-seq data

corrected to remove technology and experiment effect estimated by the regression model (Suppl Methods, section 2).

B. UMAP representation (X- and Y-axis) of individual cells as in **panel A**, with cells coloured by study (paper reporting the data), donor or sequencing technology (method).

Figure S20

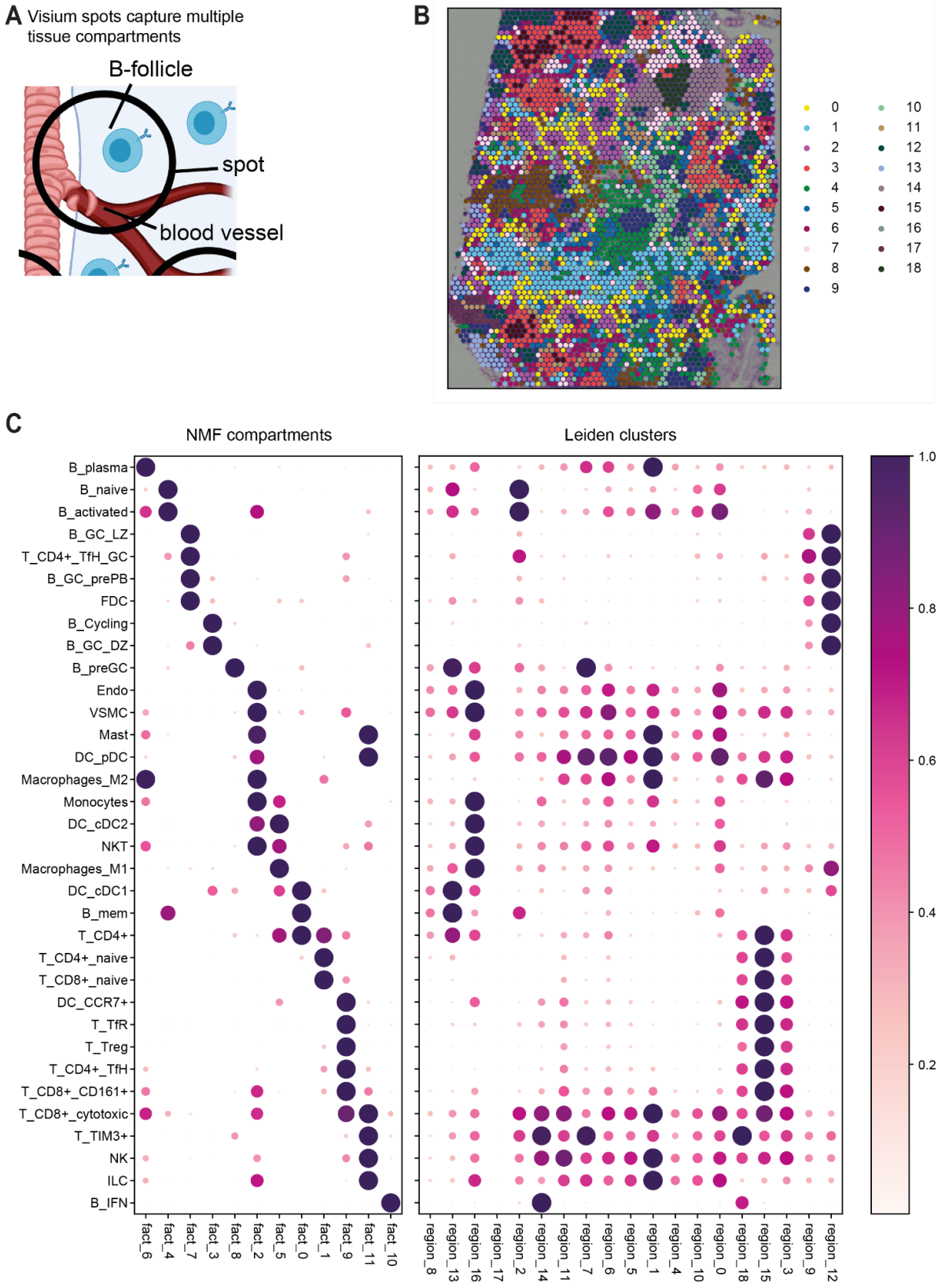


Figure S20. Comparison of NMF and location clustering for identifying spatially interlaced tissues zones in the lymph nodes.

A. Diagram illustrating that 10X Visium capture areas are not aligned with tissue compartments and hence tend to capture multiple tissue compartments, such as B-follicle zone and blood vessel zone. This motivates using additive decomposition (NMF), which naturally accounts for mixed class assignment compared to discrete clustering for identifying such spatially interlaced zones.
B. Human lymph node regions (colour) as identified using clustering of locations based on cell2location cell type abundance estimates (using Leiden clustering with the resolution parameter selected to yield 19 clusters as in panel C). Corresponding histology image is shown in Fig 4C.
C. Comparison of cell abundances (rows) across NMF zones (cf. Fig. 4D) and regions identified by Leiden clustering (matching panel B). Shown is a dot plot of the NMF weights (left) for each cell type (rows) in each factor (columns), and average abundance for each cell type in each region(right, columns). Dot size and colour correspond to NMF weights and average abundance in each cluster (normalized across components per cell type and across clusters per cell type).

Figure S21

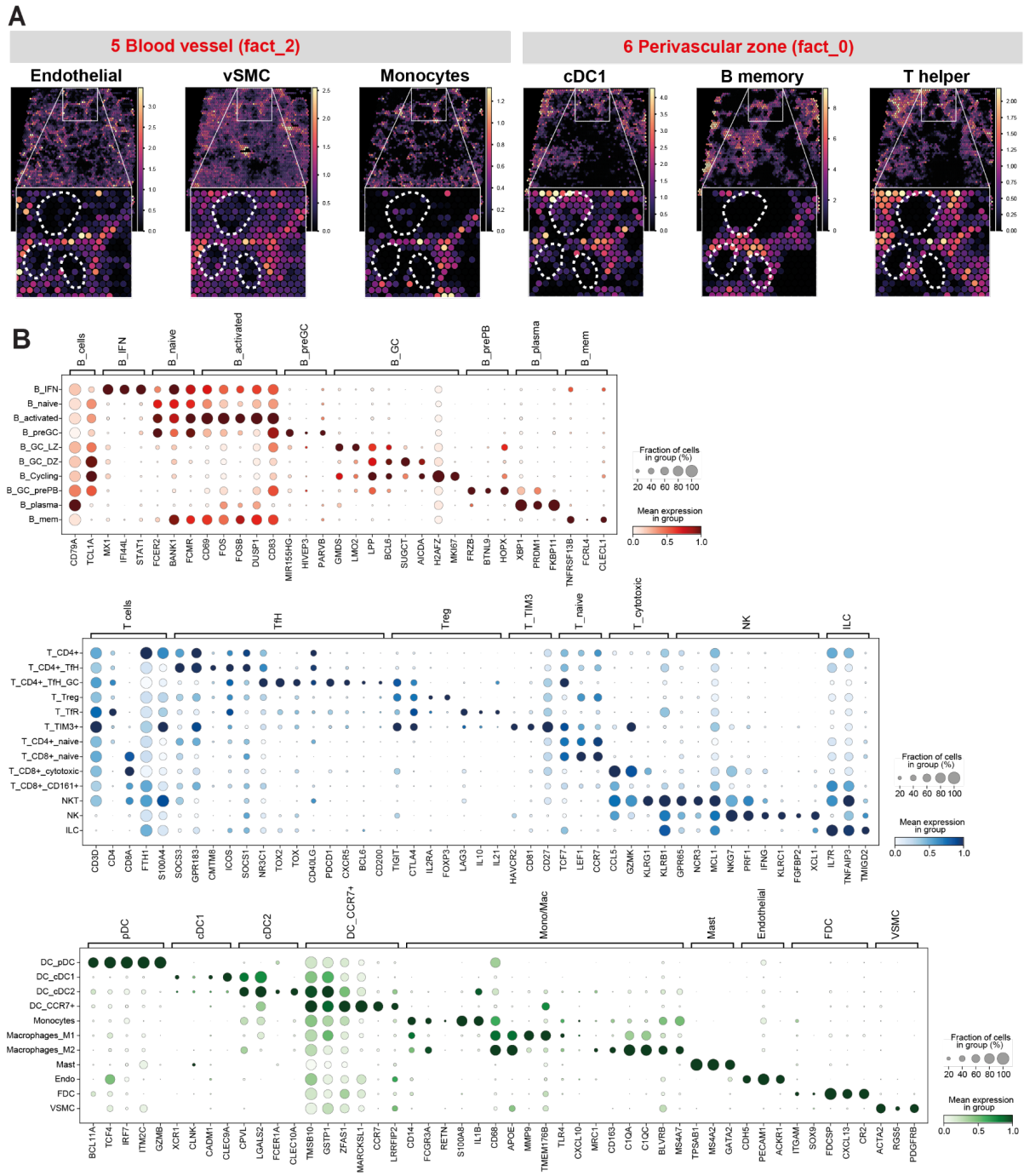


Figure S21. Identification of tissue zones in the lymph node.

A. Location patterns of the blood vessel and perivascular cell types that are associated with highlighted cell compartments (panels 5-6). Spatial plots (X, Y-axis) show cell abundance (colour) for each cell type (subpanels). Dashed boxes highlight the tissue regions shown in inset panels. Dashed circles indicate GCs.

B. Expression level of marker genes of 34 immune cell types, grouped into B-cells (top), T-cells (middle) and other populations (bottom). Dot size denotes the fraction of cells per cluster (rows) that express a given marker gene (count > 0; columns). Dot colour

denotes the relative expression of each gene in each cluster, normalised by the maximum values for each gene, subtracting minimum (scaled between 0 and 1).

Figure S22

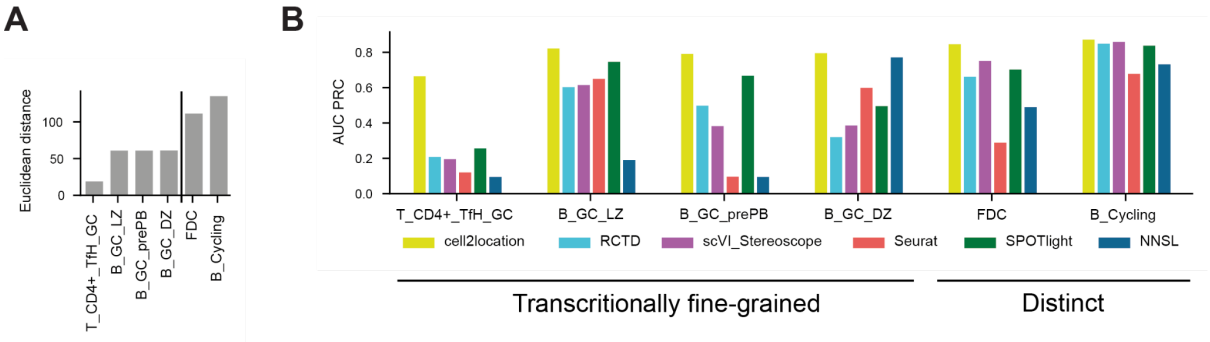


Figure S22. Definition of transcriptionally fine cell types in lymph nodes.

A. Shown is Euclidean distance (bar height) between expression signature of individual cell types (X-axis, 6 cell types expected in the germinal center) and the closest other cell type (considering all 34 cell types in the reference). Cell types are ordered by Euclidean distance from left to right. Vertical line separates transcriptionally fine from distinct cell types.

B. Shown is the precision-recall score (bar height) for each cell type (panels, ordered according to Euclidean distance as in panel A) for alternative methods (colour). Horizontal lines highlight the cell types considered transcriptionally fine and distinct.

Figure S23

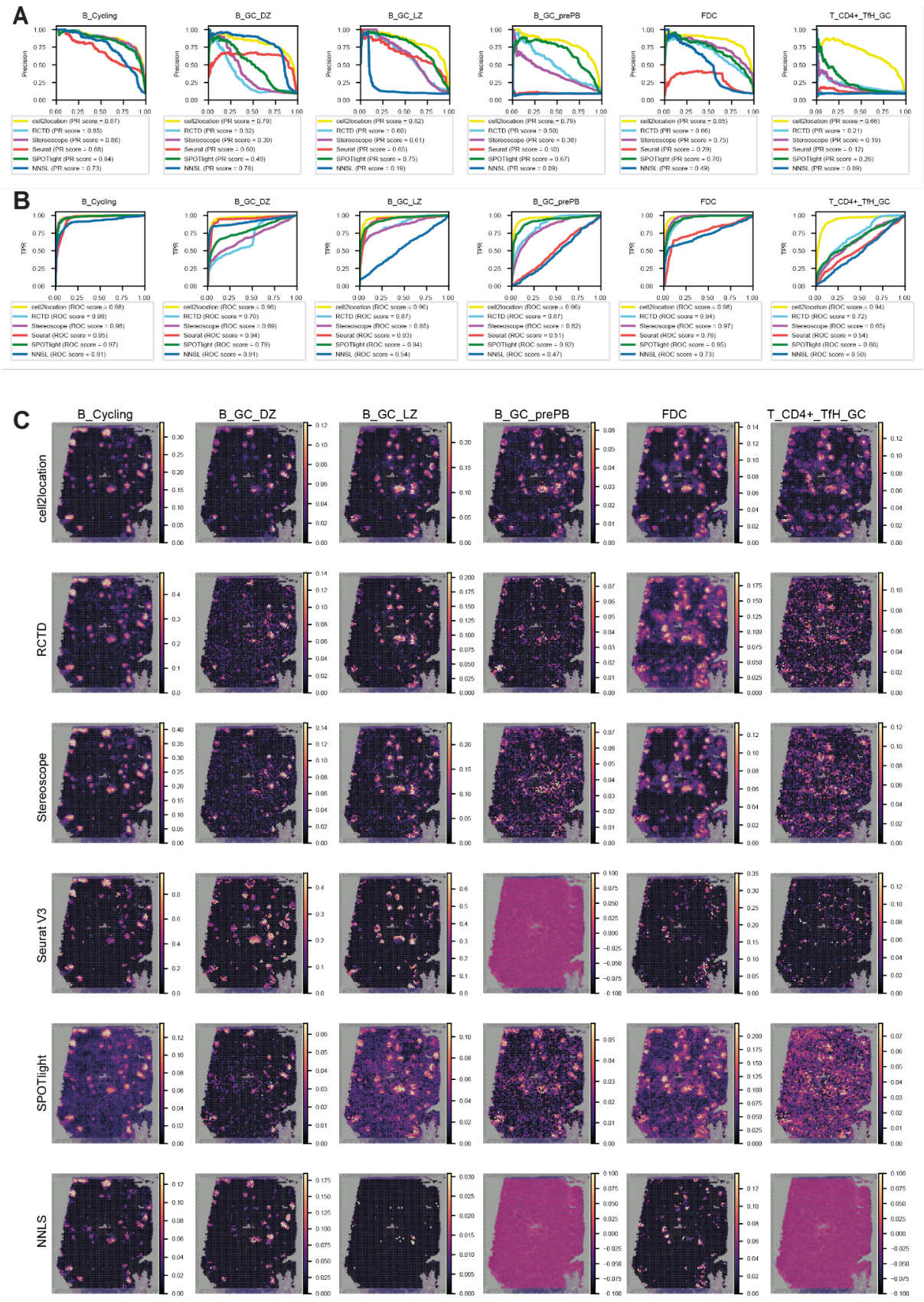


Figure S23. Accuracy of spatial mapping methods at mapping Germinal Center cell types to histology-annotated GC zones in lymph node Visium data.

C. Assessment of spatial mapping of 6 germinal center cell populations in the human lymph nodes using histology-based annotations of GC zones. Shown are the precision-recall curves for 6 cell populations (columns, B_Cycling, B_GC_DZ, B_GC_LZ, B_GC_prePB, FDC, T_CD4+_TfH_GC) that are expected to be present in annotated GC (positive set of locations) and absent from other locations (negative set) for alternative methods (colour). PR score is shown in the legend for the considered methods.
D. Similar to A, shown are the ROC (Receiver operating characteristic) curves with false positive rate (FPR, X-axis) and true positive rate (TPR, Y-axis) for 6 considered methods (colour) for 6 cell populations (columns) that are expected to be present in annotated GC and absent from other locations. ROC score is shown in the legend for the considered methods.
E. Spatial plots (X, Y-axis) show relative cell abundance (colour intensity) of 6 cell populations (columns) for each considered method (rows).

Figure S24

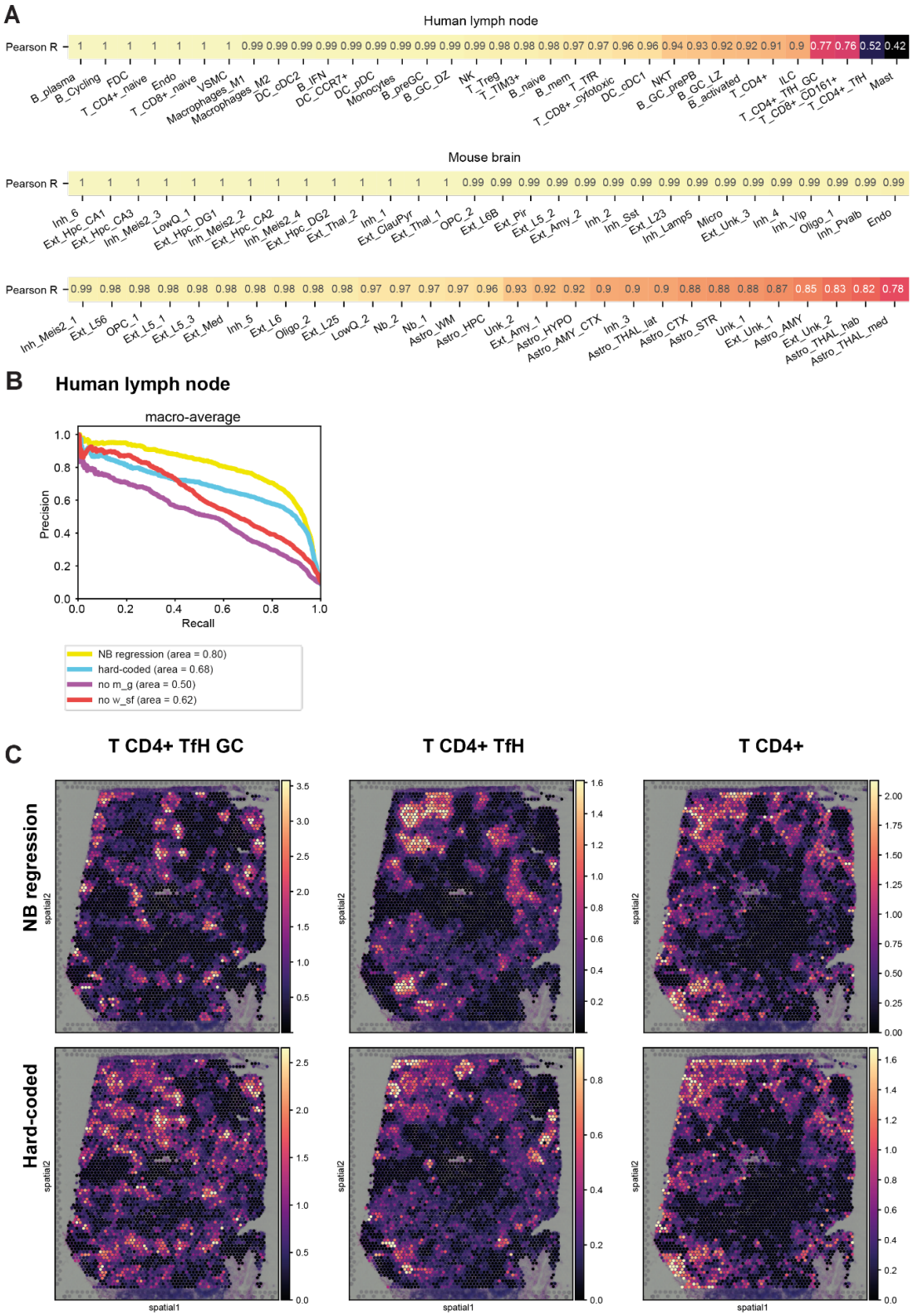


Figure S24. Comparison of alternative methods for estimating cell type reference signatures.

Compared are 1) the cell2location default Negative Binomial (NB) regression model, which accounts for batch and technology effects, and 2) a hard-coded method that computes average count of each gene in each reference cluster (see Suppl. Methods for details).

A. Consistency of cell2locaiton abundance estimates when mapping cell type signatures derived using either of the two considered methods. Shown are Pearson correlation coefficients, pooled across locations (R, colour, value in print), for each cell type (entries, columns). Top: human lymph nodes mapped using single cell RNA-seq reference composed of multiple batches and 34 cell types (3 organs, 2 technologies). Bottom: mouse brain mapped using paired single nucleus RNA-seq reference of 59 cell types. The differences in mapping are smaller for the mouse brain, which reflects that lower extent of batch variation of the corresponding reference snRNA-seq dataset.
B. Assessment of the accuracy of alternative variants of cell2location (Suppl. Methods). Considered is the full cell2location model in conjunction with two alternative approaches to estimate reference signatures (NB regression and hard-coded). Additionally, we considered a reduced variant of cell2location without specific features: prior factorisation of cell abundance w_sf and gene-specific technology scaling m_g (used in conjunction with the NB regression model, Suppl. Methods). The assessment was performed using the same procedure as shown in Fig 4F-I and Fig S21.
C. Spatial plots (X, Y-axis) display estimated relative cell abundance (colour intensity) of 3 cell populations (columns) for 2 considered reference expression signature estimation methods (rows). T_CD4+_TfH_GC is expected to be present in the GC zone whereas T_CD4+_TfH and T_CD4+ are expected to be excluded from GC, especially dark zone GC (see Fig 4E).

Figure S25

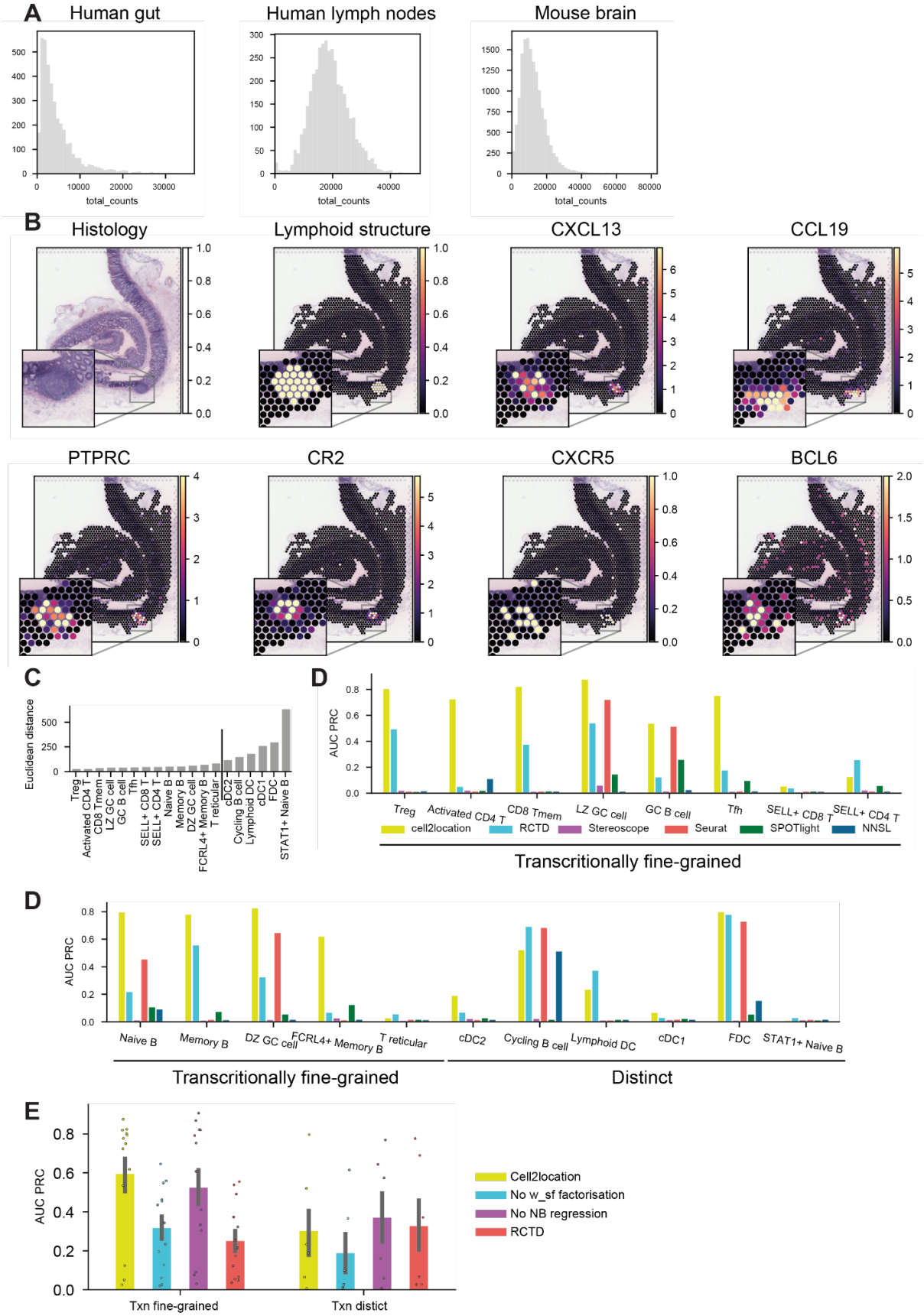


Figure S25. Transcriptionally fine grained cell types in the gut.

A. Variation of total mRNA abundance across locations for different datasets (from left to right: human gut, human lymph nodes, mouse brain). Histogram counts (Y-axis) of total RNA (UMI) counts per location (X-axis).
B. Annotation of lymphoid follicles by considering both histology and marker gene expression in Visium data (Methods). Panels in spatial coordinates (X- and Y- axis) show (left to right) histology image, annotation of lymphoid follicle (colour), and mRNA count of 6 marker genes (colour). The colour scale is cropped at 99.7% quantile of count distribution across locations. Boxes highlight the tissue regions shown in inset panels.
C. Extent of fine-grained versus distinctness of cell types. Shown is Euclidean distance (bar height) between expression signature of each cell type (X-axis, 19 cell types expected in the lymphoid follicle) and the closest other cell type (selected from all 76 cell types in the reference). Cell types are ordered left to right by Euclidean distance. Vertical line separates transcriptionally fine from distinct cell types.
D. Precision-recall score (bar height) for individual cell types (panels, ordered according to Euclidean distance as in panel C) across alternative methods (colour). Horizontal lines highlight the cell types considered transcriptionally fine and distinct.
E. Assessment of impact of cell2location model features on its performance. Shown are average AUC scores across cell types (bar height) for alternative methods: 1) cell2location model with reference signatures estimated by NB regression; 2) a reduced variant of cell2location, without prior factorisation of cell abundance w_sf (Suppl. Methods); 3) cell2location model with reference signatures estimated using hard-coded computation per cluster average expression (Suppl. Methods); 4) RCTD for reference of next best method. Error bars corresponding to standard errors. The assessment was performed using the same procedure as shown in Fig 5F (n=13, transcriptionally fine subtypes, n=6 transcriptionally distinct subtypes).

Figure S26

Figure S26. Comparison of spatial maps produced by alternative methods
Shown are cell type proportion estimates for individual cell types (rows) across 6 alternative methods (columns) in spatial coordinates. Colour scale is cropped at 99.7% quantile of count distribution across locations. Boxes highlight the tissue regions shown in inset panels.

Figure S27

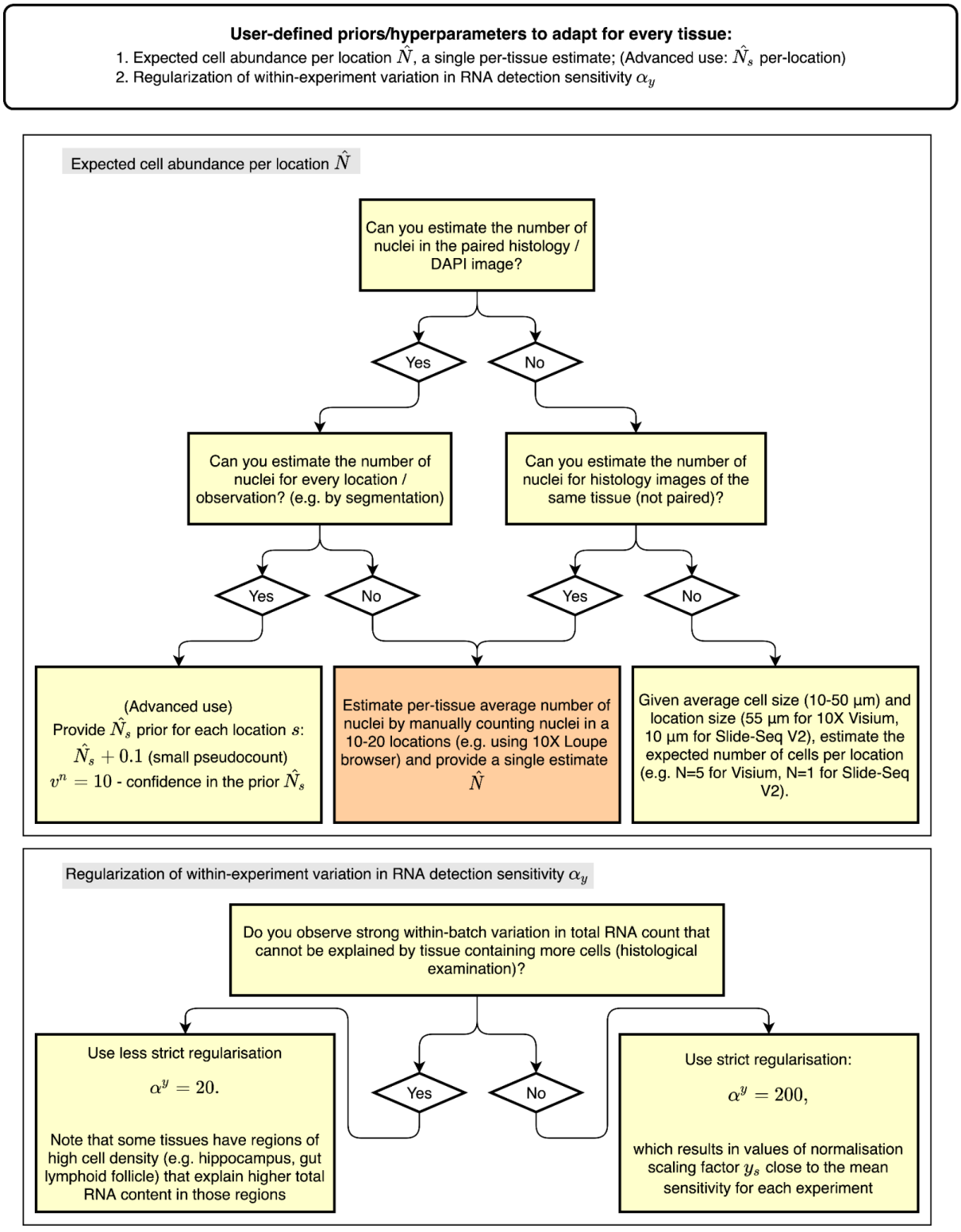


Figure S27. Flow diagram to choose cell2location hyperparameters. Cell2location has 2 hyperparameters that can be set by the users (see a separate box for each hyperparameter). The following flowchart can be used to determine these hyperparameters in the light of experimental or biological prior information. An appropriate setting of the expected cell abundance $\widehat{N}$ is essential to inform the estimation of absolute cell type abundance values. Term 'location' refers to 10X Visium spots, Slide-Seq V2 beads, WTA Nanostring regions-of-

interest (ROI) and other spatially defined voxels where transcriptomic measurements are taken. The choice of expected cell abundance $\widehat{N}$ highlighted in orange is applicable to most 10X Visium users.

Comprehensive mapping of tissue cell architecture via integrated single cell and spatial transcriptomics: cell2location model Supplementary Methods

Vitalii Kleshchevnikov, Artem Shmatko, Emma Dann, Alexander Aivazidis, Hamish W King, Tong Li, Rasa Elmentaite, Artem Lomakin, Veronika Kedlian, Adam Gayoso, Mika Sarkin Jain, Jun Sung Park, Lauma Ramona, Elizabeth Tuck, Anna Arutyunyan, Roser Vento-Tormo, Moritz Gerstung, Louisa James, Oliver Stegle, Omer Ali Bayraktar

October 27, 2021

Contents

1 Cell2location model **1**
1.1 Model derivation 1
1.2 Determining hyper parameter values 7
1.3 Inference 8
1.4 Nuclei Segmentation (optional) 9

2 Estimation of reference expression signatures of cell types **9**

3 Comparison of cell2location with alternative methods **11**

4 Downstream analyses **12**
4.1 Identification of tissue regions by clustering 13
4.2 Groups of co-located cell types 13

5 Constructing a synthetic spatial transcriptomics data set **14**

6 Appendix **17**
6.1 Complete graphical model diagram 17
6.2 Notation 17

1 Cell2location model

1.1 Model derivation

Cell2location is a Bayesian model that decomposes gene expression counts measured across spatial locations into a set of pre-defined reference cell types, thereby estimating both relative and absolute cell abundances. A representation of cell2location as graphical model is shown in Fig 1.

The input data of the method are as follows:

Let $D = \{d_{s,g}\}$ be a $S \times G$ count matrix with its entries corresponding to mRNA counts at locations $s = \{1, .., S\}$ from one or multiple batches, for genes $g = \{1, .., G\}$. The location index s iterates across unique locations across all batches $e = \{1, .., E\}$ (one location s belongs exclusively to one batch), where e denote individual experiments, such as 10X Visium chips (i.e. square capture areas) or Slide-Seq V2 pucks. Cell2location then jointly models the data across these spatial data sets $e = \{1, .., E\}$, aka batch integration, by accounting for technical effects aligned with batch (see below). When using cell2location combination with 10X Visium data, this matrix can be directly obtained from the 10X SpaceRanger software; see main text Methods.

Further, let $G = \{g_{f,g}\}$, denote an $F \times G$ matrix of reference cell type signatures, which consist of $f = \{1, .., F\}$ gene expression profiles $G_{f,:}$ for $g = \{1, .., G\}$ genes, which correspond to average expression levels of each gene in each cell type in a linear mRNA count scale. This matrix needs to be provided to cell2location and can be estimated from scRNA-seq profiles; see Section 2 for details.

We note that the matrices D and G need to be aligned such that they cover the same set of genes. The software implementation of cell2location builds on the Scanpy file formats, which provide native access to outputs of the 10X visium platform. For other technologies, such as Slide-Seq V2, files with count matrices of the appropriate formats need to be generated and imported into Scanpy data format by the user.

Cell2location models the elements of D as Negative Binomial distributed, given an unobserved expression level (rate) $\mu_{s,g}$ and a gene- and experiment- specific over-dispersion parameter $\alpha_{e,g}$, which accounts for residual variance in expression of individual genes that cannot be attributed to Poisson sampling noise:

$$d_{s,g} \sim \texttt{NB}(\mu_{s,g}, \alpha_{e,g}). \tag{1}$$

Equivalently, this negative binomial likelihood can be expressed as a Gamma-Poisson mixture with a Poisson likelihood (count measurement model) and a Gamma-distributed mean (expression dispersion model)[1]. Parameterising the NB distribution by a Poisson rate parameter μ, and over-dispersion α, which corresponds to shape parameter of Gamma distribution, the equivalent Gamma-Poisson mixture follows as:

$$d_{s,g} \sim \texttt{Poisson}(\texttt{Gamma}(\alpha_{e,g}, \alpha_{e,g}/\mu_{s,g})). \tag{2}$$

The expression of level of genes $\mu_{s,g}$ in the mRNA count space is modelled as a linear function of expression signatures of reference cell types:

$$\mu_{s,g} = \left(\underbrace{m_g}_{\text{technology sensitivity}} \cdot \underbrace{\sum_f w_{s,f}\, g_{f,g}}_{\text{cell type contributions}} + \underbrace{s_{e,g}}_{\text{additive shift}} \right) \cdot \underbrace{y_s}_{\text{per-location sensitivity}} . \tag{3}$$

Here, $w_{s,f}$ denotes regression weight of each reference signature f at location s, which can be regarded as the abundance of cells expressing reference cell-type signature f at location s (denoted absolute cell abundance in the paper); m_g denotes a gene-specific scaling parameter, which adjusts for global differences in expression estimates between technologies (platform effect); $s_{e,g}$ captures gene-specific additive shift, e.g. due to free-floating RNA; and y_s denotes a location-specific scaling parameter, which models variation in RNA detection sensitivity across locations and experiments. The last parameter scales both the RNA contributions of cell types

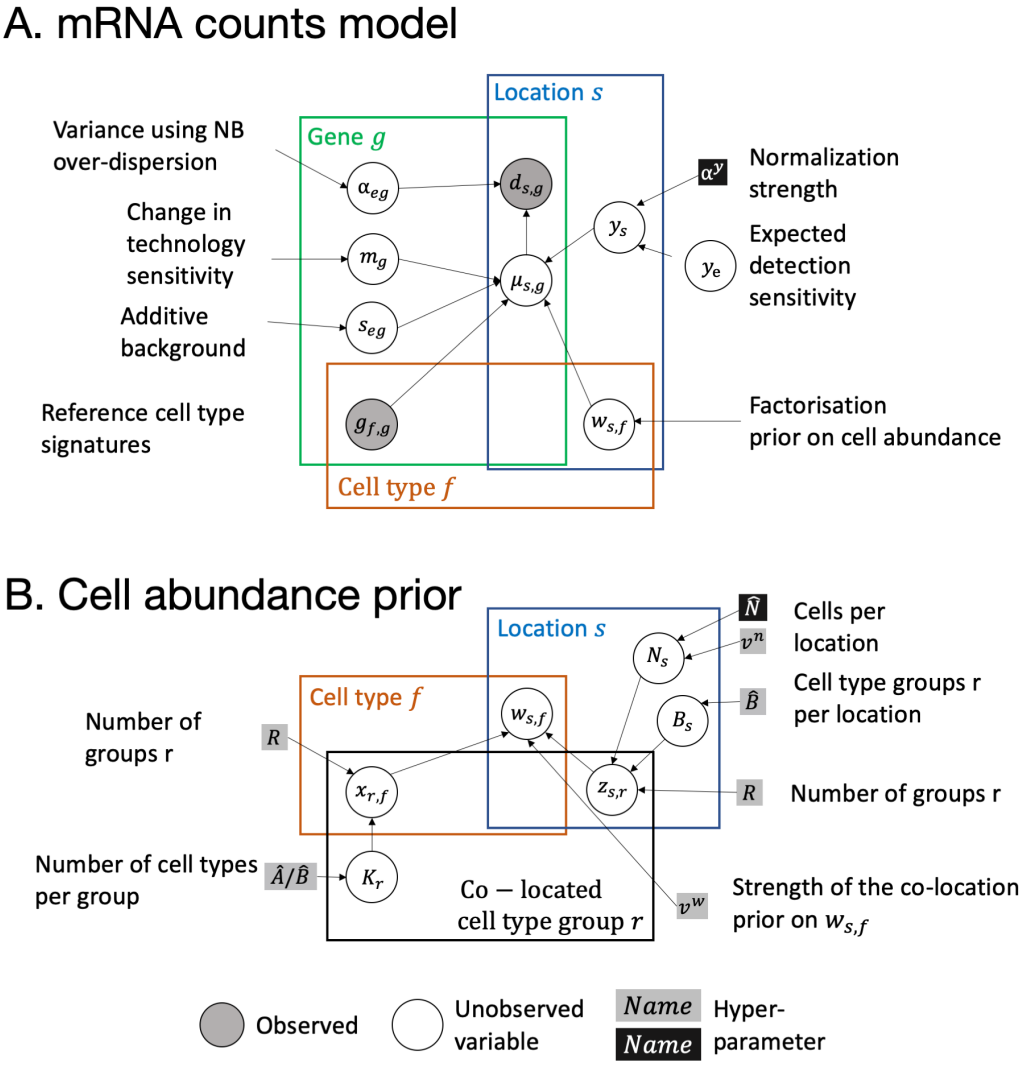


Figure 1: Graphical model illustration of the cell2location model. **A.** Main part of the model describing spatial mRNA counts. **B.** Additional layer of factorisation of $w_{s,f}$ representing similarity in cell type abundance across locations. Square boxes denote hyper-parameters, with black box highlighting the parameters that require adaptation to a specific data set (determined by tissue type and spatial technology), as mentioned in Section 1.2.

and contaminating RNA $s_{e,g}$.

The prior distributions on the unobserved parameters are as follows:

1. **Absolute abundance of cell types across locations.** The regression weights $w_{s,f}$ are modelled as Gamma distributed

$$w_{s,f} \sim \texttt{Gamma}(\mu^w_{s,f} v^w, v^w). \tag{4}$$

Here, v^w denotes a fixed hyper parameter that determines the prior strength. The prior mean parameter $\mu^w_{s,f}$ is modelled in a hierarchical fashion, decomposing the regression weights into R latent groups of cell types $r = \{1, .., R\}$, thereby accounting for linear dependencies in spatial abundance of cell types:

$$\mu^w_{s,f} = \sum_r z_{s,r}\, x_{r,f}. \tag{5}$$

The number of latent groups R is specified by the user (default $R = 50$). Intuitively, these R groups of cell types can be thought of as cellular compartments (or zones) in the tissue, which are characterized by shared cell type composition. We observed that the sensitivity of mapping cell types with fine transcriptional differences in particular increases when accounting for these dependencies (see Fig 5, Fig 4, Fig S25E, Fig S24B).

The prior distributions of $z_{s,r}$ and $x_{r,f}$ are defined to control absolute scale of the cell type abundance estimates, guiding $w_{s,f}$ to the scale of the number of the abundance of cells expressing reference cell type signature f at location s. This is necessary, because while the scaling parameter m_g facilitates the integration across technologies and y_s corrects detection sensitivity, it leads to non-identifiability between m_g, y_s and $w_{s,f}$, unless the informative priors on all three variables are used. In addition, these prior distributions help control the sparsity of how many cell types f are expected at each location s, facilitating application of cell2location to tissues and technologies with varying numbers of cells and cells types per location. The hyper-parameters controlling the $w_{s,f}$ prior can be estimated from paired histology image, if available, or literature information about the tissue. The hyper-parameters controlling y_s can estimated based on total RNA counts in the input data and the quality of the experiment (Section 1.2).

The prior distributions of $z_{s,r}$ and $x_{r,f}$ are defined as stated below:

- The total spatial abundance $z_{s,r}$ of latent cell type groups r across locations s is defined as Gamma distributed with a prior controlling the scale and sparsity of $z_{s,r}$:

$$z_{s,r} \sim \texttt{Gamma}(B_s/R, 1/(N_s/B_s)), \tag{6}$$

where N_s corresponds to the latent average number of cells in each location, and B_s is the latent number of groups r expected in each location s (See eq. (7)-(8) below). Unobserved B_s and N_s parameters are modelled as Gamma-distributed:

$$N_s \sim \texttt{Gamma}(\hat{N} \cdot v^n, v^n) \tag{7}$$

$$B_s \sim \texttt{Gamma}(\hat{B}, 1), \tag{8}$$

where $\hat{N}$ is a user-defined estimate of the expected number of cells per location; $\hat{B}$ is the expected average number of cellular compartments / zones per location (default=7); and v^n denotes the prior strength. Recommendation on setting $\hat{N}$ and v^n hyper-parameters are outlined in section 1.2 (keep $\hat{B}$ at default values).

This is constructed such that $\sum_r z_{s,r} >= N_s$, i.e. the expectation of the sum over $z_{s,r}$ equals to the expected number of cells per location $\sum_r z_{s,r} = N_s$, and that on average each location has high value of $z_{s,r}$ for B_s expected cell type groups.

- The latent variable $x_{r,f}$ represents the contribution of each latent cell type group r to the abundance of each cell type f and is Gamma distributed with a prior controlling the number of cell types f that have high values of $x_{r,f}$ in each group r:

$$x_{r,f} \sim \texttt{Gamma}(K_r/R, K_r), \tag{9}$$

where K_r represents the unobserved number of cell types for each group r. This prior controls the absolute values of $x_{r,f}$ such that on average $\sum_r x_{r,f} = 1$.

K_r is Gamma-distributed with a prior informed by default hyperparameters:

$$K_r \sim \texttt{Gamma}(\hat{A}/\hat{B}, 1), \tag{10}$$

where $\hat{A}$ is expected average number of cell types per location (default=7); and $\hat{B}$ is expected average number of cellular compartments / zones per location (default=7, see section 1.2). This prior tells that each group r has large value of $x_{r,f}$ for many cell types f when $\hat{A} > \hat{B}$, and $\hat{A} = \hat{B}$ corresponds to the prior belief that the spatial abundance of each cell type f is independent from other cell types.

2. **Gene-specific multiplicative scaling factor** m_g is modelled as Gamma-distributed (Eq. (11) - (14)) with hierarchical prior μ^m and α^m which provide regularisation:

$$m_g \sim \texttt{Gamma}(\alpha^m, \alpha^m/\mu^m) \tag{11}$$

$$\alpha^m = 1/(o^m)^2 \tag{12}$$

$$o^m \sim \texttt{Exponential}(3) \tag{13}$$

$$\mu^m \sim \texttt{Gamma}(1, 1). \tag{14}$$

3. **The prior on detection efficiency** y_s **per location** is selected to discourage over-normalisation, such that unless data has evidence of strong within-experiment variability in RNA detection sensitivity across locations, it is assumed to be small and close in values to the mean sensitivity for each experiment/batch y_e (See Fig S5, Fig S8F):

$$y_s \sim \texttt{Gamma}(\alpha^y, \alpha^y/y_e), \tag{15}$$

where α^y is a user-provided hyper-parameter that regularises within-experiment variation (Section 1.2); and y_e is unknown/latent average detection efficiency in each batch/experiment:

$$y_e \sim \texttt{Gamma}(10, 10/\mu^y), \tag{16}$$

where μ^y is estimated using data input to the model ($d_{s,g}$, $g_{f,g}$) and the expected number of cells per location $\hat{N}$ (Section 1.2).

4. **Overdispersion for each gene.** Containment prior[2] is used for modelling unobserved variance using NB over-dispersion $\alpha_{e,g}$. The idea of a containment prior is that, by the prior, NB distribution should be maximally close to Poisson distribution, which is achieved by putting most probability mass on large $\alpha_{e,g}$. So, under the following prior most genes have low over-dispersion:

$$\alpha_{e,g} = 1/o_{e,g}^2 \tag{17}$$

$$o_{e,g} \sim \texttt{Exponential}(\beta^o) \tag{18}$$

$$\beta^o \sim \texttt{Gamma}(9, 3) \tag{19}$$

where constants 9 and 3 correspond to the prior $\alpha_{e,g}$ close to that observed in previous studies in scRNA-seq data[3].

5. **Additive shift for genes.** $s_{e,g}$ represent background RNA counts for every gene g in every spatial experiment e and are modelled as Gamma distributed with hierarchical experiment-specific priors α_e^s and β_e^s which provide regularisation:

$$s_{e,g} \sim \texttt{Gamma}(\alpha_e^s, \alpha_e^s/\mu_e^s) \tag{20}$$

$$\mu_e^s \sim \texttt{Gamma}(1, 100) \tag{21}$$

$$\alpha_e^s = 1/o_e^2 \tag{22}$$

$$o_e \sim \texttt{Exponential}(\beta^s) \tag{23}$$

$$\beta^s \sim \texttt{Gamma}(9, 3) \tag{24}$$

where hierarchical prior α_e^s and μ_e^s model variation in background across experiments e; and β^s serves as hyper-prior allowing additional flexibility for the model to learn the scale of α_e^s rather than relying on user input. These modelling choices follow the containment prior principle outlined above.

Joint modelling of multiple spatial experiments/batches provides the several benefits due to normalisation and sharing of information between experiments:

1. Modelling differences in RNA detection sensitivity across experiments - via hierarchical prior y_e on normalisation parameter y_s - allows direct comparison of cell abundance across multiple tissue sections with varying sequencing depth.

2. Increasing accuracy by improving the ability of the model to distinguish low sensitivity m_g from zero cell abundance $w_{r,f}$, which is achieved by sharing the variable representing gene-specific sensitivity m_g across experiments. Similarly to common practice in single cell data analysis, this is equivalent to regressing out the effect of technology but not the effect of individual experiment.

3. Increasing sensitivity by sharing factorisation prior on cell abundance $w_{r,f}$, namely which cell types tend to co-locate across experiments represented by $x_{r,f}$.

In addition to reporting absolute cell density we also compute absolute number of mRNA molecules ($u_{s,f}$) contributed by each cell type f to each location s:

$$u_{s,f} = w_{s,f}(\sum_g m_g \, g_{f,g}) \tag{25}$$

$u_{s,f}$ are a more robust estimate than either absolute cell density $w_{s,f}$ or relative cell proportions $w_{s,f} / \sum_f w_{s,f}$. For this reason it can be used as a criteria to accurately exclude non-mapped cell types. For comparison of cell and mRNA abundance in the mouse brain see Suppl Fi1es 1 and 6.

1.2 Determining hyper parameter values

There are 3 parameters that are linked to the properties of the data. The remaining hyper-paremeters should be used at their default values as specified in the next section.

The first two priors should be specified by the user for every tissue and the appropriate settings for the third parameter are derived by the software from input data.

1. **Expected cell abundance $\hat{N}$ per location** is a tissue-level global estimate, which can be derived from histology images (H&E or DAPI), paired to the spatial expression data $d_{s,g}$, or when not available - coming from the same tissue type. An estimate can be obtained by manually counting nuclei in a 10-20 locations in the histology image (e.g. using 10X Loupe browser, Fig S8A-B). When suitable histology images are not available, the size of capture regions relative to the size of cells can be used to estimate $\hat{N}$ (Slide-Seq V2, Fig S27).

   For all analysis in this manuscript, a single tissue-level estimate was used.

2. **Hyperparameter α^y for regularising within-experiment variation in RNA detection sensitivity**. By default, we assume that the data does not have strong within-experiment variability in RNA detection sensitivity across locations ($\alpha^y = 200$), which results in values of y_s close to the mean sensitivity for each experiment y_e. However, when strong gradients in mRNA detection sensitivity are observed, a common issue in human adult 10X Visium data, we recommend less strict regularisation ($\alpha^y = 20$) to allow normalisation of such effect and improved sensitivity (data not shown).

3. **Expected detection sensitivity** μ^y is a single global estimate which is derived from average total RNA counts observed in input spatial data $d_{s,g}$, reference cell type signatures $g_{f,g}$, and the value of $\hat{N}$ as follows:

$$\mu^y = \frac{\frac{\sum_s \sum_g d_{s,g}/S}{\hat{N}}}{\sum_f \sum_g g_{f,g}/F} \tag{26}$$

   where S is the total number of locations and F is the total number of cell types.

While good choices of these hyper-parameters can have a positive impact on accuracy, the estimate of relative cell abundance and absolute mRNA abundance is robust to a range of reasonable hyper-parameters choices (Fig S5).

Slide-Seq V2 locations are expected to have fractions of cells in each location ("bead"), where we observed 1-2 cell types per location (Fig S12)[4]. However, we observed large absolute cell abundance $w_{s,f}$ of certain cell types in a fraction of location in the Slide-Seq V2 data (Fig

2H) potentially indicating that additional technical effects may need to be accounted for.

The following priors should be used at their default values unless stated otherwise:

1. The value of **the number of cell types** $\hat{A} = 7$ **per location** and **the number of tissue zones** $\hat{B} = 7$ **per location**. The value of these hyper-priors has minimal effect on model accuracy (Fig S5). Consequently, we recommend setting their values to 7, a single global estimate, which is in the middle of hyper parameter space that was assessed in the sensitivity analysis.

2. **The number of co-located cell type groups is** $R = 50$. Regardless of selected R the model tends to find much lower number of groups (3-7) r that have substantial values of $x_{r,f}$.

3. **Hyper-parameter** v^w, controlling the extent to which the co-abundance prior controls $w_{s,f}$, is set to a medium value of $v^w = 5$. Increasing v^w forces stronger linear dependencies in $w_{s,f}$ leading to over-smoothing and lower model accuracy. Decreasing v^w leads to cell types f modelled as independent, which also decreases model accuracy because of the extent of those effects in real data.

4. v^n **controlling the strength of** $\hat{N}$ **prior** in eq. (8) is set to 1 when a global rather than location-specific estimate $\hat{N}$ is specified.

Optionally, as advanced feature, cell2location can utilise the per-location number of cells that is determined by using image segmentation to estimate the number of nuclei (See Section 1.4, Fig S27). It should be provided as a location-specific $\hat{N}_s$ hyper-parameter and v^n should be increased to make the prior more informative ($v^n = 10$). When the size of cell capture areas varies substantially, such as in laser capture microscopy (LCM) method, we recommend to provide a location-specific $\hat{N}_s$ estimate.

1.3 Inference

Variational Bayesian Inference is used to approximate the posterior, building on the Automatic Differentiation Variational Inference (ADVI) framework implemented in pyro[5]. An alternative implementation in pymc3[6] is provided to simplify prototyping. Briefly, within the ADVI inference framework, the posterior distribution over unknown parameters are approximated by softplus-transformed (to ensure a positive scale) univatiate normal distributions. The parameters of these variational distributions are determined by minimizing the KL divergence between the variational approximation and the true posterior distribution, or equivalently maximizing the evidence lower bound (ELBO loss function). Training is stopped when the ELBO stops increasing, usually after 20,000 - 50,000 iterations of training using ADAM optimiser (learning rate 0.002). Posterior mean, standard deviation, 5% and 95% quantiles for each parameter are computed using 1,000 samples from the Variational posterior distribution. We observed that 5% quantile provides a slightly more accurate estimate of cell abundance compared to the mean so we used these estimates in all paper figures.

Using pymc3 implementation, we empirically observed that the estimated posterior distribution are very stable to initial conditions (training restarts). Hence, we did not consider a model selection step across restarts and did not implement this step in pyro. In principle training restarts can be used as a quality control step. Significant divergence of the solutions between restarts tends to indicate 1) poor match of the cell type reference or 2) large mismatch between the prior distribution of $d_{s,g}$ (sample from the model given hyper-parameter choices) and the

count distribution in the observed data $d_{s,g}$ (prior predictive check). Optionally, the suitability of the selected hyper-parameters can be assessed using prior predictive check (also provided within the cell2location package in pymc3).

As a quality control step, we recommend evaluation of the model fit as described in the tutorial:

1. comparing the posterior mean estimate $log10(\mu_{s,g} + 1)$ to observed data $log10(d_{s,g} + 1)$ (posterior predictive check);

2. assessing that y_s has successfully normalised technical RNA detection effect: 1) variation in estimated total cell abundance $\sum_f w_{s,f}$ should be consistent with histology (areas with more cells should have higher abundance) and 2) spatial distribution of y_s should generally resemble total RNA count $\sum_g d_{s,g}$ across locations.

1.4 Nuclei Segmentation (optional)

To perform nuclei segmentation, H&E-stained image of the Visium-profiled tissue was used. The image was acquired using Hamamatsu slide scanner as described in the section on experimental processing. Full resolution images were divided into sub-images of size about 1000 by 1000 pixels. Then a state-of-the-art convolutional neural network (CNN)-based segmentation pipeline described in Caicedo et al.[7] was used. The model is available in GitHub repository selimsef/dsb2018_topcoders and the code to use it in yozhikoff/segmentation. An ensemble of 32 pre-trained CNNs each with Unet- or FPN-like architecture was used to classify pixels to 3 classes: background, nuclei and nuclei boundaries. Predicted segmentation masks were averaged across CNNs, then a watershed algorithm was applied to refine predicted boundaries and separate individual nuclei. As a final step, a gradient boosting model over morphological features such as nuclei colour, shape and size was used to remove false-positively detected nuclei. Obtained individual nuclei masks were used to compute several morphological features: 1) area (count of pixels in a mask), 2) shape - ratio of lengths of the major and minor axis of an ellipse fitted to the mask, and 3) position - center of mass of the image containing individual mask. Kd-tree approach was used to efficiently assign nuclei to Visium locations. Kd-tree implementation from the scikit-learn package was used. Nuclei positions in the histology image are reported in the GitHub repository vitkl/cell2location_paper.

2 Estimation of reference expression signatures of cell types

We provide two methods for deriving reference expression signatures of cell types $g_{f,g}$ from scRNA-seq expression data $J = \{j_{c,g}\}$ of each gene $g = \{1..G\}$ in cell $c = \{1..C\}$. For both methods, we assume that cell type labels for each cell are provided by the user, for example as obtained using conventional cell clustering to identify putative cell types and sub-populations.

First, untransformed and unnormalised count matrix $J_{c,g}$ is filtered to select expressed genes (Fig S1) at 2 cut-offs: 1) selecting genes detected at mRNA count > 0 in many cells ($> 5\%$ of cells), 2) selecting genes detected at mRNA count > 0 in a few cells ($5\% >$ cell count > 10) but with mean expression across non-zero cells slightly larger than 1 (> 1.12). The cut-offs listed are a guideline, with values adjusted for every data set (Methods).

Next, we offer two alternative methods to derive $g_{f,g}$ from the scRNA-seq count matrix. Although method 1 is computationally more efficient, we recommend method 2 as robust general-purpose solution which provides more accurate results when the scRNA-seq data suffers from batch effects (Fig S24B):

1. **Model-based approach estimation of reference expression signatures of cell types $g_{f,g}$ using a regularised Negative Binomial regression.** This model robustly derives reference expression signatures of cell types $g_{f,g}$ using the data composed of multiple batches $e = \{1..E\}$ and technologies $t = \{1..T\}$. Adapting the assumptions of a range of computational methods for scRNA-seq[8,9], we assume that the expression count matrix $J = \{j_{c,g}\}$ follows a Negative Binomial distribution with unobserved expression levels (rates) $\mu_{c,g}$ and a gene-specific over-dispersion $\alpha_{e,g}$:

$$J_{c,g} \sim \texttt{NB}(\mu_{c,g}, 1/\alpha_g^2). \tag{27}$$

We model $\mu_{c,g}$ as a linear function of reference cell type signatures and technical effects:

$$\mu_{c,g} = (g_{f,g} + b_{e,g})\, h_e\, p_{t,g}. \tag{28}$$

Here, h_e denotes a multiplicative global scaling parameter between experiments/batches e (e.g. differences in sequencing depth); $p_{t,g}$ accounts for multiplicative gene-specific difference in sensitivity between technologies which can be also used to regress out the effect of other covariates (such as donor effect); $b_{e,g}$ accounts for additive background shift of each gene in each experiment e (proxy for free-floating RNA).

The priors on these parameters are set as described below.

2. **Hard coded approach for computing mean count each gene g in cell cluster $f = \{1..F\}$.** This method is computationally highly efficient and generally yields good mapping quality as long as the scRNA-seq reference data are obtained from a single batch (e.g. 10X experiment) or several batches with minimal technical differences (Fig S24A). We also recommend this methods when using Smart-Seq 2 scRNA-seq data, as the distributional assumptions of method 2 are not appropriate for this technology.

3. **The software can utilise user-provided reference expression signatures** $g_{f,g}$, such as estimated with alternative methods and methods that could be developed in the future to account for more complex batch structures and cell type hierarchies (Discussion).

Details of the model-based estimation of reference expression signatures of cell types. Similarly to the cell2location model, the signature model is defined in a Bayesian framework, relying on ADVI inference and implemented in pyro. The following priors are used on parameters listed in eq. 28:

$$g_{f,g} \sim \texttt{Gamma}(1, 1), \tag{29}$$

$$h_e \sim \texttt{Gamma}(1, 1), \tag{30}$$

$$p_{t,g} \sim \texttt{Gamma}(200, 200/1). \tag{31}$$

where the constant 200 is chosen to provide regularisation and penalise deviation of $p_{t,g}$ values from 1, similarly to the approach used for y_s prior in eq. 15-16.

The prior on $b_{e,g}$ is specified identically to $s_{e,g}$ in eq. 20-24. The prior on α_g is specified similarly to $\alpha_{e,g}$ in eq. 17-19 as follows:

$$\alpha_g = 1/o_g^2 \tag{32}$$

$$g \sim \texttt{Exponential}(\beta^o) \tag{33}$$

$$\beta^o \sim \texttt{Gamma}(9, 3) \tag{34}$$

where constants 9 and 3 correspond to the prior $\alpha_{e,g}$ being close to that observed in previous studies in scRNA-seq data[3].

$g_{f,g}$ is initialised at analytical average for each cell type f. Such informative initialisation leads to fast convergence. Implementation of training using mini batches makes the model fast (10min-60min on GPU) and scalable to very large data sets (> 200k cells). Training hyper-parameters are used as specified in Section 1.3 with recommended mini-batch size of 2500. The model training should stopped at the number of epochs needed for convergence, which should be determined for every data set by the user by examining the history of ELBO loss.

To verify that the model successfully accounted to non-biological sample effects h_e, $p_{t,g}$ and $b_{e,g}$, a corrected expression matrix $J_{c,g}^{corrected}$ is obtained by normalising $J_{c,g}$:

$$J_{c,g}^{corrected} = J_{c,g}/(h_e\, p_{t,g}) - b_{e,g} \tag{35}$$

$J_{c,g}^{corrected}$ matrix is subjected to standard Scanpy workflow[10] to verify the mixing of the cells from different samples and technologies within matching cells types (Fig S19B).

3 Comparison of cell2location with alternative methods

Cell2location is related to alternative methods, which have recently been proposed or developed in parallel (Table 1). While cell2location and alternative methods are all based on regression, there are important differences that set cell2location apart from existing methods. These include in particular:

1. Cell2location models linear dependencies in abundance of cell types across locations by employing a factorized prior on the cell abundance variable $w_{s,f}$ (See Section 1.1). This additional level of hierarchy in cell2location is a major contributor to the model performance (Fig S25D-E, see below), and is not considered by existing methods.

2. Cell2location allows for joint modelling of multiple batches/experiments, thereby 1) sharing statistical strength for estimating key parameters, in particular those mentioned in point 1 and the gene-specific difference in sensitivity between technologies m_g, and 2) enabling direct comparison of cell abundance across batches with varying RNA detection efficiency by utilising normalisation using latent variable y_s.

3. Cell2location is implemented within a Bayesian framework, which allows for leveraging prior information on key experimental parameters in a rigorous manner (also see Section 1.2). The modular implementation in both pymc3 and pyro frameworks will also facilitate future extensions, for example to adapt cell2location to future spatial technologies (e.g. recently developed Nanostring WTA model extension provided on GitHub).

For a tabular comparison of specific features see Table 1, which considers the features mentioned above and other criteria.

In addition, we present a direct assessment how the individual modelling choices in cell2location (points 4-6 in Table 1) contribute to the model performance (Fig S4, Fig S24, Fig S25). Specifically, we evaluate the importance of 1) prior factorisation of cell abundance $w_{s,f}$; 2) gene-specific technology sensitivity scaling m_g (in both cases we observed substantial improvement

in accuracy); 3) NB regression for estimating reference signatures of cell types compared to hard-coded method (see Section 2).

This is assessed by simplifying the model presented in Section 1.1 in the following way:

1. **No $w_{s,f}$ factorisation model.** To estimate the impact of factorisation of cell abundance $w_{s,f}$ on accuracy (Fig S4, Fig S24, Fig S25E), we define a simplified cell2location model, where prior factorisation is replaced with the following hierarchical prior designed to enable meaningful comparison and as such allowing estimation of absolute density:

$$w_{s,f} \sim \texttt{Gamma}(A/F, 1/(N/A)), \tag{36}$$

where F corresponds to the number of cell types f, N corresponds to a single global estimate of the number of cells per location, A corresponds to a single global estimate of the number of cell types per location. The priors on N and A are defined as follows:

$$N \sim \texttt{Gamma}(mu = \hat{N}, \sigma^2 = \hat{N}) \tag{37}$$

$$A \sim \texttt{Gamma}(mu = \hat{A}, \sigma^2 = \hat{A}), \tag{38}$$

where $\hat{N}$ is a user-defined estimate of the expected number of cells per location, which is set to an appropriate value for a given tissue using the guidance in Section 1.2; $\hat{A}$ is the expected number of cell types (default $\hat{A} = 7$).

2. **No m_g model.** To estimate the impact of gene-specific technology sensitivity scaling m_g (Fig S4, Fig S24, Fig S25E) on accuracy we define a simplified cell2location model which does not use m_g but utilises all other parameters as described in Section 1.1.

3. **No NB regression approach.** We compare the two alternative strategies to estimate reference signatures (see Section 2, Fig S24, Fig S25E) in the context of the quantitative benchmarking analysis in the lymph node and the gut (See Methods), which demonstrates that the regression-based model results in higher mapping accuracy than the simpler hard-coded method (Fig S24), by combining data across technologies and batches (Fig S19).

   We observed that the hard-coded method based on mean counts yields similar results to the model-based approach in the mouse brain snRNA-seq data, where the batch effects are relatively minor (Fig S24A). In contrast, the scRNA-seq data from human lymphoid organs has more substantial batch effects, as these data span 3 published studies from 3 organs, 21 experiments/batches, as well as 3' and 5' technologies (Fig S19). Consistently, we observed mapping of T CD4+ TfH GC cells, but not T CD4+ TfH and T CD4+ cells, to GC zone with model-based approach whereas hard-coded approach led substantial abundance of T CD4+ TfH and T CD4+ cells in GC dark zone where those cell types should not be located according to the prior knowledge (Fig S24C, Methods).

   These results indicate that the model-based approach is in general more accurate, in particular when the scRNA reference is composed of multiple batches and technologies.

4 Downstream analyses

The output of cell2location can be used as a starting point for a range of different downstream analyses. Here, we outline strategies how cell2location can be combined with conventional clustering, as well as non-negative matrix decomposition.

Table 1: Comparison of cell2location with other alternative methods

Criteria	cell2location	stereoscope[11]	Seurat V3[12]	SpotLIGHT[13]	NNLS (autogenes)[14]	RCTD[4]
1. Data distribution, normalisation	Negative Binomial, non-negative parameters	Negative Binomial, non-negative parameters	Normalised log(x+1)	Counts normalised by standard deviation, non-negative parameters	Counts, non-negative parameters	Negative Binomial, log-normal model
2. Inference method	Variational Inference	Penalised maximum likelihood	PCA + MNN	NNLS	NNLS	Penalised maximum likelihood
3. Method for estimating signatures	2 methods provided but user-derived signatures can be used (See Section 2)	Hard-coded method (NB regression with unusual parametrisation)	Not needed	Hard-coded method (topic model)	User-derived signatures	Hard-coded method (average gene expression)
4. Accounting for the difference in sensitivity between single cell and spatial technology	Yes	Yes	Unclear	No	No	Yes
5. Models similarity of locations between cell types (prior factorisation of $w_{s,f}$)	Yes	No	No	No	No	No
6. Method for estimating signatures accounts for technology and batch effects in scRNA data	Yes	No	Possible	No	No	No
7. Joint modelling of multiple spatial experiments	Yes	No	Unclear	No	No	No
8. Normalisation of RNA detection sensitivity	Yes, regularised using hierarchical priors	Yes, non-regularised	Unclear	No	No	Yes, non-regularised
9. GPU acceleration	Yes	Yes	No	No	No	No
10. Mini-batch training	Yes, at lower accuracy	Yes, likely at lower accuracy	No	No	No	No
11. Language	Python	Python	R	R	Autogenes in Python, other in R	R

4.1 Identification of tissue regions by clustering

The cell abundance $w_{s,f}$ estimated by cell2location model is used as input to cluster locations s into clusters representing non-overlapping regions. This can be achieved by constructing a KNN graph that represents connectivity of locations in terms of their cell composition (function scanpy.pp.neighbors). Reference implementations of Leiden clustering, such as the implementation (function scanpy.tl.leiden) in the Scanpy package[10], can utilise cell composition KNN graph to group organisation of the tissue. In the experiments presented in the paper, Leiden clustering was performed with default settings except for the resolution parameter (see main text Methods). The resulting clusters were cross-referenced with the mouse brain anatomy using corresponding histology images and similar clusters were merged to generate the broad region map.

4.2 Groups of co-located cell types

As an alternative to clustering, we considered non-negative matrix factorisation (NMF), thereby identifying cellular compartments within the tissue. Briefly, in NMF, absolute cell abundances $w_{s,f}$ of each cell type f across locations s are modelled as an additive function of the cell type groups r (cellular compartments). The factorization assumption of NMF combined with the

additivity constraint allow for identifying cellular compartments, which unlike discrete clustering can co-occupy one common location (Fig S20). Briefly, cell abundance estimates $w_{s,f}$ are modelled as an additive decomposition:

$$w_{s,f} = \sum_{r} z_{s,r}\, x_{r,f} \tag{39}$$

where $x_{r,f}$ represents NMF weights describing the contribution of each cell type group r to cell types f; $z_{s,r}$ represents NMF weights describing the abundances of each cell type group r across locations s.

Similarly to cell2location model (eq. (25)), we compute $u_{s,r}$ which can be intuitively interpreted as the number of cells from each cell type group r in each location s and use it for visualisation:

$$u_{s,r} = z_{s,r} \sum_{r} x_{r,f} \tag{40}$$

Inference is performed using standard NMF in the scikit-learn package[15]. The model is trained 2 times to evaluate stability of the identified cellular compartments. The first training iteration is always selected.

Rationale for applying the NMF to multi-cell spatial data. We hypothesise that linear dependencies in the abundance of cell types across locations reflects the interaction between these cell types, specifically, any interaction that could affect mutual location of these cell types. For this reason we tested this approach on human lymph nodes, a tissue where locations of cell types are determined by their interactions (Fig 4D). 10X Visium technology combines cell types located within the $55\mu m$ diameter of capture areas, a distance at which the action of local paracrine signals and direct adhesive contacts can be observed, likely driving linear dependencies in cell abundance. We assume that the signal does not propagate between neighbouring locations spaced by $45\mu m$ gaps, therefore using proximity information is not essential. Therefore we note that that this interpretation of NMF decomposition of cell abundances is a unique feature of grid-based technologies that aggregate multiple cells and cell types within single locations. This interpretation does not directly apply to technologies with increased resolution (Slide-Seq V2) or technologies where capture regions represent selected features.

Potential cell interactions driving linear dependencies in cell type abundance can be classified into:

- Signals promoting proliferation and survival of other cell type
- Adhesion molecules that increase the frequency of contacts
- Recruitment signals such as CXCR5 / CXCL13[16]
- Repulsive signals, inhibiting the action of chemo-taxis signals such as PD-1 / PD-L1 interactions[17]
- Interactions that induce new transcriptional states

5 Constructing a synthetic spatial transcriptomics data set

This section outlines the mathematical description and the step-by-step procedure for constructing a synthetic spatial transcriptomics data set. The proceduce is comprised of three steps: 1) simulating abundance of cell types across locations, 2) simulating expected multi-cell mRNA expression of genes across locations, 3) simulating multi-cell mRNA integer counts weighted by technology difference effect.

First, follow this step-by-step procedure to generate 1) ground truth spatial abundance $w_{s,f}$, 2) integer cell count $count_{s,f}$ and 3) fraction of mRNA captured $fraq_{s,f}$ for every cell type f across locations s:

1. Assign cell types to ubiquitous (n=8) and regional (n=41) abundance patterns (denoted as r).

2. Perform binary assignment of 41 sparse cell types to 12 tissue zones and 8 ubiquitous cell types to 8 ubiquitous patterns (total n=20), shown in Fig S2A and denoted as $x_{r,f}$.

3. Stratified by location pattern, randomly assign up to 20% of cell types to high abundance groups and all other cell types to low abundance groups. Generate per cell type average abundance d_f, which is different for 4 groups shown in Fig 1B:

   (a) Ubiquitous and low abundance: 5 cell types present in most locations at abundance:

   $$d_f \sim \texttt{Gamma}(1.0 \cdot 5.0, 5.0). \tag{41}$$

   (b) Ubiquitous and high abundance: 3 cell types present in most locations at abundance:

   $$d_f \sim \texttt{Gamma}(2.8 \cdot 5.0, 5.0). \tag{42}$$

   (c) Sparse and low abundance: 32 cell types present in sparse tissue zones at abundance:

   $$d_f \sim \texttt{Gamma}(1.0 \cdot 5.0, 5.0). \tag{43}$$

   (d) Sparse and high abundance: 9 cell types present in sparse tissue zones at abundance:

   $$d_f \sim \texttt{Gamma}(2.8 \cdot 5.0, 5.0). \tag{44}$$

   By following this procedure, sparsity and abundance parameters for each cell type were generated that produced an average total number of cells per location close to 10, mimicking cell count observed by nuclear segmentation of the mouse brain histology images (Fig S8, Suppl Methods). Per cell type maximum abundance d_f was used to scale $x_{r,f} = x_{r,f} * d_f$, thus defining the average abundance of each cell type across patterns r.

4. Next, fenerate spatial abundance $z_{s,r}$ for locations s for 20 location patterns (denoted as r) representing 12 tissue zones and 8 ubiquitous cell types. Gaussian Process in 50x50 grid of locations was used to generate spatial abundance values with randomly generated bandwidth parameters:

   $$bw_r \sim \texttt{Gamma}(8.0 \cdot 1.2, 1.2). \tag{45}$$

   Bandwidth controls how quickly cell abundance changes in spatial coordinates. Eta parameter controls the sparsity with large values of eta corresponding to high variance in Gaussian Process, putting distribution weight away from 0 and correspondingly producing sparse cell abundance maps.

   (a) 8 ubiquitous patterns with for non-zero abundance in most locations:

   $$z_{s,r} \sim \texttt{GP}(\mu = 0, eta = 0.5, bw = bw_r). \tag{46}$$

   (b) 12 tissue zones with sparse locations:

   $$z_{s,r} \sim \texttt{GP}(\mu = 0, eta = 1.5, bw = bw_r). \tag{47}$$

5. To ensure positive scale, cell abundance for each pattern $z_{s,r}$ were softmax-transformed:

$$z_{s,r} = e^{z_{s,r}} / \sum_r e^{z_{s,r}}. \tag{48}$$

Next, to ensure that maximum for each location pattern r is equal to 1 further normalisation was applied:

$$z_{s,r} = z_{s,r} / \texttt{max}_s z_{s,r}, \tag{49}$$

which is needed to use abundances established in step 3 as average value for each cell type.

6. Per cell type abundance for each location s was generated as a weighted sum of contributions of tissue zones (shown in Fig S2B):

$$w_{s,f} = (\sum_r z_{s,r}\ x_{r,f})\ q_{s,f}, \tag{50}$$

where $q_{s,f}$ introduces randomness to abundances of individual cell types:

$$log(q_{s,f}) \sim \texttt{Normal}(\mu = 0, \sigma = 0.35). \tag{51}$$

This additional variability mimics the observation that co-located cell types in real tissues do not have perfectly correlated abundance within tissue zones.

7. Cell abundance $w_{s,f}$ was used to generate integer cell count $count_{s,f}$ and fraction of mRNA captured $fraq_{s,f}$ for each location and cell type as follows:

   (a) generate integer cell count by rounding $\lceil w_{s,f} \rceil$ to the smallest integer, such that $count_{s,f} >= w_{s,f}$).

   (b) Compute the fraction of mRNA captured as $fraq_{s,f} = w_{s,f} / count_{s,f}$.

Second, follow this step-by-step procedure to generate expected multi-cell mRNA count profiles $ed_{s,g}$ for every gene g in a given location s using 1) the integer cell count $count_{s,f}$ and 2) the fraction of mRNA captured $fraq_{s,f}$ for each cell type f in a given location s (from the steps above), by combining cells c drawn from reference cell types f in the snRNA-seq data $j_{c,g}$ as follows:

1. Create a matrix $a_{s,c}$ which that defines which cells should be added to construct every location. Randomly select indices of cells c of a given cell type f ($c \subset f$) to form a subset s for every location s containing $n = count_{s,f}$ cells:

$$count_{s,f} = \sum_s a_{s,c \subset f}. \tag{52}$$

2. Construct per cell type expected mRNA abundance profiles for a given location and cell type:

$$ed_{s,f,g} = (\sum_c a_{s,c \subset f} \cdot j_{c,g}) \cdot fraq_{s,f}. \tag{53}$$

3. Construct multi-cell expected mRNA abundance profiles by adding mRNA across all cell types:

$$ed_{s,g} = \sum_f ed_{s,f,g}. \tag{54}$$

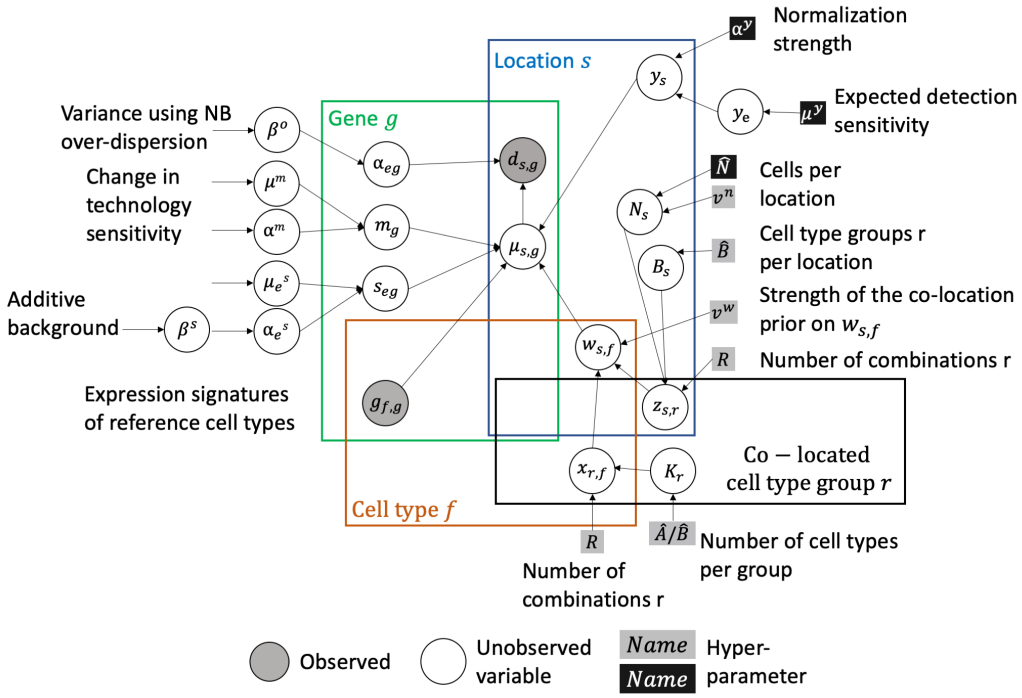


Figure 2: Illustration of the corresponding graphical model. Square boxes denote hyper-parameters, with black box highlighting the parameters that require adaptation to a specific data set (determined by tissue type and spatial technology), as mentioned in Section 1.2.

Third, follow this step-by-step procedure to generate multi-cell mRNA integer counts $d_{s,g}$. In this step, gene-specific scaling was applied (m_g) to mimic the difference in sensitivity between technologies and counts were samples from Poisson distribution:

$$d_{s,g} \sim \texttt{Poisson}(ed_{s,g}\ m_g), \tag{55}$$

where m_g characterises the difference between the mouse brain Visium data and single nucleus RNA-seq reference (Fig S2C, estimated by cell2location). Using these values makes the simulated data representative for mapping single nucleus RNA-seq derived reference cell types. Under this simulation, the total number of mRNA per location mimics that observed in mouse brain data (Fig S2D).

6 Appendix

6.1 Complete graphical model diagram

The complete illustration of the cell2location graphical model, complementing Figure 1, is shown in Figure 2.

6.2 Notation

Two alternative parameterizations of Gamma distribution are used: 1) α (also called shape or over-dispersion) and β (also called rate); 2) μ and σ^2; with the link between the two given by:

$$\mu = \alpha/\beta; \sigma^2 = \alpha/\beta^2 \tag{56}$$

$$\alpha = \mu^2/\sigma^2; \beta = \mu/\sigma^2 \tag{57}$$

By default, α and β parametrization is used; when μ and σ^2 are used their notation is shown.

References

1. Sarkar, A. & Stephens, M. *Separating measurement and expression models clarifies confusion in single cell RNA-seq analysis* en. preprint (Genomics, Apr. 2020). http://biorxiv.org/lookup/doi/10.1101/2020.04.07.030007 (2020).
2. Simpson, D., Rue, H., Riebler, A., Martins, T. G. & Sørbye, S. H. Penalising Model Component Complexity: A Principled, Practical Approach to Constructing Priors. en. *Statistical Science* **32,** 1–28. ISSN: 0883-4237. http://projecteuclid.org/euclid.ss/1491465621 (2020) (Feb. 2017).
3. Choudhary, S. & Satija, R. *Comparison and evaluation of statistical error models for scRNA-seq* en. preprint (Genomics, July 2021). http://biorxiv.org/lookup/doi/10.1101/2021.07.07.451498 (2021).
4. Cable, D. M. *et al. Robust decomposition of cell type mixtures in spatial transcriptomics* en. preprint (Genomics, May 2020). http://biorxiv.org/lookup/doi/10.1101/2020.05.07.082750 (2020).
5. Bingham, E. *et al.* Pyro: Deep Universal Probabilistic Programming. *Journal of Machine Learning Research* (2018).
6. Salvatier, J., Wiecki, T. V. & Fonnesbeck, C. Probabilistic programming in Python using PyMC3. en. *PeerJ Computer Science* **2,** e55. ISSN: 2376-5992. https://peerj.com/articles/cs-55 (2020) (Apr. 2016).
7. Caicedo, J. C. *et al.* Nucleus segmentation across imaging experiments: the 2018 Data Science Bowl. en. *Nature Methods* **16,** 1247–1253. ISSN: 1548-7091, 1548-7105. http://www.nature.com/articles/s41592-019-0612-7 (2020) (Dec. 2019).
8. Lopez, R., Regier, J., Cole, M. B., Jordan, M. I. & Yosef, N. Deep generative modeling for single-cell transcriptomics. en. *Nature Methods* **15,** 1053–1058. ISSN: 1548-7091, 1548-7105. http://www.nature.com/articles/s41592-018-0229-2 (2020) (Dec. 2018).
9. Love, M. I., Huber, W. & Anders, S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. en. *Genome Biology* **15.** ISSN: 1474-760X. http://genomebiology.biomedcentral.com/articles/10.1186/s13059-014-0550-8 (2020) (Dec. 2014).
10. Wolf, F. A., Angerer, P. & Theis, F. J. SCANPY: large-scale single-cell gene expression data analysis. en. *Genome Biology* **19.** ISSN: 1474-760X. https://genomebiology.biomedcentral.com/articles/10.1186/s13059-017-1382-0 (2020) (Dec. 2018).
11. Andersson, A. *et al. Spatial mapping of cell types by integration of transcriptomics data* en. preprint (Bioinformatics, Dec. 2019). http://biorxiv.org/lookup/doi/10.1101/2019.12.13.874495 (2020).
12. Stuart, T. *et al. Comprehensive integration of single cell data* en. preprint (Genomics, Nov. 2018). http://biorxiv.org/lookup/doi/10.1101/460147 (2020).

13. Elosua, M., Nieto, P., Mereu, E., Gut, I. & Heyn, H. *SPOTlight: Seeded NMF regression to Deconvolute Spatial Transcriptomics Spots with Single-Cell Transcriptomes* en. preprint (Bioinformatics, June 2020). http://biorxiv.org/lookup/doi/10.1101/2020.06.03.131334 (2020).

14. Aliee, H. & Theis, F. *AutoGeneS: Automatic gene selection using multi-objective optimization for RNA-seq deconvolution* en. preprint (Bioinformatics, Feb. 2020). http://biorxiv.org/lookup/doi/10.1101/2020.02.21.940650 (2020).

15. Pedregosa, F. *et al.* Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* **12,** 2825–2830 (2011).

16. Van de Pavert, S. A. *et al.* Chemokine CXCL13 is essential for lymph node initiation and is induced by retinoic acid and neuronal stimulation. en. *Nature Immunology* **10,** 1193–1199. ISSN: 1529-2908, 1529-2916. http://www.nature.com/articles/ni.1789 (2020) (Nov. 2009).

17. Shi, J. *et al.* PD-1 Controls Follicular T Helper Cell Positioning and Function. en. *Immunity* **49,** 264–274.e4. ISSN: 10747613. https://linkinghub.elsevier.com/retrieve/pii/S1074761318302942 (2020) (Aug. 2018).